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INTRODUCTION

ALGEBRAIC TOPOLOGY

$$\{x \in X \mid P(x)\}$$

denotes the subset of X for which $P(x)$ is true.

If $I = \{j\}$ is a set and $\{A_j\}$ is a family of sets indexed by I , their *union* is denoted by $\bigcup A_j$ (or by $\bigcup_{j \in I} A_j$), their *intersection* is denoted by $\bigcap A_j$ (or by $\bigcap_{j \in I} A_j$), their *cartesian product* is denoted by $\prod A_j$ (or by $\prod_{j \in I} A_j$), and their *set sum* (sometimes called their *disjoint union*) is denoted by $\bigvee A_j$ (or by $\bigvee_{j \in I} A_j$) and is defined by $\bigvee A_j = \bigcup (j \times A_j)$. In case $I = \{1, 2, \dots, n\}$, we also use the notation $A_1 \cup A_2 \cup \dots \cup A_n$, $A_1 \cap A_2 \cap \dots \cap A_n$, $A_1 \times A_2 \times \dots \times A_n$, and $A_1 \vee A_2 \vee \dots \vee A_n$, respectively, for the union, intersection, cartesian product, and set sum.

A *function* (or *map*) f from A to B is denoted by $f: A \rightarrow B$. The set of all functions from A to B is denoted by B^A . If $A' \subset A$, there is an *inclusion map* $i: A' \rightarrow A$, and we use the notation $i: A' \subset A$ to indicate that A' is a subset of A and i is the inclusion map. The inclusion map from a set A to itself is called the *identity map* of A and is denoted by 1_A . If $I' \subset I$, there is an inclusion map

$$i_{I'}: \bigvee_{j \in I'} A_j \subset \bigvee_{j \in I} A_j$$

An *equivalence relation* in a set A is a relation $\sim$ between elements of A which is *reflexive* (that is, $a \sim a$ for all $a \in A$), *symmetric* (that is, $a \sim a'$ implies $a' \sim a$ for $a, a' \in A$), and *transitive* (that is, $a \sim a'$ and $a' \sim a''$ imply $a \sim a''$ for $a, a', a'' \in A$). The *equivalence class* of $a \in A$ with respect to $\sim$ is the subset $\{a' \in A \mid a \sim a'\}$. The set of all equivalence classes of elements of A with respect to $\sim$ is denoted by $A/\sim$ and is called a *quotient set* of A . There is a *projection map* $A \rightarrow A/\sim$ which sends $a \in A$ to its equivalence class. If I' is a nonempty subset of I , there is also a *projection map*

$$pr: \prod_{j \in I} A_j \rightarrow \prod_{j \in I'} A_j$$

(which is a projection map in the sense above).

Given functions $f: A \rightarrow B$ and $g: B \rightarrow C$, their *composite* $g \circ f$ (also denoted by gf) is the function from A to C defined by $(g \circ f)(a) = g(f(a))$ for $a \in A$. If $A' \subset A$ and $f: A \rightarrow B$, the *restriction* of f to A' is the function $f|A': A' \rightarrow B$ defined by $(f|A')(a') = f(a')$ for $a' \in A'$ (thus $f|A' = f \circ i$, where $i: A' \subset A$), and the function f is called an *extension* of $f|A'$ to A .

An *injection* (or *injective function*) is a function $f: A \rightarrow B$ such that $f(a_1) = f(a_2)$ implies $a_1 = a_2$ for $a_1, a_2 \in A$. A *surjection* (or *surjective function*) is a function $f: A \rightarrow B$ such that $b \in B$ implies that there is $a \in A$ with $f(a) = b$. A *bijection* (also called a *bijective function* or a *one-to-one correspondence*) is a function which is both injective and surjective.

A *partial order* in a set A is a relation $\leq$ between elements of A which is reflexive and transitive (note that it is not assumed that $a \leq a'$ and $a' \leq a$ imply $a = a'$). A *total order* (or *simple order*) in A is a partial order in A such that for $a, a' \in A$ either $a \leq a'$ or $a' \leq a$ and which is *antisymmetric* (that is, $a \leq a'$ and $a' \leq a$ imply $a = a'$). A *partially ordered set* is a set with a partial order, and a *totally ordered set* is a set with a total order.

THE READER OF THIS BOOK IS ASSUMED TO HAVE A GRASP OF THE ELEMENTARY concepts of set theory, general topology, and algebra. Following are brief summaries of some concepts and results in these areas which are used in this book. Those listed explicitly are done so either because they may not be exactly standard or because they are of particular importance in the subsequent text.

1 SET THEORY¹

The terms “set,” “family,” and “collection” are synonyms, and the term “class” is reserved for an aggregate which is not assumed to be a set (for example, the class of all sets). If X is a set and $P(x)$ is a statement which is either true or false for each element $x \in X$, then

¹ As a general reference see P. R. Halmos, *Naïve Set Theory*, D. Van Nostrand Company, Inc., Princeton, N.J., 1960.

1 ZORN'S LEMMA A partially ordered set in which every simply ordered subset has an upper bound contains maximal elements.

A directed set Λ is a set with a partial-order relation $\leq$ such that for $\alpha, \beta \in \Lambda$ there is $\gamma \in \Lambda$ with $\alpha \leq \gamma$ and $\beta \leq \gamma$. A direct system of sets $\{A^\alpha, f_\alpha^\beta\}$ consists of a collection of sets $\{A^\alpha\}$ indexed by a directed set $\Lambda = \{\alpha\}$ and a collection of functions $f_\alpha^\beta: A^\alpha \rightarrow A^\beta$ for every pair $\alpha \leq \beta$ such that

- (a) $f_\alpha^\alpha = I_{A^\alpha}: A^\alpha \subset A^\alpha$ for all $\alpha \in \Lambda$
- (b) $f_\alpha^\gamma = f_\beta^\gamma \circ f_\alpha^\beta: A^\alpha \rightarrow A^\gamma$ for $\alpha \leq \beta \leq \gamma$ in Λ

The direct limit of the direct system, denoted by $\lim_{\rightarrow} \{A^\alpha\}$, is the set of equivalence classes of $\bigvee A^\alpha$ with respect to the equivalence relation $a^\alpha \sim a^\beta$ if there is γ with $\alpha \leq \gamma$ and $\beta \leq \gamma$ such that $f_\alpha^\gamma a^\alpha = f_\beta^\gamma a^\beta$. For each α there is a map $i_\alpha: A^\alpha \rightarrow \lim_{\rightarrow} \{A^\alpha\}$, and if $\alpha \leq \beta$, then $i_\alpha = i_\beta \circ f_\alpha^\beta$.

2 Given a direct system of sets $\{A^\alpha, f_\alpha^\beta\}$ and given a set B and for every $\alpha \in \Lambda$ a function $g_\alpha: A^\alpha \rightarrow B$ such that $g_\alpha = g_\beta \circ f_\alpha^\beta$ if $\alpha \leq \beta$, there is a unique map $g: \lim_{\rightarrow} \{A^\alpha\} \rightarrow B$ such that $g \circ i_\alpha = g_\alpha$ for all $\alpha \in \Lambda$.

3 With the same notation as in theorem 2, the map g is a bijection if and only if both the following hold:

- (a) $B = \bigcup g_\alpha(A^\alpha)$
- (b) $g_\alpha(a^\alpha) = g_\beta(a^\beta)$ if and only if there is γ with $\alpha \leq \gamma$ and $\beta \leq \gamma$ such that $f_\alpha^\gamma(a^\alpha) = f_\beta^\gamma(a^\beta)$

Let $\{A_j\}$ be a collection of sets indexed by $J = \{j\}$. Let Λ be the collection of finite subsets of J and define $\alpha \leq \beta$ for $\alpha, \beta \in \Lambda$ if $\alpha \subset \beta$. Then Λ is a directed set and there is a direct system $\{A^\alpha\}$ defined by $A^\alpha = \bigvee_{j \in \alpha} A_j$ and if $\alpha \leq \beta$, then $f_\alpha^\beta: A^\alpha \rightarrow A^\beta$ is the injection map. Let $g_\alpha: A^\alpha \rightarrow \bigvee_{j \in J} A_j$ be the injection map.

4 With the above notation, there is a bijection $g: \lim_{\rightarrow} \{A^\alpha\} \rightarrow \bigvee_{j \in J} A_j$ such that $g \circ i_\alpha = g_\alpha$ (that is, any set sum is the direct limit of its finite partial set sums).

An inverse system of sets $\{A_\alpha, f_\alpha^\beta\}$ consists of a collection of sets $\{A_\alpha\}$ indexed by a directed set $\Lambda = \{\alpha\}$ and a collection of functions $f_\alpha^\beta: A_\beta \rightarrow A_\alpha$ for $\alpha \leq \beta$ such that

- (a) $f_\alpha^\alpha = I_{A_\alpha}: A_\alpha \subset A_\alpha$ for $\alpha \in \Lambda$
- (b) $f_\alpha^\gamma = f_\alpha^\beta \circ f_\beta^\gamma: A_\gamma \rightarrow A_\alpha$ for $\alpha \leq \beta \leq \gamma$ in Λ

The inverse limit of the inverse system, denoted by $\lim_{\leftarrow} \{A_\alpha\}$, is the subset of $\bigtimes A_\alpha$ consisting of all points (a_α) such that if $\alpha \leq \beta$, then $a_\alpha = f_\alpha^\beta a_\beta$. For each α there is a map $p_\alpha: \lim_{\leftarrow} \{A_\alpha\} \rightarrow A_\alpha$ and if $\alpha \leq \beta$, then $p_\alpha = f_\alpha^\beta \circ p_\beta$.

5 Given an inverse system of sets $\{A_\alpha, f_\alpha^\beta\}$ and given a set B and for every $\alpha \in \Lambda$ a function $g_\alpha: B \rightarrow A_\alpha$ such that $g_\alpha = f_\alpha^\beta \circ g_\beta$ if $\alpha \leq \beta$, there is a unique function $g: B \rightarrow \lim_{\leftarrow} \{A_\alpha\}$ such that $g_\alpha = p_\alpha \circ g$ for all $\alpha \in \Lambda$.

6 With the same notation as in theorem 5, the map g is a bijection if and only if both the following hold:

- (a) $g_\alpha(b) = g_\alpha(b')$ for all $\alpha \in \Lambda$ implies $b = b'$
- (b) Given $(a_\alpha) \in \bigtimes A_\alpha$ such that $a_\alpha = f_\alpha^\beta a_\beta$ if $\alpha \leq \beta$, there is $b \in B$ such that $g_\alpha(b) = a_\alpha$ for all $\alpha \in \Lambda$

Let $\{A^j\}$ be a collection of sets indexed by $J = \{j\}$. Let Λ be the collection of finite nonempty subsets of J , and define $\alpha \leq \beta$ for $\alpha, \beta \in \Lambda$ if $\alpha \subset \beta$. Then Λ is a directed set and there is an inverse system $\{A_\alpha\}$ defined by $A_\alpha = \bigtimes_{j \in \alpha} A^j$, and if $\alpha \leq \beta$, $f_\alpha^\beta: A_\beta \rightarrow A_\alpha$ is the projection map. For each $\alpha \in \Lambda$ let $g_\alpha: \bigtimes_{j \in J} A^j \rightarrow A_\alpha$ be the projection map.

7 With the above notation, there is a bijection $g: \bigtimes_{j \in J} A^j \rightarrow \lim_{\leftarrow} \{A_\alpha\}$ such that $g_\alpha = p_\alpha \circ g$ (that is, any cartesian product is the inverse limit of its finite partial cartesian products).

2 GENERAL TOPOLOGY¹

A topological space, also called a space, is not assumed to satisfy any separation axioms unless explicitly stated. Paracompact, normal, and regular spaces will always be assumed to be Hausdorff spaces. A continuous map from one topological space to another will also be called simply a map.

Given a set X and an indexed collection of topological spaces $\{X_j\}_{j \in J}$ and functions $f_j: X \rightarrow X_j$, the topology induced on X by the functions $\{f_j\}$ is the smallest or coarsest topology such that each f_j is continuous.

1 The topology induced on X by functions $\{f_j: X \rightarrow X_j\}$ is characterized by the property that if Y is a topological space, a function $g: Y \rightarrow X$ is continuous if and only if $f_j \circ g: Y \rightarrow X_j$ is continuous for each $j \in J$.

A subspace of a topological space X is a subset A of X topologized by the topology induced by the inclusion map $A \subset X$. A discrete subset of a topological space X is a subset such that every subset of it is closed in X . The topological product of an indexed collection of topological spaces $\{X_j\}_{j \in J}$ is the cartesian product $\bigtimes X_j$, given the topology induced by the projection maps $p_j: \bigtimes X_j \rightarrow X_j$ for $j \in J$. If $\{X_\alpha\}_{\alpha \in \Lambda}$ is an inverse system of topological spaces (that is, X_α is a topological space for $\alpha \in \Lambda$ and $f_\alpha^\beta: X_\beta \rightarrow X_\alpha$ is continuous for $\alpha \leq \beta$) their inverse limit $\lim_{\leftarrow} \{X_\alpha\}$ is given the topology induced by the functions $p_\alpha: \lim_{\leftarrow} \{X_\alpha\} \rightarrow X_\alpha$ for $\alpha \in \Lambda$.

Given a set X and an indexed collection of topological spaces $\{X_j\}_{j \in J}$ and functions $g_j: X_j \rightarrow X$, the topology coincided on X by the functions $\{g_j\}$ is the largest or finest topology such that each g_j is continuous.

¹ As general references see J. L. Kelley, *General Topology*, D. Van Nostrand Company, Inc., Princeton, N.J., 1955, and S. T. Hu, *Elements of General Topology*, Holden-Day, Inc., San Francisco, 1964.

and $B \subset Y$, we use $(Y, B)^{(\alpha, A)}$ to denote the subspace of Y^X of continuous functions $f: X \rightarrow Y$ such that $f(A) \subset B$. Let $E: Y^X \times X \rightarrow Y$ be the evaluation map defined by $E(f, x) = f(x)$. Given a function $g: Z \rightarrow Y^X$, the composite

$$Z \times X \xrightarrow{g \times 1} Y^X \times X \xrightarrow{E} Y$$

is a function from $Z \times X$ to Y .

THEOREM OF EXPONENTIAL CORRESPONDENCE If X is a locally compact Hausdorff space and Y and Z are topological spaces, a map $g: Z \rightarrow Y^X$ is continuous if and only if $E \circ (g \times 1): Z \times X \rightarrow Y$ is continuous.

EXPONENTIAL LAW If X is a locally compact Hausdorff space, Z is a Hausdorff space, and Y is a topological space, the function $\psi: (Y^X)^Z \rightarrow Y^{Z \times X}$ defined by $\psi(g) = E \circ (g \times 1)$ is a homeomorphism.

10 If X is a compact Hausdorff space and Y is metrized by a metric d , then Y^X is metrized by the metric d' defined by

$$d'(f, g) = \sup \{d(f(x), g(x)) \mid x \in X\}$$

3 GROUP THEORY

A homomorphism is called a *monomorphism*, *epimorphism*, *isomorphism*, respectively, if it is injective, surjective, bijective. If $\{G_i\}_{i \in J}$ is an indexed collection of groups, their *direct product* is the group structure on the cartesian product $\times G_i$ defined by $(g_i)(g'_i) = (gg')$. If $\{G_\alpha\}$ is an inverse system of groups (that is, G_α is a group for each α and $f_{\alpha\beta}: G_\beta \rightarrow G_\alpha$ is a homomorphism for $\alpha \leq \beta$), their inverse limit $\lim_{\leftarrow} \{G_\alpha\}$ (which is a set) is a subgroup of $\times G_\alpha$.

Let A be a subset of a group G . G is said to be *freely generated* by A and A is said to be a *free generating set* or *free basis* for G if, given any function $f: A \rightarrow H$, where H is a group, there exists a unique homomorphism $\varphi: G \rightarrow H$ which is an extension of f . A group is said to be *free* if it is freely generated by some subset. For any set A a *free group generated by A* is a group $F(A)$ containing A as a free generating set. Such groups $F(A)$ exist, and any two are canonically isomorphic.

1 Any group is isomorphic to a quotient group of a free group.

A *presentation* of a group G consists of a set A of generators, a set $B \subset F(A)$ of relations, and a function $f: A \rightarrow G$ such that the extension of f to a homomorphism $\varphi: F(A) \rightarrow G$ is an epimorphism whose kernel is the normal closure of B .

¹As a general reference for elementary group theory see G. Birkhoff and S. MacLane, *A Survey of Modern Algebra*, The Macmillan Company, New York, 1953. For a discussion of free groups see R. H. Crowell and R. H. Fox, *Introduction to Knot Theory*, Ginn and Company, Boston, 1963.

2 The topology coincided on X by functions $\{g_i: X_j \rightarrow X\}$ is characterized by the property that if Y is any topological space, a function $f: X \rightarrow Y$ is continuous if and only if $f \circ g_i: X_j \rightarrow Y$ is continuous for each $j \in J$.

A *quotient space* of a topological space X is a quotient set X' of X topologized by the topology coincided by the projection map $X \rightarrow X'$. If $A \subset X$, then X/A will denote the quotient space of X obtained by identifying all of A to a single point. The *topological sum* of an indexed collection of topological spaces $\{X_j\}_{j \in J}$ is the set $\sum \bigvee X_j$, given the topology coincided by the injection maps $i_j: X_j \rightarrow \sum \bigvee X_j$ for $j \in J$. If $\{X^\alpha\}_{\alpha \in \Lambda}$ is a direct system of topological spaces (that is, X^α is a topological space for $\alpha \in \Lambda$ and $f_{\alpha\beta}: X^\alpha \rightarrow X^\beta$ is continuous for $\alpha \leq \beta$), their direct limit $\lim_{\rightarrow} \{X^\alpha\}$ is given the topology coincided by the functions $i_\alpha: X^\alpha \rightarrow \lim_{\rightarrow} \{X^\alpha\}$ for $\alpha \in \Lambda$.

Let $\mathfrak{A} = \{A\}$ be a collection of subsets of a topological space X . X is said to have a *topology coherent with $\mathfrak{A}$* if the topology on X is coincided from the subspaces $\{A\}$ by the inclusion maps $A \subset X$. (In the literature this topology is often called the *weak topology* with respect to $\mathfrak{A}$.)

3 A *necessary and sufficient condition* that X have a topology coherent with $\mathfrak{A}$ is that a subset B of X be closed (or open) in X if and only if $B \cap A$ is closed (or open) in the subspace A for every $A \in \mathfrak{A}$.

4 If $\mathfrak{A}$ is an arbitrary open covering or a locally finite closed covering of X , then X has a topology coherent with $\mathfrak{A}$.

5 Let X be a set and let $\{A_j\}$ be an indexed collection of topological spaces each contained in X and such that for each j and j' , $A_j \cap A_{j'}$ is a closed (or open) subset of A_j and of $A_{j'}$ and the topology induced on $A_j \cap A_{j'}$ from A_j equals the topology induced on $A_j \cap A_{j'}$ from $A_{j'}$. Then the topology coincided on X by the collection of inclusion maps $\{A_j \subset X\}$ is characterized by the properties that A_j is a closed (or open) subspace of X for each j and X has a topology coherent with the collection $\{A_j\}$.

The topology on X in theorem 5 will be called the *topology coherent with $\{A_j\}$* . A *compactly generated space* is a Hausdorff space having a topology coherent with the collection of its compact subsets (this is the same as what is sometimes referred to as a Hausdorff k -space).

6 A Hausdorff space which is either locally compact or satisfies the first axiom of countability is compactly generated.

7 If X is compactly generated and Y is a locally compact Hausdorff space, $X \times Y$ is compactly generated.

If X and Y are topological spaces and $A \subset X$ and $B \subset Y$, then $\langle A; B \rangle$ denotes the set of continuous functions $f: X \rightarrow Y$ such that $f(A) \subset B$. Y^X denotes the space of continuous functions from X to Y ; given the compact-open topology (which is the topology generated by the subbase $\{\langle K; U \rangle\}$, where K is a compact subset of X and U is an open subset of Y). If $A \subset X$

mal subgroup of $F(A)$ generated by B . If A and B are both finite sets, the presentation is said to be *finite* and G is said to be *finitely presented*.

4 MODULES¹

We are mainly interested in R modules where R is a principal ideal domain. However, we shall begin with some properties of R modules where R is a commutative ring with a unit which acts as the identity on every module. If $\varphi: A \rightarrow B$ is a homomorphism of R modules, then we have R modules

$$\begin{aligned} \ker \varphi &= \{a \in A \mid \varphi(a) = 0\} \subset A \\ \operatorname{im} \varphi &= \{b \in B \mid b = \varphi(a) \text{ for some } a \in A\} \subset B \\ \operatorname{coker} \varphi &= B / \operatorname{im} \varphi \end{aligned}$$

1 **NOETHER ISOMORPHISM THEOREM** *Let A and B be submodules of a module C and let $A + B$ be the submodule of C generated by $A \cup B$. The inclusion map $A \subset A + B$ sends $A \cap B$ into B and induces an isomorphism of $A/(A \cap B)$ with $(A + B)/B$.*

If $\{A_j\}_{j \in J}$ is an indexed collection of R modules, their direct product $\times A_j$ is an R module and their *direct sum* $\bigoplus A_j$ is an R module ($\bigoplus A_j$ is the submodule of $\times A_j$ consisting of those elements having only a finite number of nonzero coordinates). The inverse limit $\lim_{\leftarrow} \{A_\alpha\}$ of an inverse system of R modules (and homomorphisms $f_\alpha^\beta: A_\beta \rightarrow A_\alpha$ for $\alpha \leq \beta$) is an R module, and the direct limit of a direct system of R modules (and homomorphisms) is an R module.

2 *Any R module is isomorphic to the direct limit of its finitely generated submodules directed by inclusion.*

If A and B are R modules, their *tensor product* $A \otimes B$ (also written $A \otimes_R B$) is an R module. For $a \in A$ and $b \in B$, there is a corresponding element $a \otimes b \in A \otimes B$. $A \otimes B$ is generated by the elements $\{a \otimes b \mid a \in A, b \in B\}$ with the relations (for $a, a' \in A, b, b' \in B$, and $r, r' \in R$)

$$\begin{aligned} (ra + r'a') \otimes b &= r(a \otimes b) + r'(a' \otimes b) \\ a \otimes (rb + r'b') &= r(a \otimes b) + r'(a \otimes b') \end{aligned}$$

In case A or B is also an R' module, then so is $A \otimes_R B$.

3 *For any R module A the homomorphisms $a \mapsto a \otimes 1$ and $a \mapsto 1 \otimes a$ define isomorphisms of A with $A \otimes R$ and $R \otimes A$.*

¹ As general references see H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton University Press, Princeton, N.J., 1956 and S. MacLane, *Homology*, Springer-Verlag OHG, Berlin, 1963.

4 *For R modules A and B there is an isomorphism of $A \otimes B$ with $B \otimes A$ taking $a \otimes b$ to $b \otimes a$.*

5 *If A and B are R modules and B and C are R' modules, there is an isomorphism of $(A \otimes_R B) \otimes_{R'} C$ with $A \otimes_R (B \otimes_{R'} C)$ (both being regarded as R and R' modules) taking $(a \otimes b) \otimes c$ to $a \otimes (b \otimes c)$.*

If A and B are R modules, their *module of homomorphisms* $\operatorname{Hom}(A, B)$ [also written $\operatorname{Hom}_R(A, B)$] is an R module whose elements are R homomorphisms $A \rightarrow B$. In case A or B is also an R' module, then so is $\operatorname{Hom}_R(A, B)$.

6 *If A and B are R modules and B and C are R' modules, there is an isomorphism of $\operatorname{Hom}_{R'}(A \otimes_R B, C)$ with $\operatorname{Hom}_R(A, \operatorname{Hom}_{R'}(B, C))$ (both being regarded as R and R' modules) taking an R' homomorphism $\varphi: A \otimes_R B \rightarrow C$ to the R homomorphism $\varphi': A \rightarrow \operatorname{Hom}_{R'}(B, C)$ such that $\varphi'(a)(b) = \varphi(a \otimes b)$.*

A subset S of an R module A is said to be a *basis* for A (and A is said to be *freely generated* by S) if any function $f: S \rightarrow B$, where B is an R module, admits a unique extension to a homomorphism $\varphi: A \rightarrow B$. If a module has a basis, it is said to be a *free module*. For any set S the *free module generated* by S , denoted by $F_R(S)$, is the module of all finitely nonzero functions from S to R (with pointwise addition and scalar multiplication) and with $s \in S$ identified with its characteristic function. $F_R(S)$ contains S as a basis, and any module containing S as a basis is canonically isomorphic to $F_R(S)$.

7 *Any R module is isomorphic to a quotient of a free R module.*

8 *If A' is a submodule of A , with A/A' free, then A is isomorphic to the direct sum $A' \oplus (A/A')$.*

We now assume that R is a principal ideal domain (that is, it is an integral domain in which every ideal is principal). If A is an R module, its *torsion submodule* $\operatorname{Tor} A$ is defined by

$$\operatorname{Tor} A = \{a \in A \mid ra = 0 \text{ for some nonzero } r \in R\}$$

A is said to be *torsion free* or *without torsion* if $\operatorname{Tor} A = 0$.

9 *Over a principal ideal domain, a submodule of a free module is free.*

10 *Over a principal ideal domain, a finitely generated module is free if and only if it is torsion free.*

11 *Over a principal ideal domain, $A/\operatorname{Tor} A$ is torsion free.*

If A is a finitely generated module over a principal ideal domain R , its rank $\rho(A)$ is defined to be the number of elements in a basis of the quotient module $A/\operatorname{Tor} A$.

12 *If A' is a submodule of a finitely generated module A (over a principal ideal domain), then*

$$\rho(A) = \rho(A') + \rho(A/A')$$

If x and y are points of a real vector space, the *closed line segment* joining them, denoted by $[x, y]$, is the set of points of the form $tx + (1 - t)y$ for $0 \leq t \leq 1$ (thus $I = [0, 1]$). If $x \neq y$, the *line* determined by them is the set $\{tx + (1 - t)y \mid t \in \mathbf{R}\}$. A subset C of a real vector space is said to be an *affine variety* if whenever $x, y \in C$, with $x \neq y$, then the line determined by x and y is also in C . A subset C is said to be *convex* if $x, y \in C$ imply $[x, y] \subset C$. A *convex body*¹ in $\mathbf{R}^n$ is a convex subset of $\mathbf{R}^n$ containing a non-empty open subset of $\mathbf{R}^n$ (thus I^n and E^n are convex bodies in $\mathbf{R}^n$).

1 If C is a convex body in $\mathbf{R}^n$ and C' is a convex body in $\mathbf{R}^m$, then $C \times C'$ is a convex body in $\mathbf{R}^n \times \mathbf{R}^m = \mathbf{R}^{n+m}$.

2 Any two compact convex bodies in $\mathbf{R}^n$ are homeomorphic.

A subset S of a real vector space is said to be *affinely independent* if, given a finite number of distinct elements $x_0, x_1, \dots, x_m \in S$ and $t_0, t_1, \dots, t_m \in \mathbf{R}$ such that $\sum t_i = 0$ and $\sum t_i x_i = 0$, then $t_i = 0$ for $0 \leq i \leq m$ (this is equivalent to the condition that

$$x_1 - x_0, x_2 - x_0, \dots, x_m - x_0$$

be linearly independent).

3 There exist affinely independent subsets of $\mathbf{R}^n$ containing $n + 1$ points, but no subset of $\mathbf{R}^n$ containing more than $n + 1$ points is affinely independent.

4 Given points $x_0, x_1, \dots, x_m \in \mathbf{R}^n$, the convex set generated by them is the set of all points of the form $\sum t_i x_i$, with $0 \leq t_i \leq 1$ and $\sum t_i = 1$. The set $\{x_0, x_1, \dots, x_m\}$ is *affinely independent* if and only if every point x in the convex set generated by this set has a *unique representation* in the form $x = \sum t_i x_i$, with $0 \leq t_i \leq 1$ for $0 \leq i \leq m$ and $\sum t_i = 1$.

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¹ For general properties of convex sets see F. A. Valentine, *Convex Sets*, McGraw-Hill Book Company, New York, 1964.

Let $\varphi: A \rightarrow A$ be an endomorphism of a finitely generated module (over a principal ideal domain R). The *trace* of φ , $\text{Tr } \varphi$, is the element of R which is the trace of the endomorphism φ' induced by φ on the free module $A/\text{Tor } A$ [that is, if $A/\text{Tor } A$ has a basis $a_1, \dots, a_n$, then $\varphi'(a_i) = \sum r_{ij} a_j$ and $\text{Tr } \varphi = \sum r_{ii}$].

1.3 Let φ be an endomorphism of a finitely generated module A and let A' be a submodule of A such that $\varphi(A') \subset A'$. Then $\varphi \mid A'$ is an endomorphism of A' and there is induced an endomorphism φ'' of A/A' . Their traces satisfy the relation

$$\text{Tr } \varphi = \text{Tr } (\varphi \mid A') + \text{Tr } \varphi''$$

A module with a single generator is said to be *cyclic*. Over a principal ideal domain R such a module A is characterized, up to isomorphism, by the element $r_A \in R$ which generates the ideal of elements annihilating every element of A (r_A is unique up to multiplication by invertible elements of R).

1.4 STRUCTURE THEOREM FOR FINITELY GENERATED MODULES Over a principal ideal domain every finitely generated module is the direct sum of a free module and cyclic modules $A_1, \dots, A_q$ whose corresponding elements $r_1, \dots, r_q \in R$ have the property that r_i divides r_{i+1} for $1 \leq i \leq q$. The elements $r_1, \dots, r_q$ are unique up to multiplication by invertible elements of R and, together with the rank of the module, characterize the module up to isomorphism.

5 EUCLIDEAN SPACES

We use the following fixed notations:

- $\emptyset$ = empty set
- $\mathbf{Z}$ = ring of integers
- $\mathbf{Z}_m$ = ring of integers modulo m
- $\mathbf{R}$ = field of real numbers
- $\mathbf{C}$ = field of complex numbers
- $\mathbf{Q}$ = division ring of quaternions
- $\mathbf{R}^n$ = euclidean n -space, with $\|x\| = \sqrt{\sum x_i^2}$ and $\langle x, y \rangle = \sum x_i y_i$
- 0 = origin of $\mathbf{R}^n$
- I = closed unit interval
- $\bar{I} = \{0, 1\} \subset I$
- I^n = n -cube = $\{x \in \mathbf{R}^n \mid 0 \leq x_i \leq 1 \text{ for } 1 \leq i \leq n\}$
- $I^n_n = \{x \in I^n \mid \text{for some } i, x_i = 0 \text{ or } x_i = 1\}$
- E^n = n -ball = $\{x \in \mathbf{R}^n \mid \|x\| \leq 1\}$
- S^{n-1} = $(n - 1)$ -sphere = $\{x \in \mathbf{R}^n \mid \|x\| = 1\}$
- $\mathbf{P}^n$ = projective n -space = quotient space of S^n with x and $-x$ identified for all $x \in S^n$

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CHAPTER ONE

HOMOTOPY AND THE

FUNDAMENTAL GROUP

The first functor considered in detail is the fundamental group functor, introduced and discussed in Secs. 1.7 and 1.8. This is an intuitively appealing example of the kind of functor considered in algebraic topology. Some applications of this functor are presented in the exercises at the end of the chapter. In Chapter Two this functor is used in a systematic study and classification of covering spaces.

I . . . CATEGORIES

An algebraic representation of topology is a mapping from topology to algebra. Such a representation converts a topological problem into an algebraic one to the end that, with sufficiently many representations, the topological problem will be solvable if (and only if) all the corresponding algebraic problems are solvable.

The definition of a representation, formally called a functor, is given in the next section. This section is devoted to the concept of category, because functors are functions, with certain naturality properties, from one or several categories to another.

A category may be thought of intuitively as consisting of sets, possibly with additional structure, and functions, possibly preserving additional structure. More precisely, a *category* $\mathcal{C}$ consists of

- (a) A class of *objects*
- (b) For every ordered pair of objects X and Y , a set $\text{hom}(X, Y)$ of *morphisms* with *domain* X and *range* Y ; if $f \in \text{hom}(X, Y)$, we write $f: X \rightarrow Y$ or $X \xrightarrow{f} Y$
- (c) For every ordered triple of objects X, Y , and Z , a function associating to a pair of morphisms $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ their *composite*

$$gf = g \circ f: X \rightarrow Z$$

These satisfy the following two axioms:

Associativity. If $f: X \rightarrow Y$, $g: Y \rightarrow Z$, and $h: Z \rightarrow W$, then

$$h(gf) = (hg)f: X \rightarrow W$$

Identity. For every object Y there is a morphism $1_Y: Y \rightarrow Y$ such that if $f: X \rightarrow Y$, then $1_Y f = f$, and if $h: Y \rightarrow Z$, then $h 1_Y = h$.

If the class of objects is a set, the category is said to be *small*. For most of our purposes we could restrict our attention to small categories, but it would be inconvenient to have to specify a set of objects before obtaining a category. For example, we should like to consider categories whose objects are sets or groups, and we prefer to consider the class of all sets or groups, rather than some suitable set of sets or groups in each instance.

From the two axioms it follows that 1_Y is unique (see lemma 1 below),

TOPOLOGY IS THE STUDY OF TOPOLOGICAL SPACES AND CONTINUOUS FUNCTIONS between them. A standard problem is the classification of such spaces and functions up to homeomorphism. A weaker equivalence relation, based on continuous deformation, leads to another classification problem. This latter classification problem is of fundamental importance in algebraic topology, since it is the one where the tools available seem to be most successful.

As a working definition for our purposes, algebraic topology may be regarded as the study of topological spaces and continuous functions by means of algebraic objects such as groups, rings, homomorphisms. The link from topology to algebra is by means of mappings, called functors. For this reason, Secs. 1.1 and 1.2 are devoted to the basic concepts of category and functor.

In Secs. 1.3 and 1.4 the concept of continuous deformation, known technically as homotopy, is introduced. We then define the homotopy category and certain functors on this category, all of which are important for the subject. Sections 1.5 and 1.6 are devoted to a study of conditions under which these functors on the homotopy category take values in the category of groups. As examples, the homotopy group functors are briefly mentioned.

and it is called the *identity morphism* of Y . Given morphisms $f: X \rightarrow Y$ and $g: Y \rightarrow X$ such that $gf = 1_X$, g is called a *left inverse* of f and f is called a *right inverse* of g . A *two-sided inverse* (or simply an *inverse*) of f is a morphism which is both a left inverse of f and a right inverse of f . A morphism $f: X \rightarrow Y$ is called an *equivalence*, denoted by $f: X \approx Y$, if there is a morphism $g: Y \rightarrow X$ which is a two-sided inverse of f . If $g': Y \rightarrow X$ is a left inverse of f and $g'': Y \rightarrow X$ is a right inverse of f , then

$$g' = g'1_Y = g'(fg'') = (g'f)g'' = 1_Y g'' = g''$$

showing that $g' = g''$. Therefore we have the following lemma.

1 **LEMMA** If $f: X \rightarrow Y$ has a *left inverse* and a *right inverse*, they are equal, and f is an *equivalence*. ■

In particular, it follows that an equivalence $f: X \approx Y$ has a unique inverse, denoted by f^{-1} : $Y \rightarrow X$, and f^{-1} is an equivalence. If there is an equivalence $f: X \approx Y$, X and Y are said to be *equivalent*, denoted by $X \approx Y$. Because the composite of equivalences is easily seen to be an equivalence, the relation $X \approx Y$ is an equivalence relation in any set of objects of $\mathcal{C}$.

We list some examples of categories.

- 2** The category of sets and functions [that is, the class of objects is the class of all sets, and for sets X and Y , $\text{hom}(X, Y)$ equals the set of functions from X to Y]
- 3** The category of topological spaces and continuous maps
- 4** The category of groups and homomorphisms
- 5** The category of R modules and homomorphisms
- 6** The category of normed rings (over $\mathbf{R}$) and continuous homomorphisms
- 7** The category of sets and injections (or surjections or bijections)
- 8** The category of *pointed sets* (a pointed set is a nonempty set with a distinguished element) and functions preserving distinguished elements
- 9** The category of *pointed topological spaces* (a pointed topological space is a nonempty topological space with a *base point*) and continuous maps preserving base points
- 10** The category of finite sets and functions
- 11** Given a partial order $\leq$ in X , there is a category whose objects are the elements of X and such that $\text{hom}(x, x')$ is either the singleton consisting of the ordered pair (x, x') or empty, according to whether $x \leq x'$ or $x \not\leq x'$
- 12** The category of groups and conjugacy classes of homomorphisms (that is, a morphism $G \rightarrow G'$ is an equivalence class of homomorphisms from G to G' , two homomorphisms being equivalent if they differ by an inner automorphism of G')

A *subcategory* $\mathcal{C}' \subset \mathcal{C}$ is a category such that

- (a) The objects of $\mathcal{C}'$ are also objects of $\mathcal{C}$
- (b) For objects X' and Y' of $\mathcal{C}'$, $\text{hom}_{\mathcal{C}'}(X', Y') \subset \text{hom}_{\mathcal{C}}(X, Y)$
- (c) If $f': X' \rightarrow Y'$ and $g': Y' \rightarrow Z'$ are morphisms of $\mathcal{C}'$, their composite in $\mathcal{C}'$ equals their composite in $\mathcal{C}$

$\mathcal{C}'$ is called a *full subcategory* of $\mathcal{C}$ if $\mathcal{C}'$ is a subcategory of $\mathcal{C}$ and for objects X' and Y' in $\mathcal{C}'$, $\text{hom}_{\mathcal{C}'}(X', Y') = \text{hom}_{\mathcal{C}}(X, Y)$. The category in example 7 above is a subcategory of the one in example 2, and the category in example 10 is a full subcategory of the one in example 2. The categories in examples 3, 4, 5, 6, and 8 are not subcategories of the category of sets, because each object of one of these categories consists of a set, together with an additional structure on it (hence, different objects in these categories may have the same underlying sets). In examples 11 and 12, the morphisms in the respective categories are not functions, and so neither of these categories is a subcategory of the category of sets.

A diagram of morphisms such as the square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \varepsilon \downarrow & & \downarrow h \\ X' & \xrightarrow{f'} & Y' \end{array}$$

is said to be *commutative* if any two composites of morphisms in the diagram beginning at the same place and ending at the same place are equal. This square is commutative if and only if $hf = f'g$.

Following are descriptions of some categories which are associated to a given category. Given a category $\mathcal{C}$, there is an associated category called the *category of morphisms* of $\mathcal{C}$. Its objects are morphisms $X \xrightarrow{f} Y$, and its morphisms with domain $X \xrightarrow{f} Y$ and range $X' \xrightarrow{f'} Y'$ are pairs of morphisms $g: X \rightarrow X'$ and $h: Y \rightarrow Y'$ such that the square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \varepsilon \downarrow & & \downarrow h \\ X' & \xrightarrow{f'} & Y' \end{array}$$

is commutative. In a similar way, diagrams of morphisms in $\mathcal{C}$ more general than $X \xrightarrow{f} Y$ are the objects of a suitable category associated to $\mathcal{C}$.

Let $\mathcal{C}$ be a category whose objects are sets with additional structures (such as distinguished elements or topologies) and whose morphisms are functions preserving the additional structures. For example, $\mathcal{C}$ might be any of the categories in examples 2 through 10. There is a category associated to $\mathcal{C}$, called the *category of pairs* of $\mathcal{C}$, whose objects are injective morphisms $i: A \rightarrow X$ (because each morphism in such a category is a function, it is meaningful to consider those which are injective) and whose morphisms are commutative squares

$$\begin{array}{ccc} A & \xrightarrow{i} & X \\ \varepsilon \downarrow & & \downarrow h \\ B & \xrightarrow{j} & Y \end{array}$$

Thus the category of pairs of $\mathcal{C}$ is a full subcategory of the category of morphisms of $\mathcal{C}$. The notation (X, A) will denote the pair consisting of X and $i: A \subset X$, and the notation $f: (X, A) \rightarrow (Y, B)$ will mean that $f: X \rightarrow Y$ is a morphism of $\mathcal{C}$ such that $f(i(A)) \subset i(B)$. The category of pairs of $\mathcal{C}$, therefore, has as objects the pairs (X, A) and has as morphisms the morphisms $f: (X, A) \rightarrow (Y, B)$.

If $\mathcal{C}_1$ and $\mathcal{C}_2$ are categories, their *product* $\mathcal{C}_1 \times \mathcal{C}_2$ is the category whose objects are ordered pairs (Y_1, Y_2) of objects Y_1 in $\mathcal{C}_1$ and Y_2 in $\mathcal{C}_2$ and whose morphisms $(X_1, X_2) \rightarrow (Y_1, Y_2)$ are ordered pairs of morphisms (f_1, f_2) , where $f_1: X_1 \rightarrow Y_1$ in $\mathcal{C}_1$ and $f_2: X_2 \rightarrow Y_2$ in $\mathcal{C}_2$. Similarly, there is a product of an arbitrary indexed family of categories.

Given a category $\mathcal{C}$, there is an *opposite* category $\mathcal{C}^*$ whose objects $Y^*: Y^* \rightarrow X^*$ are in one-to-one correspondence with the objects Y of $\mathcal{C}$ and whose morphisms $f^*: Y^* \rightarrow X^*$ are in one-to-one correspondence with the morphisms $f: X \rightarrow Y$ [with f^*g^* defined to equal $(gf)^*$ for $X \xrightarrow{g} Y \xrightarrow{f} Z$ in $\mathcal{C}$]. We identify $(\mathcal{C}^*)^*$ with $\mathcal{C}$, so that $(X^*)^* = X$ and $(f^*)^* = f$.

We next show how to interpret sums and products, as well as direct and inverse limits in arbitrary categories. An object X in a category $\mathcal{C}$ is said to be an *initial object* if for each object Y in $\mathcal{C}$ the set $\text{hom}(X, Y)$ contains exactly one element. Dually, an object Z of $\mathcal{C}$ is said to be a *terminal object* if for each Y of $\mathcal{C}$ the set $\text{hom}(Y, Z)$ contains exactly one element. Note that any two initial objects of $\mathcal{C}$ are equivalent and any two terminal objects of $\mathcal{C}$ are equivalent. In examples 2 and 3 the empty set is an initial object and any one-point set is a terminal object. In example 4 the trivial group is both an initial and a terminal object. In example 7 the category of sets and bijections has neither an initial object nor a terminal object.

Let $\{Y_j\}_{j \in J}$ be an indexed collection of objects of a category $\mathcal{C}$. Let $\mathcal{S}\{Y_j\}$ be the category whose objects are indexed collections of morphisms $\{f_j\}_{j \in J}$ of $\mathcal{C}$ having the same range and whose morphisms with domain $\{f_j: Y_j \rightarrow Z\}$ and range $\{g_j: Y_j \rightarrow Z'\}$ are morphisms $g: Z \rightarrow Z'$ of $\mathcal{C}$ such that $gf_j = f_j$ for every $j \in J$. An initial object of $\mathcal{S}\{Y_j\}$ is called a *sum* of the collection $\{Y_j\}$. A given collection may or may not have a sum in $\mathcal{C}$. The set sum is a sum in the category of sets, the topological sum is a sum in the category of topological spaces, the free product is a sum in the category of groups, and the direct sum is a sum in the category of R modules. In the category of finite sets, in general only finite collections have a sum. Similarly, in the category of finitely generated R modules, in general only finite collections have a sum.

Dually, given an indexed collection of objects $\{Y_j\}_{j \in J}$ in $\mathcal{C}$, let $\mathcal{P}\{Y_j\}$ be the category whose objects are indexed collections of morphisms $\{g_j\}_{j \in J}$ of $\mathcal{C}$ having the same domain and whose morphisms with domain $\{g_j: X \rightarrow Y_j\}$ and range $\{g'_j: X' \rightarrow Y_j\}$ are morphisms $f: X \rightarrow X'$ of $\mathcal{C}$ such that $gf_j = g'_j$ for every $j \in J$. A terminal object of $\mathcal{P}\{Y_j\}$ is called a *product* of the collection $\{Y_j\}$. The cartesian product of sets is a product in the category of sets, the topological product is a product in the category of topological spaces, and the direct product is a product in the category of groups, or R modules. In the category of finite sets (or finitely generated R modules), in general only finite collections have a product.

A *direct system* $\{Y^\alpha f_{\alpha\beta}\}$ in a category $\mathcal{C}$ consists of a collection of objects $\{Y^\alpha\}$ indexed by a directed set $\Lambda = \{\alpha\}$ and a collection of morphisms $\{f_{\alpha\beta}: Y^\alpha \rightarrow Y^\beta\}$ in $\mathcal{C}$ for $\alpha \leq \beta$ in Λ such that

- (a) $f_{\alpha\alpha} = 1_{Y^\alpha}$ for $\alpha \in \Lambda$
- (b) $f_{\alpha\gamma} = f_{\beta\gamma} f_{\alpha\beta}: Y^\alpha \rightarrow Y^\gamma$ for $\alpha \leq \beta \leq \gamma$ in Λ

There is then a category $\text{dir}\{Y^\alpha f_{\alpha\beta}\}$ whose objects are indexed collections of morphisms $\{g_\alpha: Y^\alpha \rightarrow Z\}_{\alpha \in \Lambda}$ such that $g_\alpha = g_\beta f_{\alpha\beta}$ if $\alpha \leq \beta$ in Λ and whose morphisms with domain $\{g_\alpha: Y^\alpha \rightarrow Z\}$ and range $\{g'_\alpha: Y^\alpha \rightarrow Z'\}$ are morphisms $h: Z \rightarrow Z'$ such that $hg_\alpha = g'_\alpha$ for $\alpha \in \Lambda$. An initial object of $\text{dir}\{Y^\alpha f_{\alpha\beta}\}$ is called a *direct limit* of the direct system $\{Y^\alpha f_{\alpha\beta}\}$. The direct limits of sets, topological spaces, groups, and R modules are examples of direct limits in their respective categories.

Dually, an *inverse system* $\{Y_\alpha f_{\alpha\beta}\}$ in $\mathcal{C}$ consists of a collection of objects $\{Y_\alpha\}$ indexed by a directed set $\Lambda = \{\alpha\}$ and a collection of morphisms $\{f_{\alpha\beta}: Y_\beta \rightarrow Y_\alpha\}$ in $\mathcal{C}$ for $\alpha \leq \beta$ in Λ such that

- (a) $f_{\alpha\alpha} = 1_{Y_\alpha}$ for $\alpha \in \Lambda$
- (b) $f_{\alpha\gamma} = f_{\beta\gamma} f_{\alpha\beta}: Y_\gamma \rightarrow Y_\alpha$ for $\alpha \leq \beta \leq \gamma$ in Λ

There is then a category $\text{inv}\{Y_\alpha f_{\alpha\beta}\}$ whose objects are indexed collections of morphisms $\{g_\alpha: X \rightarrow Y_\alpha\}_{\alpha \in \Lambda}$ such that $g_\alpha = f_{\alpha\beta} g_\beta$ if $\alpha \leq \beta$ in Λ and whose morphisms with domain $\{g_\alpha: X \rightarrow Y_\alpha\}$ and range $\{g'_\alpha: X' \rightarrow Y_\alpha\}$ are morphisms $h: X \rightarrow X'$ of $\mathcal{C}$ such that $g'_\alpha h = g_\alpha$ for $\alpha \in \Lambda$. A terminal object of $\text{inv}\{Y_\alpha f_{\alpha\beta}\}$ is called an *inverse limit* of the inverse system $\{Y_\alpha f_{\alpha\beta}\}$. The inverse limits of sets, topological spaces, groups, and R modules are examples of inverse limits in their respective categories.

By similar considerations it is possible to define a direct or inverse limit for an arbitrary indexed collection of objects in a category $\mathcal{C}$ and an indexed collection of morphisms in $\mathcal{C}$ between these objects. We omit the details.

2 FUNCTORS

Our main interest in categories is in the maps from one category to another. Those maps which have the natural properties of preserving identities and composites are called functors. This section is devoted to the definition of functors of one or more variables, some examples and applications, and the definition of natural transformations between functors.

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *covariant functor* (or *contravariant functor*) T from $\mathcal{C}$ to $\mathcal{D}$ consists of an object function which assigns to every object X of $\mathcal{C}$ an object $T(X)$ of $\mathcal{D}$ and a morphism function which assigns to every morphism $f: X \rightarrow Y$ of $\mathcal{C}$ a morphism $T(f): T(X) \rightarrow T(Y)$ [or $T(f): T(Y) \rightarrow T(X)$] of $\mathcal{D}$ such that

- (a) $T(1_X) = 1_{T(X)}$
- (b) $T(gf) = T(g)T(f)$ [or $T(gf) = T(f)T(g)$]

SEC. 2 FUNCTORS

We list some examples of functors.

1 There is a covariant functor from the category of topological spaces and continuous maps to the category of sets and functions which assigns to every topological space its underlying set. This functor is called a *forgetful functor* because it “forgets” some of the structure of a topological space.

2 There is a covariant functor from the category of sets and functions to the category of R modules and homomorphisms which assigns to every set the free R module generated by it.

3 Given a fixed R module M_0 , there is a covariant functor (or contravariant functor) from the category of R modules and homomorphisms to itself which assigns to an R module M the R module $\text{Hom}_R(M_0, M)$ [or $\text{Hom}_R(M, M_0)$].

4 For any category $\mathcal{C}$ and object Y of $\mathcal{C}$ there is a covariant functor π_Y (or contravariant functor π^Y) from $\mathcal{C}$ to the category of sets and functions which assigns to an object Z (or X) of $\mathcal{C}$ the set $\pi_Y(Z) = \text{hom}(Y, Z)$ [or $\pi^Y(X) = \text{hom}(X, Y)$] and to a morphism $h: Z \rightarrow Z'$ [or $f: X \rightarrow X'$] the function

$$h_{\#}: \text{hom}(Y, Z) \rightarrow \text{hom}(Y, Z') \quad [\text{or } f_{\#}: \text{hom}(X', Y) \rightarrow \text{hom}(X, Y)]$$

defined by $h_{\#}(g) = h \circ g$ for $g: Y \rightarrow Z$ [or $f_{\#}(g') = g' \circ f$ for $g': X' \rightarrow Y$]

5 There is a contravariant functor C from the category of compact Hausdorff spaces and continuous maps to the category of normed rings over $\mathbf{R}$ and continuous homomorphisms which assigns to X its normed ring of continuous real-valued functions.

6 There is a covariant functor H_0 from the category of topological spaces and continuous maps to the category of abelian groups and homomorphisms such that $H_0(X)$ is the free abelian group generated by the set of components of X , and if $f: X \rightarrow Y$, then $H_0(f): H_0(X) \rightarrow H_0(Y)$ is the homomorphism such that if C is a component of X and C' is the component of Y containing $f(C)$, then $H_0(f)C = C'$.

7 A direct system (or inverse system) in a category $\mathcal{C}$ is a covariant functor (or contravariant functor) from the category of a directed set (defined as in example 1.1.11) to $\mathcal{C}$.

8 For any category $\mathcal{C}$ there is a contravariant functor to its opposite category $\mathcal{C}^*$ which assigns to an object X of $\mathcal{C}$ the object X^* of $\mathcal{C}^*$ and to a morphism $f: X \rightarrow Y$ of $\mathcal{C}$ the morphism $f^*: Y^* \rightarrow X^*$.

Note that any contravariant functor on $\mathcal{C}$ corresponds to a covariant functor on $\mathcal{C}^*$, and vice versa. Therefore any functor can be regarded as covariant on a suitable category. Despite this, we shall find it convenient to consider contravariant as well as covariant functors on $\mathcal{C}$, rather than consider only covariant functors on two categories.

Any functor from the category of topological spaces and continuous maps to an algebraic category (such as the category of abelian groups and

homomorphisms) is a representation of the topological category by an algebraic one. Algebraic topology is the study of such functors; we show that simple remarks about functors can be used to obtain necessary conditions for the solvability of topological problems.

9 **THEOREM** *Let T be a functor from a category $\mathcal{C}$ to a category $\mathfrak{A}$. Then T maps equivalences in $\mathcal{C}$ to equivalences in $\mathfrak{A}$.*

PROOF Assume that T is a covariant functor (the argument is similar if T is contravariant). Let $f: X \rightarrow Y$ be an equivalence in $\mathcal{C}$. Then $f^{-1}f = 1_X$. Therefore

$$1_{T(X)} = T(1_X) = T(f^{-1}f)T(f)$$

Similarly, $T(f)T(f^{-1}) = 1_{T(Y)}$. Therefore $T(f^{-1})$ is a two-sided inverse of $T(f)$, and $T(f)$ is an equivalence in $\mathfrak{A}$. ■

In particular, if T is an algebraic functor on the category of topological spaces and continuous maps, a necessary condition that X be homeomorphic to Y is that $T(X)$ be equivalent to $T(Y)$. Thus the functor H_0 of example 6 shows that the real line $\mathbf{R}$ and the real plane $\mathbf{R}^2$ are not homeomorphic [if they were homeomorphic, then $\mathbf{R} - 0$ would be homeomorphic to $\mathbf{R}^2 - p$ for some $p \in \mathbf{R}^2$, but $H_0(\mathbf{R} - 0)$ is a free abelian group on two generators, while $H_0(\mathbf{R}^2 - p)$ is a free abelian group on one generator]. This is a trivial example. However, the homology functors H_n defined in Chapter 4 generalize H_0 and can be used in much the same way to prove that $\mathbf{R}^n$ and $\mathbf{R}^m$ are not homeomorphic if $n \neq m$.

In applications of algebraic functors to topological problems the algebra will frequently play an essential role. For example, let $T_0(X)$ be the functor obtained by composing the functor H_0 with the forgetful functor, which assigns to every abelian group its underlying set. The functor T_0 contains less information than the functor H_0 and does not give as strong a necessary condition for homeomorphism [for example, $T_0(\mathbf{R} - 0)$ and $T_0(\mathbf{R}^2 - p)$ are both countably infinite sets and are equivalent in the category of sets and functions]. For this reason it is important to provide functors with as much algebraic structure as possible. Later we shall consider functors which depend on a chosen topological space. These functors take values in the category of sets and functions, but some of them, depending on properties of the particular spaces which define them, are functors to the category of groups and homomorphisms. The added algebraic structure in such cases will prove useful.

To show how functors can be applied to another problem, let A be a subspace of a topological space X and let $f: A \rightarrow Y$ be continuous. The extension problem is to determine whether f has a continuous extension to X —that is, whether the dotted arrow in the triangle

$$\begin{array}{ccc} A & \subset & X \\ f \searrow & & \swarrow \\ & Y & \end{array}$$

corresponds to a continuous map making the diagram commutative.

1.0 THEOREM Let T be a covariant functor (or contravariant functor) from the category of topological spaces and continuous maps to a category $\mathfrak{C}$. A necessary condition that a map $f: A \rightarrow Y$ be extendable to X (where $i: A \subset X$) is that there exist a morphism $q: T(X) \rightarrow T(Y)$ [or $q: T(Y) \rightarrow T(X)$] such that $q \circ T(i) = T(f)$ [or $T(f) = T(i) \circ q$].

PROOF Assume that $f': X \rightarrow Y$ is an extension of f . Then $f'i = f$. Therefore $T(f') \circ T(i) = T(f)$ [or $T(f) = T(i) \circ T(f')$], and $T(f')$ can be taken as the morphism q . ■

The above result can be applied to prove that the identity map of I cannot be extended to a continuous map $I \rightarrow I$. We use the functor H_0 and obtain the necessary condition that there must exist a homomorphism $q: H_0(I) \rightarrow H_0(I)$ such that $q \circ H_0(i) = H_0(1_I)$ (where $i: I \subset I$). Because $H_0(I)$ is a free abelian group on two generators and $H_0(I)$ is a free abelian group on one generator, there is no such homomorphism q . Again, this is a trivial example, but it illustrates the method, and the general homology functors H_q defined later can be used in the same way to show that there is no continuous map $E^{n+1} \rightarrow S^n$ that is the identity map on S^n .

Thus we see that a functor yields necessary conditions for the solvability of topological problems. There are situations in which these necessary conditions are also sufficient. For example, the functor C of example 5 gives a necessary and sufficient condition for homeomorphism—that is, two compact Hausdorff spaces X and Y are homeomorphic if and only if $C(X)$ and $C(Y)$ are isomorphic.¹ This is not a particularly useful result, however, because it seems to be no easier to determine whether or not two normed rings are isomorphic than it is to determine whether or not two compact Hausdorff spaces are homeomorphic. We seek functors to categories that are somewhat simpler than the category of topological spaces, so that the algebraic problems that arise in these categories can be effectively solved. One big problem of algebraic topology is to find, and compute, sufficiently many such functors that the solvability of a particular topological problem is equivalent to the solvability of the corresponding (and simpler) algebraic problems.

We shall also have occasion to compare functors with each other. This is done by means of a suitable definition of a map between functors. Let T_1 and T_2 be functors of the same variance (either both covariant or both contravariant) from a category $\mathfrak{C}$ to a category $\mathfrak{Q}$. A *natural transformation* q from T_1 to T_2 is a function from the objects of $\mathfrak{C}$ to morphisms of $\mathfrak{Q}$ such that for every morphism $f: X \rightarrow Y$ of $\mathfrak{C}$ the appropriate one of the following diagrams is commutative:

$$\begin{array}{ccccc} T_1(X) & \xrightarrow{T_1(f)} & T_1(Y) & & T_1(X) & \xleftarrow{T_1(f)} & T_1(Y) \\ \downarrow q(X) & & \downarrow q(Y) & & \downarrow q(X) & & \downarrow q(Y) \\ T_2(X) & \xrightarrow{T_2(f)} & T_2(Y) & & T_2(X) & \xleftarrow{T_2(f)} & T_2(Y) \\ & & T_1, T_2 \text{ covariant} & & & & T_1, T_2 \text{ contravariant} \end{array}$$

¹ See Theorem D on page 330 of G. F. Simmons, *Introduction to Topology and Modern Analysis*, McGraw-Hill Book Company, New York, 1963.

If φ is a natural transformation from T_1 to T_2 such that $q(X)$ is an equivalence in $\mathfrak{Q}$ for each object X in $\mathfrak{C}$, then φ is called a *natural equivalence*.

As an example of a natural transformation, let Y_1 and Y_2 be objects of a category $\mathfrak{C}$ and let $g: Y_1 \rightarrow Y_2$ be a morphism in $\mathfrak{C}$. There is a natural transformation $g\#$ from the covariant functor π_{Y_2} to the covariant functor π_{Y_1} , and a natural transformation $g\#$ from the contravariant functor π^{Y_1} to the contravariant functor π^{Y_2} . If g is an equivalence in $\mathfrak{C}$, both these natural transformations are natural equivalences.

It is also of interest to consider functors of several variables. Thus, if $\mathfrak{C}_1$, $\mathfrak{C}_2$, and $\mathfrak{Q}$ are categories, a covariant functor from $\mathfrak{C}_1 \times \mathfrak{C}_2$ to $\mathfrak{Q}$ is called a *functor of two arguments covariant* in each. A covariant functor from $\mathfrak{C}_1 \times \mathfrak{C}_2^*$ to $\mathfrak{Q}$, regarded as a function from ordered pairs (X_1, X_2) , where X_1 is an object of $\mathfrak{C}_1$ and X_2 is an object of $\mathfrak{C}_2$, is called a *functor of two arguments covariant* in the first and *contravariant* in the second. In a similar fashion, functors of more arguments with mixed variance are defined.

If $\mathfrak{C}$ is any category, there is a functor of two arguments in $\mathfrak{C}$ to the category of sets and functions which is contravariant in the first argument and covariant in the second. This functor assigns to an ordered pair of objects X and Y of $\mathfrak{C}$ the set $\text{hom}(X, Y)$ and to an ordered pair of morphisms $f: X' \rightarrow X$ and $g: Y \rightarrow Y'$ in $\mathfrak{C}$ the function $f\#g\# = g\#f\#: \text{hom}(X, Y) \rightarrow \text{hom}(X', Y')$.

3 HOMOTOPY

The problem of classifying topological spaces and continuous maps up to topological equivalence does not seem to be amenable to attack directly by computable algebraic functors, as described in Sec. 1.2. Many of the computable functors, because they are computable, are invariant under continuous deformation. Therefore they cannot distinguish between spaces (or maps) that can be continuously deformed from one to the other; the most that can be hoped for from such functors is that they characterize the space (or map) up to continuous deformation.

The intuitive concept of a continuous deformation will be made precise in this section in the concept of homotopy. This leads to the homotopy category which is fundamental for algebraic topology. Its objects are topological spaces and its morphisms are equivalence classes of continuous maps (two maps being equivalent if one can be continuously deformed into the other). For technical reasons we consider not just the homotopy category of topological spaces, but rather the larger homotopy category of pairs.

A *topological pair* (X, A) consists of a topological space X and a subspace $A \subset X$. If A is empty, denoted by $\emptyset$, we shall not distinguish between the pair $(X, \emptyset)$ and the space X . A *subpair* $(X', A') \subset (X, A)$ consists of a pair with $X' \subset X$ and $A' \subset A$. A map $f: (X, A) \rightarrow (Y, B)$ between pairs is a continuous function f from X to Y such that $f(A) \subset B$, and as in Sec. 1.1, there is

a category of topological pairs and maps between them which contains as full subcategories the category of topological spaces and continuous maps, as well as the category of pointed topological spaces and continuous maps.

Given a pair (X, A) , we let $(X, A) \times I$ denote the pair $(X \times I, A \times I)$. Let $X' \subset X$ and suppose that $f_0, f_1: (X, A) \rightarrow (Y, B)$ agree on X' (that is, $f_0|_{X'} = f_1|_{X'}$). Then f_0 is *homotopic to f_1 relative to X'* , denoted by $f_0 \simeq f_1 \text{ rel } X'$, if there exists a map

$$F: (X, A) \times I \rightarrow (Y, B)$$

such that $F(x, 0) = f_0(x)$ and $F(x, 1) = f_1(x)$ for $x \in X$ and $F(x, t) = f_0(x)$ for $x \in X'$ and $t \in I$. Such a map F is called a *homotopy relative to X'* from f_0 to f_1 and is denoted by $F: f_0 \simeq f_1 \text{ rel } X'$. If $X' = \emptyset$, we omit the phrase 'relative to $\emptyset$ '. Clearly, $f_0 \simeq f_1 \text{ rel } X'$ implies $f_0 \simeq f_1 \text{ rel } X''$ for any $X'' \subset X'$. A map from X to Y is said to be *null homotopic*, or *inessential*, if it is homotopic to some constant map.

For $t \in I$ define $h_t: (X, A) \rightarrow (X, A) \times I$ by $h_t(x) = (x, t)$. If $F: f_0 \simeq f_1 \text{ rel } X'$, then $Fh_0 = f_0$, $Fh_1 = f_1$, and $Fh_t|_{X'} = f_0|_{X'}$ for all $t \in I$. Therefore the collection $\{Fh_t\}_{t \in I}$ is a continuous one-parameter family of maps from (X, A) to (Y, B) , agreeing on X' , which connects $f_0 = Fh_0$ to $f_1 = Fh_1$. Hence $f_0 \simeq f_1 \text{ rel } X'$ corresponds to the intuitive idea of continuously deforming f_0 into f_1 by maps all of which agree on X' . Note that if $f_0 \simeq f_1 \text{ rel } X'$ there will usually be many maps F which are homotopies relative to X' from f_0 to f_1 (see example 3 below).

1 EXAMPLE Let $X = Y = \mathbf{R}^n$ and define $f_0(x) = x$ and $f_1(x) = 0$ for $x \in \mathbf{R}^n$ (that is, $f_0 = 1_{\mathbf{R}^n}$ and f_1 is the constant map of $\mathbf{R}^n$ to its origin). If $F: \mathbf{R}^n \times I \rightarrow \mathbf{R}^n$ is defined by

$$F(x, t) = (1 - t)x$$

then $F: f_0 \simeq f_1 \text{ rel } 0$.

2 EXAMPLE Let $X = Y = I$ and define $f_0(t) = t$ and $f_1(t) = 0$ for $t \in I$. If $F: I \times I \rightarrow I$ is defined by

$$F(t, t') = (1 - t')t$$

then $F: f_0 \simeq f_1 \text{ rel } 0$.

3 EXAMPLE Let $X = Y = E^2 = \{z \in \mathbf{C} \mid |z| \leq r \leq 1\}$ and let $A = B = S^1 = \{z \in \mathbf{C} \mid |z| = e^{i\theta}\}$. Define $f_0: (E^2, S^1) \rightarrow (E^2, S^1)$ to be the identity map and $f_1: (E^2, S^1) \rightarrow (E^2, S^1)$ to be the reflection in the origin [that is, $f_1(re^{i\theta}) = re^{i(\theta+\pi)}$]. Define a homotopy $F: f_0 \simeq f_1 \text{ rel } 0$ by $F(re^{i\theta}, t) = re^{i(\theta+t\pi)}$. Another homotopy F' from f_0 to f_1 rel 0 is defined by $F'(re^{i\theta}, t) = re^{i(\theta-t\pi)}$.

¹ A one-parameter family $f_t: (X, A) \rightarrow (Y, B)$ for $t \in I$ is continuous if $f_t(x)$ is jointly continuous in t and x , in which case the function $(x, t) \rightarrow f_t(x)$ is a homotopy from f_0 to f_1 . The corresponding function $t \rightarrow f_t$ from I to $(Y, B)^{(X, A)}$ is always continuous [where $(Y, B)^{(X, A)} = \{g: (X, A) \rightarrow (Y, B)\}$ topologized by the compact-open topology]. Conversely, in case X is a locally compact Hausdorff space, it follows from theorem 2.8 in the Introduction that for any continuous map $q: I \rightarrow (Y, B)^{(X, A)}$ the one-parameter family $q(t)$ is continuous and defines a homotopy from $q(0)$ to $q(1)$.

4 EXAMPLE Let X be an arbitrary space and let Y be a convex subset of $\mathbf{R}^n$. Let $f_0, f_1: X \rightarrow Y$ be maps which agree on some subspace $X' \subset X$. Then $f_0 \simeq f_1 \text{ rel } X'$ because the map $F: X \times I \rightarrow Y$ defined by

$$F(x, t) = tf_1(x) + (1 - t)f_0(x)$$

is a homotopy relative to X' from f_0 to f_1 .

Example 4 is a generalization of examples 1 and 2. In example 3 the space E^2 is convex, but the homotopy between f_0 and f_1 cannot be taken to be a particular case of the homotopy in example 4, because it must keep S^1 mapped into itself at all stages, and S^1 is not convex.

To define the homotopy category we need the following easy results.

5 THEOREM *Homotopy relative to X' is an equivalence relation in the set of maps from (X, A) to (Y, B) .*

PROOF *Reflexivity.* For $f: (X, A) \rightarrow (Y, B)$ define $F: f \simeq f \text{ rel } X$ by $F(x, t) = f(x)$.

Symmetry. Given $F: f_0 \simeq f_1 \text{ rel } X'$, define $F': f_1 \simeq f_0 \text{ rel } X'$ by $F'(x, t) = F(x, 1 - t)$.

Transitivity. Given $F: f_0 \simeq f_1 \text{ rel } X'$ and $G: f_1 \simeq f_2 \text{ rel } X'$, define $H: f_0 \simeq f_2 \text{ rel } X'$ by

$$H(x, t) = \begin{cases} F(x, 2t) & 0 \leq t \leq \frac{1}{2} \\ G(x, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

Note that H is continuous because its restriction to each of the closed sets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$ is continuous. ■

It follows that the set of maps from (X, A) to (Y, B) is partitioned into disjoint equivalence classes by the relation of homotopy relative to X' . These equivalence classes are called *homotopy classes relative to X'* . We use $[X, A; Y, B]_{X'}$ to denote this set of homotopy classes. Given $f: (X, A) \rightarrow (Y, B)$, we use $[f]_{X'}$ to denote the element of $[X, A; Y, B]_{X'}$ determined by f . Homotopy classes relative to the empty set will be denoted by omitting the subscript X' .

6 THEOREM *Composites of homotopic maps are homotopic.*

PROOF Let $f_0, f_1: (X, A) \rightarrow (Y, B)$ be homotopic relative to X' and let $g_0, g_1: (Y, B) \rightarrow (Z, C)$ be homotopic relative to Y' , where $f_1(X') \subset Y'$. To show that $g_0f_0, g_1f_1: (X, A) \rightarrow (Z, C)$ are homotopic relative to X' , let $F: f_0 \simeq f_1 \text{ rel } X'$ and $G: g_0 \simeq g_1 \text{ rel } Y'$. Then the composite

$$\begin{aligned} (X, A) \times I &\xrightarrow{F} (Y, B) \xrightarrow{g_0} (Z, C) \\ (X, A) \times I &\xrightarrow{f_1 \times 1_Y} (Y, B) \times I \xrightarrow{G} (Z, C) \end{aligned}$$

is a homotopy relative to X' from g_0f_0 to g_1f_1 , and the composite

is a homotopy relative to $f_1^{-1}(Y')$ from g_0f_1 to g_1f_1 . Since $X' \subset f_1^{-1}(Y')$, we have shown that $g_0f_0 \simeq g_1f_1 \text{ rel } X'$ and $g_0f_1 \simeq g_1f_1 \text{ rel } X'$. The result now follows from theorem 5. ■

The last result shows that there is a *homotopy category of pairs* whose objects are topological pairs and whose morphisms are homotopy classes (relative to $\emptyset$). This category contains as full subcategories the *homotopy category of topological spaces* (also shortened to *homotopy category*) and the *homotopy category of pointed topological spaces*. There is a covariant functor from the category of pairs and maps to the homotopy category of pairs whose object function is the identity map and whose mapping function sends a map f to its homotopy class $[f]$. As pointed out at the beginning of the section, most of the algebraic functors we consider will be defined from the appropriate homotopy category. A diagram of topological pairs and maps is said to be *homotopy commutative* if it can be made a commutative diagram in the homotopy category (that is, when each map is replaced by its homotopy class).

As in example 1.2.4, for any pair (P, Q) there is a covariant functor $\pi_{(P, Q)}$ (or a contravariant functor $\pi^{(P, Q)}$) from the homotopy category of pairs to the category of sets and functions defined by $\pi_{(P, Q)}(X, A) = [P, Q; X, A]$ (or $\pi^{(P, Q)}(X, A) = [X, A; P, Q]$), and if $f: (X, A) \rightarrow (Y, B)$, then $\pi_{(P, Q)}([f]) = f_{\#}$ (or $\pi^{(P, Q)}([f]) = f^{\#}$), where $f_{\#}[g] = [fg]$ for $g: (P, Q) \rightarrow (X, A)$ (or $f^{\#}[h] = [hf]$ for $h: (Y, B) \rightarrow (P, Q)$). If $\alpha: (P, Q) \rightarrow (P', Q')$, there is a natural transformation $\alpha_{\#}$ from $\pi_{(P, Q)}$ to $\pi_{(P', Q')}$ and a natural transformation $\alpha^{\#}$ from $\pi^{(P, Q)}$ to $\pi^{(P', Q')}$.

A map $f: (X, A) \rightarrow (Y, B)$ is called a *homotopy equivalence* if $[f]$ is an equivalence in the homotopy category of pairs. A map $g: (Y, B) \rightarrow (X, A)$ is called a *homotopy inverse* of f if $[g] = [f]^{-1}$ in the homotopy category. Pairs (X, A) and (Y, B) are said to have the *same homotopy type* if they are equivalent in the homotopy category.

The simplest nonempty space is a one-point space. We characterize the homotopy type of such a space as follows. A topological space X is said to be *contractible* if the identity map of X is homotopic to some constant map of X to itself. A homotopy from 1_X to the constant map of X to $x_0 \in X$ is called a *contraction* of X to x_0 . Examples 1 and 2 show that $\mathbf{R}^n$ and I are contractible, and example 4 shows that any convex subset of $\mathbf{R}^n$ is contractible. The following lemma may be regarded as a generalization of the result of example 4.

7 LEMMA Any two maps of an arbitrary space to a contractible space are homotopic.

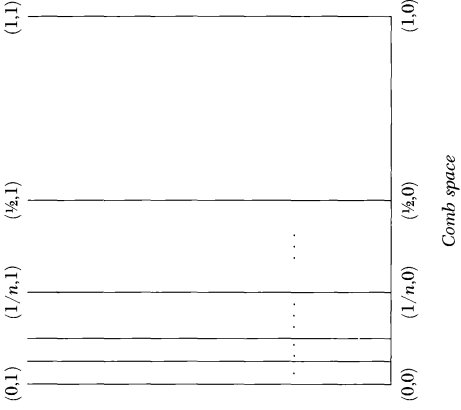
PROOF Let Y be a contractible space and suppose $1_Y \simeq c$, where c is a constant map of Y to itself. Let $f_0, f_1: X \rightarrow Y$ be arbitrary. By theorem 6, $f_0 = 1_Y f_0 \simeq c f_0$, and similarly, $f_1 \simeq c f_1$. Since $c f_0 = c f_1$, it follows from theorem 5 that $f_0 \simeq f_1$. ■

8 COROLLARY If Y is contractible, any two constant maps of Y to itself are homotopic, and the identity map is homotopic to any constant map of Y to itself. ■

It is interesting to observe that lemma 7 cannot be strengthened to the case of relative homotopy. That is, if f_0 and f_1 are maps of X into a contract-

ible space Y which agree on $X' \subset X$, it need not be true that $f_0 \simeq f_1$ rel X' (although example 4 shows this to be true for convex subsets of $\mathbf{R}^n$). The following example illustrates this and will be referred to again later.

► **EXAMPLE** The comb space Y illustrated in the diagram



is defined by

$$Y = \{(x, y) \in \mathbf{R}^2 \mid 0 \leq y \leq 1, x = 0, 1/n \text{ or } y = 0, 0 \leq x \leq 1\}$$

Let $F: Y \times I \rightarrow Y$ be defined by $F((x, y), t) = (x, (1 - t)y)$. Then F is a homotopy from 1_Y to the projection of Y to the x axis. Since the latter map is homotopic to a constant map, Y is contractible. Let $c: Y \rightarrow Y$ be the constant map of Y to the point $(0, 1)$. By corollary 8, $1_Y \simeq c$, but even though these two maps agree on $(0, 1)$, there is no homotopy relative to $(0, 1)$ between them.

The following theorem shows that contractible spaces are homotopically as simple as possible.

► **THEOREM** A space is contractible if and only if it has the same homotopy type as a one-point space.

PROOF Assume that X is contractible and let $F: X \times I \rightarrow X$ be a contraction of X to a point $x_0 \in X$. Let P be the one-point space consisting of x_0 and let $f: X \rightarrow P$ and $j: P \subset X$. Then $fj = 1_P$ and $F: 1_X \simeq jf$. Therefore $[f] = [f]^{-1}$, and f is a homotopy equivalence from X to P .

Conversely, if X has the same homotopy type as a one-point space P , let $f: X \rightarrow P$ be a homotopy equivalence with homotopy inverse $g: P \rightarrow X$. Then $1_X \simeq g f$. Because $g f$ is a constant map, X is contractible. ■

11 COROLLARY Two contractible spaces have the same homotopy type, and any continuous map between contractible spaces is a homotopy equivalence.

PROOF The first part follows from theorem 10 and the transitivity of the relation of having the same homotopy type. The second part follows from the first part and lemma 7 (and from the obvious fact that any map homotopic to a homotopy equivalence is itself a homotopy equivalence). ■

The next result establishes an important relation between homotopy and the extendability of maps.

12 THEOREM Let p_0 be any point of S^n and let $f: S^n \rightarrow Y$. The following are equivalent:

- (a) f is null homotopic
- (b) f can be continuously extended over E^{n+1}
- (c) f is null homotopic relative to p_0

PROOF (a) $\Rightarrow$ (b). Let $F: f \simeq c$, where c is the constant map of S^n to $y_0 \in Y$. Define an extension f' of f over E^{n+1} by

$$f'(x) = \begin{cases} y_0 & 0 \leq \|x\| \leq \frac{1}{2} \\ F(x/\|x\|, 2 - 2\|x\|) & \frac{1}{2} \leq \|x\| \leq 1 \end{cases}$$

Since $F(x, 1) = y_0$ for all $x \in S^n$, the map f' is well-defined. f' is continuous because its restriction to each of the closed sets $\{x \in E^{n+1} | 0 \leq \|x\| \leq \frac{1}{2}\}$ and $\{x \in E^{n+1} | \frac{1}{2} \leq \|x\| \leq 1\}$ is continuous. Since $F(x, 0) = f(x)$ for $x \in S^n$, $f'|_{S^n} = f$ and f' is a continuous extension of f to E^{n+1} .

(b) $\Rightarrow$ (c). If f has the continuous extension $f': E^{n+1} \rightarrow Y$, define $F: S^n \times I \rightarrow Y$ by

$$F(x, t) = f'((1 - t)x + tp_0)$$

Then $F(x, 0) = f'(x) = f(x)$ and $F(x, 1) = f'(p_0)$ for $x \in S^n$. Since $F(p_0, t) = f'(p_0)$ for $t \in I$, F is a homotopy relative to p_0 from f to the constant map to $f'(p_0)$. (c) $\Rightarrow$ (a). This is obvious. ■

Combining theorem 12 with lemma 7, we obtain the following result.

13 COROLLARY Any continuous map from S^n to a contractible space has a continuous extension over E^{n+1} . ■

4 RETRACTION AND DEFORMATION

This section is concerned mainly with inclusion maps. We consider whether such a map has a left inverse, a right inverse, or a two-sided inverse in either the category of topological spaces and continuous maps or the homotopy category.¹

¹ Many of the results in this section can be found in R. H. Fox, On homotopy type and deformation retracts, *Annals of Mathematics*, vol. 44, pp. 40–50, 1943 (see also H. Samelson, Remark on a paper by R. H. Fox, *Annals of Mathematics*, vol. 45, pp. 448–449, 1944).

A subspace A of X is called a *retract* of X if the inclusion map $i: A \subset X$ has a left inverse in the category of topological spaces and continuous maps. Hence A is a retract of X if and only if there is a continuous map $r: X \rightarrow A$ such that $ri = 1_A$ [that is, $r(x) = x$ for $x \in A$]. Such a map r is called a *retraction* of X to A .

A subspace A of X is called a *weak retract* of X if the inclusion map $i: A \subset X$ has a left homotopy inverse (that is, a left inverse in the homotopy category). Thus A is a weak retract of X if and only if there is a continuous map $r: X \rightarrow A$ such that $ri \simeq 1_A$. Such a map r is called a *weak retraction* of X to A .

Any one-point subspace is a retract of any larger space containing it. A discrete space with more than one point is never a weak retract of a connected space containing it. If A is a retract of X , it is a weak retract of X . The converse is not true, as is shown by the following example.

EXAMPLE Let X be the closed unit square I^2 in $\mathbb{R}^2$ and let $A \subset X$ be the comb space of example 1.3.9. Then A and X are both contractible, and by corollary 1.3.11, the inclusion map $A \subset X$ is a homotopy equivalence. Therefore A is a weak retract of X . However, it can be shown that A is not a retract of X .

Despite the fact that, in general, a weak retract need not be a retract, these concepts do coincide when A is a suitable subspace of X . This occurs frequently enough to warrant special consideration and will prove of use later. Let (X, A) be a pair and Y be a space. (X, A) is said to have the *homotopy extension property with respect to* Y if, given maps $g: X \rightarrow Y$ and $G: A \times I \rightarrow Y$ such that $g(x) = G(x, 0)$ for $x \in A$, there is a map $F: X \times I \rightarrow Y$ such that $F(x, 0) = g(x)$ for $x \in X$ and $F|_{A \times I} = G$. If g is regarded as a map of $X \times 0$ to Y , the existence of F is equivalent to the existence of a map represented by the dotted arrow which makes the following diagram commutative:

$$\begin{array}{ccccc} A \times 0 & \subset & A \times I & & \\ & & \searrow g & & \\ & & Y & \cap & \\ & & \nearrow g & & \\ X \times 0 & \subset & X \times I & & \end{array}$$

If (X, A) has the homotopy extension property with respect to Y and $f_0, f_1: A \rightarrow Y$ are homotopic, then if f_0 has an extension to X , so does f_1 ; for if $g: X \rightarrow Y$ is an extension of f_0 and $G: A \times I \rightarrow Y$ is a homotopy from f_0 to f_1 , the homotopy extension property implies the existence of a map $F: X \times I \rightarrow Y$ which is an extension of G , therefore $F(x, 1)$ is an extension of f_1 . It follows that whether or not a map $A \rightarrow Y$ can be extended over X is a property of the homotopy class of that map. Therefore the homotopy extension property implies that the extension problem for maps $A \rightarrow Y$ is a problem in the homotopy category.

Of particular importance is the case when (X, A) has the homotopy extension property with respect to any space. More generally, a map $f: X' \rightarrow X$ is called a *cofibration* if, given maps $g: X \rightarrow Y$ and $G: X' \times I \rightarrow Y$ (where Y is arbitrary) such that $g(f(x')) = G(x', 0)$ for $x' \in X'$, there is a map $F: X \times I \rightarrow Y$ such that $F(x, 0) = g(x)$ for $x \in X$ and $F(f(x'), t) = G(x', t)$ for $x' \in X'$ and $t \in I$. If g is regarded as a map of $X \times 0$ to Y , the existence of F is equivalent to the existence of a map represented by the dotted arrow which makes the following diagram commutative:

$$\begin{array}{ccc} X' \times 0 & \subset & X' \times I \\ \downarrow f \times 1_0 & \nearrow G & \downarrow f \times 1_I \\ Y & & X \times I \end{array}$$

Thus an inclusion map $i: A \subset X$ is a cofibration if and only if (X, A) has the homotopy extension property with respect to any space.

2 THEOREM *If (X, A) has the homotopy extension property with respect to A , then A is a weak retract of X if and only if A is a retract of X .*

PROOF We show that any weak retraction $r: X \rightarrow A$ is, in fact, homotopic to a retraction. Let $i: A \subset X$; then $ri \simeq 1_A$. Let $G: A \times I \rightarrow A$ be a homotopy from ri to 1_A ; then $G(x, 0) = r(x)$ for $x \in A$. Because (X, A) has the homotopy extension property with respect to A , there is a map $F: X \times I \rightarrow A$ which extends G such that $F(x, 0) = r(x)$ for $x \in X$. If $r': X \rightarrow A$ is defined by $r'(x) = F(x, 1)$, then r' is a retraction of X to A , and F is a homotopy from r to r' . ■

We can just as well consider inclusion maps with right homotopy inverses as those with left homotopy inverses. This leads to the following definitions. Given $X' \subset X$, a *deformation D of X' in X* is a homotopy

$$D: X' \times I \rightarrow X$$

such that $D(x', 0) = x'$ for $x' \in X'$. If, moreover, $D(X' \times 1)$ is contained in a subspace A of X , D is said to be a *deformation of X' into A* and X' is said to be *deformable in X into A* . A space X is said to be *deformable* into a subspace A if it is deformable in itself into A . Thus a space X is contractible if and only if it is deformable into one of its points.

3 LEMMA *A space X is deformable into a subspace A if and only if the inclusion map $i: A \subset X$ has a right homotopy inverse.*

PROOF If i has a right homotopy inverse $f: X \rightarrow A$, then $if \simeq 1_X$. Let $F: X \times I \rightarrow X$ be a homotopy from 1_X to if ; then $F(x, 0) = x$, so F is a deformation of X , and $F(X \times 1) = if(X) \subset A$, so X is deformable into A .

Conversely, if X is deformable into A , let $D: X \times I \rightarrow X$ be a deformation such that $D(X \times 1) \subset A$. Let $f: X \rightarrow A$ be defined by the equation

$$if(x) = D(x, 1) \quad x \in X$$

Then $D: 1_X \simeq if$, showing that f is a right homotopy inverse of i . ■

Note that an inclusion map $i: A \subset X$ never has a right inverse in the category of topological spaces and continuous maps except in the trivial case $A = X$.

We now consider inclusion maps which are homotopy equivalences. A subspace $A \subset X$ is called a *weak deformation retract* of X if the inclusion map $i: A \subset X$ is a homotopy equivalence. From lemma 1.1.1 and lemma 3 above we obtain the following result.

4 LEMMA *A is a weak deformation retract of X if and only if A is a weak retract of X and X is deformable into A .* ■

As was the case with the concept of weak retract, there are more useful concepts than that of weak deformation retract. The subspace A is a *strong deformation retract* of X if there is a retraction r of X to A such that if $i: A \subset X$, then $1_X \simeq ir$ rel A . If $F: 1_X \simeq ir$ rel A , F is called a *strong deformation retraction* of X to A .

There is an intermediate concept useful in comparing the weak and strong forms already defined. A subspace A is called a *deformation retract* of X if there is a retraction r of X to A such that if $i: A \subset X$, then $1_X \simeq ir$. If $F: 1_X \simeq ir$, F is called a *deformation retraction* of X to A . A homotopy $F: X \times I \rightarrow X$ is a deformation retraction if and only if $F(x, 0) = x$ for $x \in X$, $F(X \times 1) \subset A$, and $F(x, 1) = x$ for $x \in A$. It is a strong deformation retraction if and only if it also satisfies the condition $F(x, t) = x$ for $x \in A$ and $t \in I$.

5 EXAMPLE It follows from example 1.3.4 that any one-point subset of a convex subset of $\mathbf{R}^n$ is a strong deformation retract of the convex set.

6 EXAMPLE S^n is a strong deformation retract of $\mathbf{R}^{n+1} - 0$. In fact the map $F: (\mathbf{R}^{n+1} - 0) \times I \rightarrow \mathbf{R}^{n+1} - 0$ defined by

$$F(x, t) = (1 - t)x + \frac{tx}{\|x\|} \quad x \in \mathbf{R}^{n+1} - 0, t \in I$$

is a strong deformation retraction of $\mathbf{R}^{n+1} - 0$ to S^n .

It is clear that a strong deformation retract is a deformation retract, and a deformation retract is a weak deformation retract. The following examples show that neither of these implications is reversible.

7 EXAMPLE As in example 1 above, let X be the closed unit square and A be the comb space. As pointed out in example 1, the inclusion map $A \subset X$ is a homotopy equivalence, but A is not a retract of X . Therefore A is a weak deformation retract of X which is not a deformation retract of X .

8 EXAMPLE Let X be the comb space and A be the one-point subspace of X consisting of the point $(0, 1)$. Because X is contractible, there is a homotopy F from 1_X to the constant map of X to A . Such a map F is a deformation re-

traction of X to A . However, as was remarked in example 1.3.9, there is no homotopy relative to A from 1_X to the constant map to A ; therefore A is a deformation retract of X which is not a strong deformation retract of X .

In the presence of suitable homotopy extension properties the three concepts of deformation retract coincide, and we shall now prove this.

9 LEMMA *If X is deformable into a retract A , then A is a deformation retract of X .*

PROOF Let $r: X \rightarrow A$ be a retraction and let $i: A \subset X$. Then r is a left homotopy inverse of i . Because X is deformable into A , it follows from lemma 3 that i has a right homotopy inverse. By lemma 1.1.1, r is also a right homotopy inverse of i . Since $1_X \simeq ir$, A is a deformation retract of X . ■

Combining lemma 9 with theorem 2 yields the following corollary.

10 COROLLARY *If $\langle X, A \rangle$ has the homotopy extension property with respect to A , then A is a weak deformation retract of X if and only if A is a deformation retract of X .* ■

11 THEOREM *If $\langle X \times I, (X \times 0) \cup (A \times I) \cup (X \times 1) \rangle$ has the homotopy extension property with respect to X and A is closed in X , then A is a deformation retract of X if and only if A is a strong deformation retract of X .*

PROOF If A is a deformation retract of X , let $F: X \times I \rightarrow X$ be a homotopy from 1_X to ir , where $r: X \rightarrow A$ is a retraction and $i: A \subset X$. A homotopy

$$\begin{aligned} G: [(X \times 0) \cup (A \times I) \cup (X \times 1)] \times I &\rightarrow X \\ G((x, 0), t') &= x & x \in X, t' \in I \\ G((x, t), t') &= F(x, (1 - t')t) & x \in A; t, t' \in I \\ G((x, 1), t') &= F(r(x), 1 - t') & x \in X, t' \in I \end{aligned}$$

G is well-defined, because for $x \in A$

$$G((x, 0), t') = x = F(x, 0)$$

by the first two equations and

$$G((x, 1), t') = F(x, 1 - t') = F(r(x), 1 - t')$$

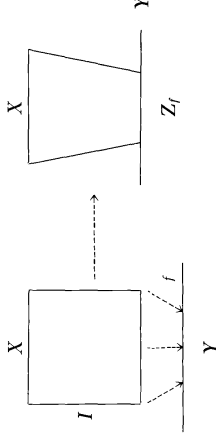
by the last two equations. G is continuous because its restriction to each of the closed sets $(X \times 0) \times I$, $(A \times I) \times I$, and $(X \times 1) \times I$ is continuous. For $(x, t) \in (X \times 0) \cup (A \times I) \cup (X \times 1)$, $G((x, t), 0) = F(x, 0)$ [because $F(x, 0) = x$, and since r is a retraction, $F(r(x), 1) = ir(r(x)) = r(x) = F(x, 1)$]. Therefore G restricted to $[(X \times 0) \cup (A \times I) \cup (X \times 1)] \times 0$ can be extended to $(X \times I) \times 0$. From the homotopy extension property in the hypothesis, G restricted to $[(X \times 0) \cup (A \times I) \cup (X \times 1)] \times 1$ can be extended to $(X \times I) \times 1$. Let $G': (X \times I) \times 1 \rightarrow X$ be such an extension, and define $H: X \times I \rightarrow X$

by $H(x, t) = G'((x, t), 1)$. Then we have the equations

$$\begin{aligned} H(x, 0) &= G'((x, 0), 1) = G((x, 0), 1) = x & x \in X \\ H(x, 1) &= G((x, 1), 1) = F(r(x), 0) = r(x) & x \in X \\ H(x, t) &= G((x, t), 1) = F(x, 0) = x & x \in A, t \in I \end{aligned}$$

Therefore H is a homotopy relative to A from 1_X to ir , so A is a strong deformation retract of X . ■

The next result asserts that any map is equivalent in the homotopy category to an inclusion map that is a cofibration. Let $f: X \rightarrow Y$ and let Z_f denote the quotient space obtained from the topological sum of $X \times I$ and Y by identifying $(x, 1) \in X \times I$ with $f(x) \in Y$. Z_f is called the *mapping cylinder* of f and is depicted in the diagram



Mapping cylinder

We use $[x, t]$ to denote the point of Z_f corresponding to $(x, t) \in X \times I$ under the identification map and $[y]$ to denote the point of Z_f corresponding to $y \in Y$ (thus $[x, 1] = [f(x)]$ for $x \in X$). There is an imbedding $i: X \rightarrow Z_f$ with $i(x) = [x, 0]$ and an imbedding $j: Y \rightarrow Z_f$ with $j(y) = [y]$. X and Y are regarded as subspaces of Z_f by means of these imbeddings. A retraction $r: Z_f \rightarrow Y$ is defined by $r[x, t] = [f(x)]$ for $x \in X$ and $t \in I$ and $r[y] = [y]$ for $y \in Y$.

12 THEOREM *Given a map $f: X \rightarrow Y$, there is a commutative diagram*

$$\begin{array}{ccc} X & \xrightarrow{i} & Z_f \\ & \searrow i & \downarrow r \\ & & Y \end{array}$$

such that

- (a) $1_{Z_f} \simeq jr \text{ rel } Y$
- (b) i is a cofibration

PROOF By definition, $ir = f$, and the triangle is commutative.

(a) A homotopy $F: Z_f \times I \rightarrow Z_f$ is defined by

$$\begin{aligned} F([x, t], t') &= [x, (1 - t')t + t'] & x \in X; t, t' \in I \\ F([y], t') &= [y] & y \in Y, t' \in I \end{aligned}$$

Then $F: 1_{Z_f} \simeq jr \text{ rel } Y$.

(b) Let $g: Z_I \rightarrow W$ and $G: X \times I \rightarrow W$ be such that $g(x, 0) = G(x, 0)$ for $x \in X$. If $H: Z_I \times I \rightarrow W$ is defined by the equations

$$\begin{aligned} H([y], t') &= g[y] & y \in Y, t' \in I \\ H([x, t], t') &= \begin{cases} g[x, (2t - t')/(2 - t')] & 0 \leq t' \leq 2t \leq 2, x \in X \\ G(x, (t' - 2t)/(1 - t)) & 0 \leq 2t \leq t' \leq 1, x \in X \end{cases} \end{aligned}$$

then $H([x, t], 0) = g[x, t]$ and $H([y], 0) = g[y]$, and $H|X \times I = G$. ■

It follows that the map $i: X \subset Z_I$ is a cofibration equivalent in the homotopy category to the map $f: X \rightarrow Y$. The mapping cylinder can be used to prove the following amusing result.

1.3 THEOREM *Two spaces X and Y have the same homotopy type if and only if they can be imbedded as weak deformation retracts of the same space Z .*

PROOF If X and Y can be imbedded as weak deformation retracts of the same space Z , then X and Y each have the same homotopy type as Z . Therefore X and Y have the same homotopy type.

Conversely, if $f: X \rightarrow Y$ is a homotopy equivalence, it follows from theorem 12 that if Z_I is the mapping cylinder of f , then the composite $X \xrightarrow{i} Z_I \xrightarrow{j} Y$ is a homotopy equivalence. Because r is a homotopy equivalence, this implies that i is a homotopy equivalence. By theorem 12a, $j: Y \rightarrow Z_I$ is a homotopy equivalence. Therefore X and Y are imbedded as weak deformation retracts in Z_I . ■

All the foregoing concepts can also be considered for pairs. For example, a pair $(X', A') \subset (X, A)$ is a *strong deformation retract* if there is a map $F: (X, A) \times I \rightarrow (X, A)$ such that $F(x, 0) = x$ for $x \in X$, $F(X \times 1) \subset X'$, $F(A \times 1) \subset A'$, and $F(x', t) = x'$ for $x' \in X'$ and $t \in I$. The *mapping cylinder* of a map $f: (X, A) \rightarrow (Y, B)$, where A is closed in X , is the pair $(Z_f, Z_{f|A})$, where Z_f is the mapping cylinder of the map $f_1: X \rightarrow Y$ defined by f and $Z_{f|A}$ is the mapping cylinder of the map $f_2: A \rightarrow B$ defined by f . A map $f: (X', A') \rightarrow (X, A)$ is a *cofibration* if, given maps $g: (X, A) \rightarrow (Y, B)$ and $G: (X', A') \times I \rightarrow (Y, B)$ [where (Y, B) is arbitrary] such that $G(x', 0) = gf(x')$ for $x' \in X'$, there exists a map $F: (X, A) \times I \rightarrow (Y, B)$ such that $F(x, 0) = g(x)$ for $x \in X$ and $G(x', t) = F(f(x'), t)$ for $x' \in X'$ and $t \in I$. All the results remain valid when suitably formulated for pairs.

5 H SPACES

In some cases it is possible to introduce a natural group structure in the set of homotopy classes of maps from one space (or pair) to another. In this section we consider spaces P such that $[X; P]$ admits a group structure for all X . It is not surprising that there is a close relation between natural group structures on $[X; P]$ for all X and "grouplike" structures on P .

We shall work in the homotopy category of pointed topological spaces, although much of what we do is also valid in the homotopy category of topological spaces. If X and Y are pointed topological spaces, $[X; Y]$ will denote the set of base-point-preserving homotopy classes of continuous maps $X \rightarrow Y$ (with all homotopies understood to be relative to the base point). Thus $[X; Y]$ is the set of morphisms from X to Y in the homotopy category of pointed topological spaces.

One method of obtaining a group structure on $[X; P]$ is to start with a group structure on P . Thus, let P be a topological group with identity element as base point. There is a law of composition in the set of all base-point-preserving continuous maps from X to P defined by pointwise multiplication of functions. That is, if $g_1, g_2: X \rightarrow P$, then $g_1 g_2: X \rightarrow P$ is defined by $g_1 g_2(x) = g_1(x) g_2(x)$, where the right-hand side is the group product in P . With this law of composition, the set of base-point-preserving continuous maps from X to P is a group (which is abelian if P is abelian). The law of composition carries over to give an operation on homotopy classes such that $[g_1][g_2] = [g_1 g_2]$, and we have the following theorem.

1 THEOREM *If P is a topological group, π^P is a contravariant functor from the homotopy category of pointed topological spaces to the category of groups and homomorphisms.* ■

We give two examples.

2 S^1 is an abelian topological group (the multiplicative group of complex numbers of norm 1). Therefore $[X; S^1]$ is an abelian group, and if $f: X \rightarrow Y$, then $f^\#: [Y; S^1] \rightarrow [X; S^1]$ is a homomorphism.

3 S^3 is a topological group (the multiplicative group of quaternions of norm 1). Therefore $[X; S^3]$ is a group, and if $f: X \rightarrow Y$, then $f^\#: [Y; S^3] \rightarrow [X; S^3]$ is a homomorphism.

This group structure on $[X; P]$ was deduced from a group structure on the set of base-point-preserving continuous maps from X to P . There are situations in which $[X; P]$ admits a natural group structure, but the set of base-point-preserving continuous maps from X to P has no group structure. For example, if P is a pointed space having the same homotopy type as some topological group P' , then π^P is naturally equivalent to $\pi^{P'}$. Therefore π^P can be regarded as a functor to the category of groups. The following definitions will be used to describe the additional structure needed on a pointed space P in order that π^P take values in the category of groups and homomorphisms.

If $f: X \rightarrow Y$ and $g: X \rightarrow Z$, we define

$$(f, g): X \rightarrow Y \times Z$$

to be the map $(f, g)(x) = (f(x), g(x))$ for $x \in X$.

An H space consists of a pointed topological space P together with a continuous multiplication

$$\mu: P \times P \rightarrow P$$

for which the (unique) constant map $c: P \rightarrow P$ is a *homotopy identity*, that is, each composite

$$P \xrightarrow{(c,1)} P \times P \xrightarrow{\mu} P \quad \text{and} \quad P \xrightarrow{(1,c)} P \times P \xrightarrow{\mu} P$$

is homotopic to 1_P . The multiplication μ is said to be *homotopy associative* if the square

$$P \times P \times P \xrightarrow{\mu \times 1} P \times P$$

$$\begin{array}{ccc} 1 \times \mu \downarrow & & \downarrow \mu \\ P \times P \times P & \xrightarrow{\mu \times 1} & P \times P \end{array}$$

is homotopy commutative, that is, $\mu \circ (\mu \times 1) \simeq \mu \circ (1 \times \mu)$. A continuous function $\varphi: P \rightarrow P$ is called a *homotopy inverse* for P and μ if each of the composites

$$P \xrightarrow{(1,\varphi)} P \times P \xrightarrow{\mu} P \quad \text{and} \quad P \xrightarrow{(\varphi,1)} P \times P \xrightarrow{\mu} P$$

is homotopic to $c: P \rightarrow P$.

A homotopy-associative H space with a homotopy inverse satisfies the group axioms up to homotopy. Such a pointed space is called an H group. Clearly, any topological group is an H group.

A multiplication μ in an H space is said to be *homotopy abelian* if the triangle

$$\begin{array}{ccc} P \times P & \xrightarrow{T} & P \times P \\ & \searrow \mu & \swarrow \mu \\ & P & \end{array}$$

where $T(p_1, p_2) = (p_2, p_1)$, is homotopy commutative. An H group with homotopy-abelian multiplication is called an *abelian H group*.

If P and P' are H spaces with multiplications μ and μ' , respectively, a continuous map $\alpha: P \rightarrow P'$ is called a *homomorphism* if the square

$$\begin{array}{ccc} P \times P & \xrightarrow{\mu} & P \\ \alpha \times \alpha \downarrow & & \downarrow \alpha \\ P' \times P' & \xrightarrow{\mu'} & P' \end{array}$$

is homotopy commutative.

4 THEOREM A pointed space having the same homotopy type as an H space (or an H group) is itself an H space (or an H group) in such a way that the homotopy equivalence is a homomorphism.

PROOF Let $f: P \rightarrow P'$ and $g: P' \rightarrow P$ be homotopy inverses and let P be an H space with multiplication $\mu: P \times P \rightarrow P$. If $\mu': P' \times P' \rightarrow P'$ is defined to be the composite

$$P' \times P' \xrightarrow{g \times g} P \times P \xrightarrow{\mu} P \xrightarrow{f} P'$$

then μ' is a continuous multiplication in P' and the composite $P' \xrightarrow{(1,c')} P' \times P' \xrightarrow{\mu'} P'$ equals the composite $P' \xrightarrow{g} P \xrightarrow{(1,c)} P \times P \xrightarrow{\mu} P \xrightarrow{f} P'$, which is homotopic to the composite $P' \xrightarrow{g} P \xrightarrow{f} P'$. Because $fg \simeq 1_P$, the map $\mu' \circ (1, c')$ is homotopic to $1_{P'}$. Similarly, the map $\mu' \circ (c', 1)$ is homotopic to $1_{P'}$. Therefore P' is an H space. Because the square

$$\begin{array}{ccc} P' \times P' & \xrightarrow{\mu'} & P' \\ g \times g \downarrow & & \downarrow g \\ P \times P & \xrightarrow{\mu} & P \end{array}$$

is homotopy commutative, g is a homomorphism (and so is f). If μ is homotopy associative or homotopy abelian, so is μ' , and if $\varphi: P \rightarrow P$ is a homotopy inverse for P , then $f\varphi g: P' \rightarrow P'$ is a homotopy inverse for P' . ■

Given an H space P , for any pointed space X there is a law of composition in $[X; P]$ defined by $[g_1][g_2] = [\mu \circ (g_1, g_2)]$. If P is an H group, $[X; P]$ becomes a group with this law of composition, and if $f: X \rightarrow Y$, then $f\# : [Y; P] \rightarrow [X; P]$ is a homomorphism. Therefore we have the following theorem.

5 THEOREM If P is an H group, π^P is a contravariant functor from the homotopy category of pointed topological spaces with values in the category of groups and homomorphisms. If P is an abelian H group, this functor takes values in the category of abelian groups. ■

It is interesting that the following converse of theorem 5 is also valid.

6 THEOREM If P is a pointed space such that π^P takes values in the category of groups, then P is an H group (abelian if π^P takes values in the category of abelian groups). Furthermore, for any pointed space X , the group structure on $\pi^P(X)$ is the same as that given by theorem 5.

PROOF Let $p_1: P \times P \rightarrow P$ and $p_2: P \times P \rightarrow P$ be the projections, and let $\mu: P \times P \rightarrow P$ be a map such that $[\mu] = [p_1] * [p_2]$, where $*$ is the law of composition in the group $[P \times P; P]$. For any maps $f, g: X \rightarrow P$, $(f, g)\# : [P \times P; P] \rightarrow [X; P]$ is a homomorphism and

$$\begin{aligned} [\mu \circ (f, g)] &= (f, g)\#[\mu] = (f, g)\#([p_1] * [p_2]) \\ &= (f, g)\#[p_1] * (f, g)\#[p_2] = [f] * [g] \end{aligned}$$

This shows that the multiplication in $[X; P]$ is induced by the multiplication map μ .

Let X be a one-point space. The unique map $X \rightarrow P$ represents the identity element of the group $[X; P]$. Because the unique map $P \rightarrow X$ induces

a homomorphism $[X; P] \rightarrow [P; P]$, it follows that the composite $P \rightarrow X \rightarrow P$, which is the constant map $c: P \rightarrow P$, represents the identity element of $[P; P]$. It follows that $\mu^\circ(1_P, c) \simeq 1_P$ and $\mu^\circ(c, 1_P) \simeq 1_P$. Therefore P is an H space.

To prove that μ is homotopy associative, let $q_1, q_2, q_3: P \times P \times P \rightarrow P$ be the projections. Then

$$\begin{aligned} [\mu^\circ(1 \times \mu)] &= (1 \times \mu)^\#[\mu] = (1 \times \mu)^\#[p_1] * (1 \times \mu)^\#[p_2] \\ &= [q_1] * [\mu(q_2, q_3)] = [q_1] * ([q_2] * [q_3]) \end{aligned}$$

Similarly,

$$[\mu^\circ(\mu \times 1)] = ([q_1] * [q_2]) * [q_3]$$

Because $[P \times P \times P; P]$ has an associative multiplication, $\mu^\circ(1 \times \mu) \simeq \mu^\circ(\mu \times 1)$.

To show that P has a homotopy inverse, let $q: P \rightarrow P$ be such that $[1_P] * [q] = [c]$; then $\mu(1_P, q) \simeq c$. Also, $[q] * [1_P] = [c]$, and so $\mu(q, 1_P) \simeq c$. Therefore q is a homotopy inverse for P .

This proves that P is an H group and that the multiplication in π^P is induced from that on P . If $[P \times P; P]$ is an abelian group, a similar argument shows that P is an abelian H group. ■

The following complement to theorems 5 and 6 is easily established by similar methods.

7 THEOREM Let $\alpha: P \rightarrow P'$ be a map between H groups. Then $\alpha_\#$ is a natural transformation from π^P to $\pi^{P'}$ in the category of groups if and only if α is a homomorphism. ■

We describe a particularly useful example of an H group. Let Y be a pointed topological space with base point y_0 . The loop space of Y (based at y_0), denoted by ΩY [or by $\Omega(Y, y_0)$], is defined to be the space of continuous functions $\omega: (I, \dot{I}) \rightarrow (Y, y_0)$ topologized by the compact-open topology. ΩY is regarded as a pointed space with base point ω_0 equal to the constant map of I to y_0 . There is a map

$$\mu: \Omega Y \times \Omega Y \rightarrow \Omega Y$$

defined by

$$\mu(\omega, \omega')(t) = \begin{cases} \omega(2t) & 0 \leq t \leq \frac{1}{2} \\ \omega'(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

To prove that μ is continuous, let $E: \Omega Y \times I \rightarrow Y$ be the evaluation map. By theorem 2.8 in the Introduction, it suffices to show that the composite

$$\Omega Y \times \Omega Y \times I \xrightarrow{\mu \times I} \Omega Y \times I \xrightarrow{E} Y$$

is continuous. The formula which defines μ shows that this composite is continuous on each of the closed sets $\Omega Y \times \Omega Y \times [0, \frac{1}{2}]$ and $\Omega Y \times \Omega Y \times [\frac{1}{2}, 1]$.

We construct a number of homotopies to show that ΩY is an H group.

Similar formulas will be used again in Sec. 1.7 to define homotopies of (non-closed) paths in a topological space.

To prove that the map $\omega \rightarrow \mu(\omega, \omega_0)$ is homotopic to the identity map of ΩY , define $F: \Omega Y \times I \rightarrow \Omega Y$ by

$$F(\omega, t)(t') = \begin{cases} \omega\left(\frac{2t'}{t+1}\right) & 0 \leq t' \leq \frac{t+1}{2} \\ y_0 & \frac{t+1}{2} \leq t' \leq 1 \end{cases}$$

This formula shows that $E(F \times 1): (\Omega Y \times I) \times I \rightarrow Y$ is continuous; therefore F is continuous and is a homotopy from the map $\omega \rightarrow \mu(\omega, \omega_0)$ to $1_{\Omega Y}$. Similarly, the map $\omega \rightarrow \mu(\omega_0, \omega)$ is homotopic to $1_{\Omega Y}$. Therefore ΩY is an H space with multiplication μ .

To show that μ is homotopy associative, define

$$G: \Omega Y \times \Omega Y \times \Omega Y \times I \rightarrow \Omega Y$$

by the formula

$$E(G \times 1)(\omega, \omega', \omega'', t, t') = \begin{cases} \omega\left(\frac{4t'}{t+1}\right) & 0 \leq t' \leq \frac{t+1}{4} \\ \omega'(4t' - t - 1) & \frac{t+1}{4} \leq t' \leq \frac{t+2}{4} \\ \omega''\left(\frac{4t' - 2 - t}{2 - t}\right) & \frac{t+2}{4} \leq t' \leq 1 \end{cases}$$

Then $G: \mu^\circ(\mu \times 1_{\Omega Y}) \simeq \mu^\circ(1_{\Omega Y} \times \mu)$, showing that μ is homotopy associative.

We define a homotopy inverse $\varphi: \Omega Y \rightarrow \Omega Y$ by $\varphi(\omega)(t) = \omega(1 - t)$. Then we define $H: \Omega Y \times I \rightarrow \Omega Y$ by

$$E(H \times 1)(\omega, t, t') = \begin{cases} y_0 & 0 \leq t' \leq \frac{t}{2} \\ \omega(2t' - t) & \frac{t}{2} \leq t' \leq \frac{1}{2} \\ \omega(2 - 2t' - t) & \frac{1}{2} \leq t' \leq 1 - \frac{t}{2} \\ y_0 & 1 - \frac{t}{2} \leq t' \leq 1 \end{cases}$$

H is a homotopy from the map $\omega \rightarrow \mu(\omega, \varphi(\omega))$ to the constant map of ΩY to itself. Similarly, there is a homotopy from the map $\omega \rightarrow \mu(\varphi(\omega), \omega)$ to the constant map of ΩY . Therefore φ is a homotopy inverse for ΩY , and ΩY is an H group.

If $h: Y \rightarrow Y'$ preserves base points, there is a map

$$\Omega h: \Omega Y \rightarrow \Omega Y'$$

defined by $\Omega h(\omega)(t) = h(\omega(t))$. Clearly, Ωh is a homomorphism, and we summarize these remarks about loop spaces as follows.

8 THEOREM *The loop functor Ω is a covariant functor from the category of pointed topological spaces and continuous maps to the category of H groups and continuous homomorphisms. ■*

The functor Ω also preserves homotopies. That is, if $h_0, h_1: Y \rightarrow Y'$ are homotopic by a homotopy h_t , then $\Omega h_0, \Omega h_1: \Omega Y \rightarrow \Omega Y'$ are homotopic by a homotopy Ωh_t , which is a continuous homomorphism for each $t \in I$.

6 SUSPENSION

This section deals primarily with results dual to those of Sec. 1.5. We consider pointed spaces Q such that π_Q is a covariant functor from the homotopy category of pointed spaces to the category of groups and homomorphisms, and this leads to the concept of H cogroup, dual to that of H group. An important example of an H cogroup is the suspension of a pointed space, a concept dual to that of the loop space. The homotopy groups of a space defined in the section are examples of groups of homotopy classes of maps from suspensions to the space.

If X and Y are pointed topological spaces, their sum in the category of pointed topological spaces will be denoted by $X \vee Y$. If X has base point x_0 and Y has base point y_0 , $X \vee Y$ may be regarded as the subspace $X \times Y_0 \cup X_0 \times Y$ of $X \times Y$. If $f: X \rightarrow Z$ and $g: Y \rightarrow Z$, we let $(f, g): X \vee Y \rightarrow Z$ be the map defined by the characteristic property of the sum [that is, $(f, g)|X = f$ and $(f, g)|Y = g$].

An H cogroup consists of a pointed topological space Q together with a continuous comultiplication

$$v: Q \rightarrow Q \vee Q$$

such that the following properties hold:

Existence of homotopy identity. If $c: Q \rightarrow Q$ is the (unique) constant map, each composite

$$Q \xrightarrow{v} Q \vee Q \xrightarrow{(c, 1)} Q \quad \text{and} \quad Q \xrightarrow{v} Q \vee Q \xrightarrow{(1, c)} Q$$

is homotopic to 1_Q .

Homotopy associativity. The square

$$\begin{array}{ccc} Q & \xrightarrow{v} & Q \vee Q \\ v \downarrow & & \downarrow 1 \vee v \\ Q \vee Q & \xrightarrow{v \vee 1} & Q \vee Q \vee Q \end{array}$$

is homotopy commutative.

Existence of homotopy inverse. There exists a map $\psi: Q \rightarrow Q$ such that each composite

$$Q \xrightarrow{v} Q \vee Q \xrightarrow{(1, \psi)} Q \quad \text{and} \quad Q \xrightarrow{v} Q \vee Q \xrightarrow{(\psi, 1)} Q$$

is homotopic to $c: Q \rightarrow Q$.

If X is any pointed space and Q is an H cogroup, there is a law of composition in $[Q; X]$ defined by $[(f_1, f_2) \circ v]$ which makes $[Q; X]$ a group.

An H cogroup is said to be *abelian* if the triangle

$$\begin{array}{ccc} & Q & \\ v' \swarrow & & \searrow v \\ Q \vee Q & \xrightarrow{\tau} & Q \vee Q \end{array}$$

where $\tau(q_1, q_2) = (q_2, q_1)$ for $q_i \in Q$, is homotopy commutative.

If Q and Q' are H cogroups with comultiplications v and v' , respectively, a continuous map $\beta: Q \rightarrow Q'$ is called a *homomorphism* if the square

$$\begin{array}{ccc} Q & \xrightarrow{v} & Q \vee Q \\ \beta \downarrow & & \downarrow \beta \vee \beta \\ Q' & \xrightarrow{v'} & Q' \vee Q' \end{array}$$

is homotopy commutative.

The proofs of the following theorems are dual to the proofs of the corresponding statements about H groups (see theorems 1.5.4, 1.5.5, 1.5.6, and 1.5.7) and are omitted.

1 THEOREM *A pointed space having the same homotopy type as an H cogroup is itself an H cogroup in such a way that the homotopy equivalence is a homomorphism. ■*

2 THEOREM *If Q is an H cogroup, π_Q is a covariant functor from the homotopy category of pointed spaces with values in the category of groups and homomorphisms. If Q is an abelian H cogroup, this functor takes values in the category of abelian groups. ■*

3 THEOREM *If Q is a pointed space such that π_Q takes values in the category of groups, then Q is an H cogroup (abelian if π_Q takes values in the category of abelian groups). Furthermore, the group structure on $\pi_Q(X)$ is identical with that determined by the H cogroup structure of Q as in theorem 2. ■*

4 THEOREM *If $\beta: Q \rightarrow Q'$ is a map between H cogroups, then $\beta\#$ is a natural transformation from π_Q to $\pi_{Q'}$ in the category of groups if and only if β is a homomorphism. ■*

We describe an example of an H cogroup dual to the loop-space example of an H group. Let Z be a pointed topological space with base point z_0 . The

suspension of Z , denoted by SZ , is defined to be the quotient space of $Z \times I$ in which $(Z \times 0) \cup (z_0 \times I) \cup (Z \times 1)$ has been identified to a single point. This is sometimes called the *reduced suspension* in the literature, the term “suspension” being used for the suspension in the category of spaces (no base points). The latter is defined to be the quotient space of $Z \times I$ in which $Z \times 0$ is identified to one point and $Z \times 1$ is identified to another point.

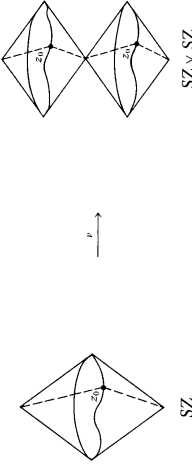
If $(z, t) \in Z \times I$, we use $[z, t]$ to denote the corresponding point of SZ under the quotient map $Z \times I \rightarrow SZ$. Then $[z, 0] = [z_0, t] = [z', 1]$ for all $z, z' \in Z$ and $t \in I$. The point $[z_0, 0] \in SZ$ is also denoted by z_0 , and SZ is a pointed space with base point z_0 . If $f: Z \rightarrow Z'$, then $Sf: SZ \rightarrow SZ'$ is defined by $Sf[z, t] = [f(z), t]$. Thus S is a covariant functor from the category of pointed spaces and continuous maps. To show that it is a covariant functor to the category of H cogroups and homomorphisms, we define a comultiplication

$$\nu: SZ \rightarrow SZ \vee SZ$$

by the formula

$$\nu([z, t]) = \begin{cases} ([z, 2t], z_0) & 0 \leq t \leq \frac{1}{2} \\ (z_0, [z, 2t - 1]) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

and illustrate it in the diagram (where the dotted lines are collapsed to one point).



The map ν provides SZ with the structure of an H cogroup such that if $f: Z \rightarrow Z'$, then Sf is a homomorphism. This can be verified directly or deduced from properties of loop spaces already established. We follow the latter course.

The functors Ω and S defined from the category of pointed spaces and continuous maps to itself are examples of *adjoint functors*. This means that for pointed spaces Z and Y there is an equivalence

$$\text{hom}(SZ, Y) \approx \text{hom}(Z, \Omega Y)$$

where both sides are interpreted as the set of morphisms in the category of pointed spaces and continuous maps. This equivalence results from theorem 2.8 in the Introduction, and if $g: Z \rightarrow \Omega Y$, the corresponding $g': SZ \rightarrow Y$ is defined by $g'[z, t] = g(z)(t)$ for $z \in Z$ and $t \in I$. It is obvious that if $h: Y \rightarrow Y'$, then $(\Omega h \circ g')' = h \circ g'$, and if $f: Z' \rightarrow Z$, then $(g \circ f)' = g' \circ Sf$. Therefore the equivalence $g \leftrightarrow g'$ comes from a natural equivalence from the functor $\text{hom}(S \cdot, \cdot)$ to the functor $\text{hom}(\cdot, \Omega \cdot)$.

This natural equivalence passes to morphisms in the homotopy category of pointed spaces. For pointed spaces a homotopy $G: Z \times I \rightarrow Y$ must map $z_0 \times I$ into y_0 . Therefore it defines a map $F: Z \times I/z_0 \times I \rightarrow Y$. Because $S(Z \times I/z_0 \times I)$ can be identified with $SZ \times I/z_0 \times I$ by the homeomorphism

$$[(z, t), t'] \leftrightarrow ([z, t'], t) \quad z \in Z; t, t' \in I$$

it follows that homotopies $F: Z \times I/z_0 \times I \rightarrow \Omega Y$ correspond bijectively to homotopies $F': SZ \times I/z_0 \times I \rightarrow Y$. Therefore the equivalence above gives rise to an equivalence

$$[SZ, Y] \approx [Z, \Omega Y]$$

such that if the maps $g: Z \rightarrow \Omega Y$ and $g': SZ \rightarrow Y$ are related by $g'[z, t] = g(z)(t)$, then $[g']$ corresponds to $[g]$. Hence there is a natural equivalence from the functor $[S \cdot; \cdot]$ to the functor $[\cdot; \Omega \cdot]$.

It follows from these remarks that for a fixed pointed space Z the functor π_{SZ} is naturally equivalent to the composite functor $\pi_Z \circ \Omega$. Here Ω is regarded as a covariant functor to the homotopy category of H groups and homomorphisms. Then the composite $\pi_Z \circ \Omega$ takes values in the category of groups and homomorphisms. By theorem 3, SZ is an H cogroup, and the map $\nu: SZ \rightarrow SZ \vee SZ$ defined above is the one which is the comultiplication in the H cogroup SZ (or is homotopic to it). In similar fashion, if $f: Z \rightarrow Z'$, the natural transformation $f^\#$ from π_{SZ} to π_{SZ} corresponds to the natural transformation $f^\#$ from the composite $\pi_{Z'} \circ \Omega$ to the composite $\pi_Z \circ \Omega$. Because the latter is a natural transformation in the category of groups, so is $(Sf)^\#$, and by theorem 4, Sf is a homomorphism of the H cogroup SZ to the H cogroup SZ' .

These statements are summarized as follows.

5 THEOREM *The suspension functor S is a covariant functor from the category of pointed spaces and maps to the category of H cogroups and continuous homomorphisms. ■*

The functor S also preserves homotopies. That is, if $f_0, f_1: Z \rightarrow Z'$ are homotopic by a homotopy f_t , then Sf_0, Sf_1 are homotopic by a homotopy Sf_t , which is a continuous homomorphism for each $t \in I$.

We now show that for $n \geq 1$ the sphere S^n is homeomorphic to a suspension, and thus obtain an interesting family of H cogroups. The corresponding functors are known as the homotopy group functors and are particularly important.

6 LEMMA *For $n \geq 0$, $S(S^n)$ is homeomorphic to S^{n+1} .*

PROOF Let $p_0 = (1, 0, \dots, 0)$ be the base point of S^n . We regard $\mathbf{R}^{n+1}$ as imbedded in $\mathbf{R}^{n+2}$ as the set of points in $\mathbf{R}^{n+2}$ whose $(n+2)$ nd coordinate is 0. Then S^n is imbedded as an equator in S^{n+1} .

$$S^n = \{z \in \mathbf{R}^{n+2} \mid \|z\| = 1 \text{ and } z_{n+2} = 0\}$$

and E^{n+1} is also imbedded in E^{n+2} ;

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$$E^{n+1} = \{z \in \mathbf{R}^{n+2} \mid \|z\| \leq 1 \text{ and } z_{n+2} = 0\}$$

Let H_+ and H_- be the two closed hemispheres of S^{n+1} defined by the equation S^n . Then

$$H_+ = \{z \in S^{n+1} \mid z_{n+2} \geq 0\} \text{ and } H_- = \{z \in S^{n+1} \mid z_{n+2} \leq 0\}$$

and $S^{n+1} = H_+ \cup H_-$ and $S^n = H_+ \cap H_-$. Furthermore, the projection map $\mathbf{R}^{n+2} \rightarrow \mathbf{R}^{n+1}$ defines projection maps $p_+ : H_+ \rightarrow E^{n+1}$ and $p_- : H_- \rightarrow E^{n+1}$, which are homeomorphisms. A map $f : S(S^n) \rightarrow S^{n+1}$ is defined by

$$f[z, t] = \begin{cases} p_+^{-1}(2tz + (1 - 2t)p_0) & 0 \leq t \leq \frac{1}{2} \\ p_-^{-1}((2 - 2t)z + (2t - 1)p_0) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

and is verified to be a homeomorphism $f : S(S^n) \approx S^{n+1}$. ■

For $n \geq 1$ the n th homotopy group functor π_n is the covariant functor on the homotopy category of pointed spaces defined by $\pi_n = \pi_{S^n}$. It follows from theorems 6 and 5 that these functors take values in the category of groups and homomorphisms.

In the last two sections of this chapter we give another definition of π_1 and study it in more detail. In Chapter 7 we return to the study of the higher homotopy groups π_n .

The following necessary and sufficient condition for a map $S^n \rightarrow X$ to represent the trivial element of $\pi_n(X)$ is an immediate consequence of theorem 1.3.12.

7 THEOREM A map $\alpha : S^n \rightarrow X$ represents the trivial element of $\pi_n(X)$ for $n \geq 1$ if and only if α can be continuously extended over E^{n+1} . ■

Before leaving this section let us consider the interplay between two possible group structures on the set $[X; Y]$ for particular pointed spaces X and Y (for example, if X is an H cogroup and Y is an H group, this set can be given a group structure in two ways). It is a fact that under rather general circumstances two laws of composition on $\text{hom}(X, Y)$ in a category are equal, and we establish this result.

8 THEOREM Let X and Y be objects in a category and let $*$ and $\star$ be two laws of composition in $\text{hom}(X, Y)$ such that

- (a) $*$ and $\star$ have a common two-sided identity element
- (b) $*$ and $\star$ are mutually distributive

Then $*$ and $\star$ are equal, and each is commutative and associative.

PROOF Statement (a) means there is a map $f_0 : X \rightarrow Y$ such that for any $f : X \rightarrow Y$

$$f * f_0 = f_0 * f = f = f *' f_0 = f_0 *' f$$

Statement (b) means that for $f_1, f_2, g_1, g_2 : X \rightarrow Y$

$$(f_1 * f_2) *' (g_1 * g_2) = (f_1 *' g_1) * (f_2 *' g_2)$$

If $f, g : X \rightarrow Y$, then

$$f * g = (f *' f_0) * (f_0 *' g) = (f *' f_0) *' (f_0 *' g) = f *' g$$

$$\text{and } g * f = (f_0 *' g) * (f *' f_0) = (f_0 *' f) *' (g *' f_0) = f *' g$$

Therefore $f * g = f *' g = g * f$. For associativity we have

$$(f * g) * h = (f *' g) *' (f_0 *' h) = (f *' f_0) *' (g *' h) = f *' (g * h) \quad \blacksquare$$

9 COROLLARY If P is an H space and Q is any H cogroup, then $[Q; P]$ is an abelian group and the group structure is defined by the multiplication map in P .

PROOF This follows on observing that the two laws of composition defined in $[Q; P]$ by using the comultiplication in Q or the multiplication in P satisfy the hypotheses of theorem 8. ■

Note that if P is just an H space (but not an H group), the law of composition in $[X; P]$ defined by the multiplication in P is in general not a group structure on $[X; P]$. However, if X is an H cogroup (for instance, a suspension), it follows from corollary 9 that this law of composition is a group structure on $[X; P]$, and in this case the resulting group structure on $[X; P]$ is the same no matter what multiplication map P is given (so long as it is an H space).

10 COROLLARY If P is an H space, $\pi_n(P)$ is abelian for all $n \geq 1$ and the group structure in $\pi_n(P)$ is defined by the multiplication map in P . ■

For a double suspension $S(SZ)$ whose points are represented in the form $[[z, t], t']$, with $z \in Z$ and $t, t' \in I$, there are two laws of composition in the set of maps $S(SZ) \rightarrow X$. If $f, g : S(SZ) \rightarrow X$, we define

$$(f * g)[[z, t], t'] = \begin{cases} f[[z, 2t], t'] & 0 \leq t \leq \frac{1}{2} \\ g[[z, 2t - 1], t'] & \frac{1}{2} \leq t \leq 1 \end{cases}$$

and

$$(f \star g)[[z, t], t'] = \begin{cases} f[[z, t], 2t'] & 0 \leq t' \leq \frac{1}{2} \\ g[[z, t], 2t' - 1] & \frac{1}{2} \leq t' \leq 1 \end{cases}$$

The corresponding operations in $[S(SZ); X]$ satisfy the hypotheses of theorem 8. Therefore they are equal, and $[S(SZ); X]$ is an abelian group. In particular, we have the following corollary.

11 COROLLARY For $n \geq 2$, π_n is a functor to the category of abelian groups. ■

A similar argument can be applied to the loop space ΩP , where P is itself an H space. There is a multiplication map in ΩP , because it is a loop space, and another multiplication obtained from the original multiplication in P . The corresponding laws of composition in $[X; \Omega P]$ satisfy theorem 8. Therefore it follows that if P is an H space, $\pi_n \Omega P$ is a contravariant functor to the category of abelian groups.

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7 THE FUNDAMENTAL GROUPOID

This section concerns paths in a topological space. This leads to another description (in Sec. 1.8) of the first homotopy group π_1 , introduced in Sec. 1.6. We shall have occasion to define a number of homotopies between paths in a topological space. These homotopies are generalizations (to nonclosed paths) of those used in Sec. 1.5 to prove that a loop space is an H group and are defined by the same formulas (except that the t and t' arguments are interchanged). It is clear that this repetition of formulas could have been eliminated by proving a suitably general result about path spaces instead of merely considering loop spaces in Sec. 1.5. However, each usage has its own value, and it is hoped that the repetition may be an aid to understanding the formulas.

A *groupoid* is a small category in which every morphism is an equivalence. We list without proof a number of facts about groupoids which are easy consequences of general properties of categories.

1 The relation between objects A and B of a groupoid defined by the condition $\text{hom}(A, B) \neq \emptyset$ is an equivalence relation. ■

The equivalence classes of this equivalence relation are called the *components* of the groupoid. The groupoid is said to be *connected* if it has just one component.

2 For any object A of a groupoid, the law of composition which sends $f, g: A \rightarrow A$ to $f \circ g: A \rightarrow A$ is a group operation in $\text{hom}(A, A)$. ■

3 There is a covariant functor from any groupoid to the category of groups and homomorphisms which assigns to an object A the group $\text{hom}(A, A)$ and to a morphism $f: A \rightarrow B$ the homomorphism

$$h_f: \text{hom}(A, A) \rightarrow \text{hom}(B, B)$$

defined by $h_f(g) = f \circ g \circ f^{-1}$ for $g: A \rightarrow A$. ■

Because any morphism $f: A \rightarrow B$ in a groupoid is an equivalence, $h_f: \text{hom}(A, A) \rightarrow \text{hom}(B, B)$ is an isomorphism. The following statement describes the collection of isomorphisms obtained by taking all possible morphisms $f: A \rightarrow B$.

4 If A and B are in the same component of a groupoid, the collection of isomorphisms $\{h_f | f: A \rightarrow B\}$ is a conjugacy class of isomorphisms $\text{hom}(A, A) \rightarrow \text{hom}(B, B)$. ■

5 Let F be a covariant functor from one groupoid $\mathcal{C}$ to another $\mathcal{C}'$. Then F maps each component of $\mathcal{C}$ into some component of $\mathcal{C}'$, and there is a natural transformation $F_*(A)$ from the covariant functor $\text{hom}_{\mathcal{C}}(A, A)$ on $\mathcal{C}$ to the covariant functor $\text{hom}_{\mathcal{C}'}(F(A), F(A))$ on $\mathcal{C}'$ defined by

$$F_*(A)(f) = F(f): F(A) \rightarrow F(A) \quad f: A \rightarrow A \quad \blacksquare$$

With these general remarks about groupoids out of the way, we proceed to define the fundamental groupoid. A *path* ω in a topological space is defined to be a continuous map $\omega: I \rightarrow X$ [note that the path is the map, not just the image set $\omega(I)$]. The *origin* of the path is the point $\omega(0)$, and the *end* of the path is the point $\omega(1)$. We also say that ω is a *path from* $\omega(0)$ *to* $\omega(1)$. A *closed path*, or *loop*, at $x_0 \in X$ is a path ω such that $\omega(0) = x_0 = \omega(1)$. If ω and ω' are paths in X such that end $\omega = \text{orig } \omega'$, there is a *product path* $\omega * \omega'$ in X defined by the formula

$$(\omega * \omega')(t) = \begin{cases} \omega(2t) & 0 \leq t \leq \frac{1}{2} \\ \omega'(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

Then $\text{orig}(\omega * \omega') = \text{orig } \omega$ and $\text{end}(\omega * \omega') = \text{end } \omega'$.

We should like to form a category whose objects are the points of X , whose morphisms from x_1 to x_0 are the paths from x_0 to x_1 , and with the composite defined to be the product path. With these definitions, neither axiom of a category is satisfied. That is, there need not be an identity morphism for each point, and it is generally not true that the associative law for product paths holds [that is, $\omega * (\omega' * \omega'')$ is usually different from $(\omega * \omega') * \omega''$]. A category can be obtained, however, if the morphisms are defined not to be the paths themselves, but instead, homotopy classes of paths.

Two paths ω and ω' in X are briefly said to be *homotopic*, denoted by $\omega \simeq \omega'$, if they are homotopic relative to I . Thus a necessary condition that $\omega \simeq \omega'$ is that $\omega(0) = \omega'(0)$ and $\omega(1) = \omega'(1)$. For any $x_0, x_1 \in X$ the relation $\omega \simeq \omega'$ is an equivalence relation in the set of paths from x_0 to x_1 . The resulting equivalence classes are called *path classes*, and if ω is a path in X , the path class containing it is denoted by $[\omega]$. Since two paths in the same path class have the same origin and the same end, we can speak of the origin and the end of a path class.

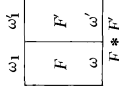
We shall construct a category whose objects are the points of X and whose morphisms from x_1 to x_0 are the path classes with x_0 as origin and x_1 as end. The following lemma shows that the path class of the product of two paths depends only on the path classes of the factors, and it will be used to define the composite in the category.

6 **LEMMA** Let $[\omega]$ and $[\omega']$ be path classes in X with end $[\omega] = \text{orig } [\omega']$. There is a well-defined path class $[\omega] * [\omega'] = [\omega * \omega']$ with $\text{orig } ([\omega] * [\omega']) = \text{orig } [\omega]$ and $\text{end } ([\omega] * [\omega']) = \text{end } [\omega']$.

PROOF To prove that $\omega \simeq \omega_1$ and $\omega' \simeq \omega'_1$ imply $\omega * \omega' \simeq \omega_1 * \omega'_1$, let $F: I \times I \rightarrow X$ be a homotopy relative to I from ω to ω_1 and let $F': I \times I \rightarrow X$ be a homotopy relative to I from ω' to ω'_1 . A homotopy $F * F': I \times I \rightarrow X$ is defined by the formula

$$(F * F')(t, t') = \begin{cases} F(2t, t') & 0 \leq t \leq \frac{1}{2} \\ F'(2t - 1, t') & \frac{1}{2} \leq t \leq 1 \end{cases}$$

and illustrated in the diagram



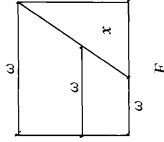
Then $F * F': \omega * \omega' \simeq \omega_1 * \omega'_1 \text{ rel } \dot{I}$. ■

7 THEOREM For each topological space X there is a category $\mathfrak{P}(X)$ whose objects are the points of X , whose morphisms from x_1 to x_0 are the path classes with x_0 as origin and x_1 as end, and whose composite is the product of path classes.

PROOF To prove the existence of identity morphisms, let $\varepsilon_x: I \rightarrow X$ be the constant map of I to x for any $x \in X$. We show that $[\varepsilon_x] = 1_x$. If ω is a path with $\omega(1) = x$, we must prove that $\omega * \varepsilon_x \simeq \omega$ (with a similar property for paths with origin at x). Such a homotopy $F: I \times I \rightarrow X$ is defined by the formula

$$F(t, t') = \begin{cases} \omega\left(\frac{2t}{t' + 1}\right) & 0 \leq t \leq \frac{t' + 1}{2} \\ x & \frac{t' + 1}{2} \leq t \leq 1 \end{cases}$$

and pictured in the diagram

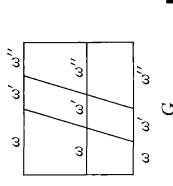


A similar homotopy shows that if $\omega(0) = x$, then $\varepsilon_x * \omega \simeq \omega$.

To prove the associativity of the composite of morphisms, let $\omega, \omega',$ and ω'' be paths such that end $\omega = \text{orig } \omega'$ and end $\omega' = \text{orig } \omega''$. We must prove that $(\omega * \omega') * \omega'' \simeq \omega * (\omega' * \omega'')$. Such a homotopy $G: I \times I \rightarrow X$ is defined by the formula

$$G(t, t') = \begin{cases} \omega\left(\frac{4t}{t' + 1}\right) & 0 \leq t \leq \frac{t' + 1}{4} \\ \omega'(4t - t' - 1) & \frac{t' + 1}{4} \leq t \leq \frac{t' + 2}{4} \\ \omega''\left(\frac{4t - 2 - t'}{2 - t'}\right) & \frac{t' + 2}{4} \leq t \leq 1 \end{cases}$$

and pictured in the diagram



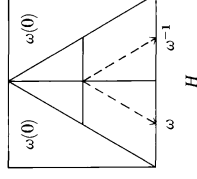
The category $\mathfrak{P}(X)$ is called the *category of path classes* of X , or the *fundamental groupoid* of X , the latter because of the following theorem.

8 THEOREM $\mathfrak{P}(X)$ is a groupoid.

PROOF Given a path ω in X , let $\omega^{-1}: I \rightarrow X$ be the path defined by $\omega^{-1}(t) = \omega(1 - t)$. To prove that $[\omega^{-1}] = [\omega]^{-1}$ in $\mathfrak{P}(X)$, we must show that $\omega * \omega^{-1} \simeq \varepsilon_{\omega(0)}$ [and also that $\omega^{-1} * \omega \simeq \varepsilon_{\omega(1)}$, which follows, however, from the first homotopy, because $\omega = (\omega^{-1})^{-1}$]. Such a homotopy $H: I \times I \rightarrow X$ is defined by the formula

$$H(t, t') = \begin{cases} \omega(0) & 0 \leq t \leq \frac{t'}{2} \\ \omega(2t - t') & \frac{t'}{2} \leq t \leq \frac{1}{2} \\ \omega(2 - 2t - t') & \frac{1}{2} \leq t \leq 1 - \frac{t'}{2} \\ \omega(0) & 1 - \frac{t'}{2} \leq t \leq 1 \end{cases}$$

and pictured in the diagram



This completes the construction of the fundamental groupoid. The components of the fundamental groupoid are called *path components* of X . It is clear that x_0 and x_1 are in the same path component of X if and only if there is a path ω in X from x_0 to x_1 . X is said to be *path connected* if its fundamental groupoid is connected. The following is an alternate characterization of the path components.

9 THEOREM The path components of X are the maximal path-connected subspaces of X .

PROOF Let A be a path component of X and let ω be a path in X such that $\omega(0) \in A$. We show that ω is a path in A . For each $t \in I$ define a path $\omega_t: I \rightarrow X$ by $\omega_t(t') = \omega(t't)$ for $t' \in I$. Then ω_t is a path in X from $\omega(0)$ to $\omega(t)$. Therefore $\omega(t)$ is in the same path component of X as x_0 , namely A . Since this is so for every $t \in I$, ω is a path in A .

A is path connected because if $x_0, x_1 \in A$ there is a path ω in X from x_0 to x_1 . By the above result, ω is a path in A . Therefore any two points of A can be joined by a path in A , and A is path connected. Since any path in X that starts in A stays in A , A is a maximal path-connected subset of X . ■

1.0 LEMMA *A path-connected space is connected.*

PROOF If ω is a path in X , then $\omega(I)$, being a continuous image of the connected space I , is connected. Therefore $\omega(0)$ and $\omega(1)$ lie in the same component of X . If X is path connected, any two points of X lie in the same component, and X is connected. ■

The converse of lemma 1.0 is false, as is shown by the following example.

1.1 EXAMPLE Let X be the subspace of $\mathbb{R}^2$ defined by

$$X = \{(x, y) \in \mathbb{R}^2 \mid x > 0, y = \sin \frac{1}{x} \text{ or } x = 0, -1 \leq y \leq 1\}$$

Then X is connected, but not path connected.

Given a map $f: X \rightarrow Y$, there is a covariant functor $f_\#$ from $\mathcal{Q}(X)$ to $\mathcal{Q}(Y)$ which sends an object x of $\mathcal{Q}(X)$ to the object $f(x)$ of $\mathcal{Q}(Y)$ and the morphism $[\omega]$ of $\mathcal{Q}(X)$ to the morphism $f_\#[\omega] = [f \circ \omega]$ of $\mathcal{Q}(Y)$. The functorial properties of $f_\#$ are easily verified. From the first part of statement 5, or by direct verification, it follows that f maps each path component of X into some path component of Y . Therefore there is a covariant functor π_0 from the category of topological spaces and maps to the category of sets and functions such that $\pi_0(X)$ equals the set of path components of X , and

$$\pi_0(f) = f_\#: \pi_0(X) \rightarrow \pi_0(Y)$$

maps the path component of x in X to the path component of $f(x)$ in Y . If $F: J_0 \simeq f_1$, then for any $x \in X$ there is a path ω_x in Y from $f_0(x)$ to $f_1(x)$ defined by $\omega_x(t) = F(x, t)$ for $t \in I$. Therefore $f_0(x)$ and $f_1(x)$ belong to the same path component of Y , and $f_{0\#} = f_{1\#}$. It follows that π_0 can be regarded as a covariant functor from the homotopy category to the category of sets and functions. This functor characterizes the functor π_X for a contractible space X as follows.

1.2 THEOREM *If X is a contractible space, then π_X and π_0 are naturally equivalent functors on the homotopy category.*

PROOF If X and X' have the same homotopy type, then π_X and $\pi_{X'}$ are naturally equivalent. It follows from corollary 1.3.11 that if P is a one-point space, π_X is naturally equivalent to π_P . It therefore suffices to prove that π_P

is naturally equivalent to π_0 . $\pi_0(P)$ consists of the single path component P , and a natural transformation

$$\psi: \pi_P \rightarrow \pi_0$$

is defined by $\psi[f] = f_\#(P)$ for $[f] \in [P, X]$. Because X^P is in one-to-one correspondence with X in such a way that homotopies $P \times I \rightarrow X$ correspond to paths $I \rightarrow X$, it follows that ψ is a natural equivalence. ■

The functor π_0 is closely related to the functor H_0 of example 1.2.6. In fact, for spaces X whose components and path components coincide, H_0 is the composite of π_0 with the covariant functor which assigns to every set the abelian group generated by that set. In particular, π_0 could have been used to obtain the results of Sec. 1.2 that were obtained by using H_0 .

8 THE FUNDAMENTAL GROUP

By choosing a fixed point $x_0 \in X$ and considering the path classes in X with x_0 as origin and end, a group called the fundamental group is obtained. We show now that this group is naturally equivalent to the first homotopy group π_1 , defined in Sec. 1.6. The section closes with a calculation of the fundamental group of the circle.

Let X be a topological space and let $x_0 \in X$. The *fundamental group of X based at x_0* , denoted by $\pi(X, x_0)$, is defined to be the group of path classes with x_0 as origin and end. It follows from theorem 1.7.8 and statement 1.7.2 that this is a group, and if $f: (X, x_0) \rightarrow (Y, y_0)$, then $f_\#$ is a homomorphism from $\pi(X, x_0)$ to $\pi(Y, y_0)$. If, $f, f': (X, x_0) \rightarrow (Y, y_0)$ are homotopic, then

$$f_\# = f'_\#: \pi(X, x_0) \rightarrow \pi(Y, y_0).$$

Therefore, we have the following theorem.

1 THEOREM *There is a covariant functor from the homotopy category of pointed spaces to the category of groups which assigns to a pointed space its fundamental group and to a map f the homomorphism $f_\#$.* ■

We show that the fundamental group functor π is naturally equivalent to π_1 , defined in Sec. 1.6. Let $\lambda: I \rightarrow S(S^0)$ be defined by $\lambda(t) = [-1, t]$, where S^0 consists of the two points -1 and 1 is its basepoint. Then λ induces a bijection $\lambda^\#$ between the homotopy classes of maps $(S(S^0), 1) \rightarrow (X, x_0)$ and the path classes of closed paths in X at x_0 defined by

$$\lambda^\#[\xi] = [\xi\lambda] \quad g: (S(S^0), 1) \rightarrow (X, x_0)$$

From the definition of the law of composition in $[S(S^0), X]$ and in $\pi(X, x_0)$, $\lambda^\#$ is seen to be a group isomorphism. Given a map $f: (X, x_0) \rightarrow (Y, y_0)$, $\lambda^\#$ commutes with $f_\#$. By lemma 1.6.6, $S(S^0)$ is homeomorphic to S^1 .

2 THEOREM The map $\tilde{\pi}^{-1}$ is a natural equivalence of the first homotopy group functor π_1 with the fundamental group functor π . ■

It will sometimes be convenient to regard the elements of $\pi(X, x_0)$ as homotopy classes of maps $(S^1, p_0) \rightarrow (X, x_0)$, rather than as path classes.

Because any closed path at x_0 (and any homotopy between such paths) must lie in the path component A of X containing x_0 , it follows that $\pi(A, x_0) \approx \pi(X, x_0)$. Hence the fundamental group can give information only about the path component of X containing x_0 . From general groupoid considerations (see statements 1.7.3 and 1.7.4), if $[\omega]$ is a path class in X from x_0 to x_1 , then $h_{[\omega]}$ is an isomorphism from $\pi(X, x_1)$ to $\pi(X, x_0)$.

3 THEOREM The fundamental groups of a path-connected space based at different points are isomorphic by an isomorphism determined up to conjugacy. ■

Even though the fundamental groups based at different points of a path-connected space are isomorphic, we cannot identify them, because the isomorphism between them is not unique. If the fundamental group at some point (and hence all points) is abelian, the isomorphism is unique. In general, the fundamental group need not be abelian; however, the following consequence of theorem 2 and corollary 1.6.10 is a general result about the commutativity of fundamental groups.

4 THEOREM The fundamental group of a path-connected H space is abelian, and if ω and ω' are closed paths at the base point, then

$$[\omega] * [\omega'] = [\mu \circ (\omega, \omega')]$$

where μ is the multiplication map in the H space. ■

A space X is said to be n -connected for $n \geq 0$ if every continuous map $f: S^k \rightarrow X$ for $k \leq n$ has a continuous extension over E^{k+1} . A 1-connected space is also said to be *simply connected*. Note that if $0 \leq m \leq n$, an n -connected space is m -connected. It follows from theorem 1.6.7 that a space X is n -connected if and only if it is path connected and $\pi_k(X, x)$ is trivial for every base point $x \in X$ and $1 \leq k \leq n$. From corollary 1.3.13 we have the following result.

5 LEMMA A contractible space is n -connected for every $n \geq 0$. ■

Note that a space is 0-connected if and only if it is path connected, and a space is simply connected if and only if it is path connected, and $\pi(X, x_0) = 0$ for some (and hence all) points $x_0 \in X$.

From theorem 1 we know that two pointed spaces having the same homotopy type as pointed spaces have isomorphic fundamental groups. To prove a similar result for two path-connected spaces which have the same

homotopy type as spaces (no base-point condition) we need some preliminary results.

6 LEMMA Let $h: I \times I \rightarrow X$ and let $\alpha_0, \alpha_1, \beta_0$, and β_1 be the paths in X defined by restricting h to the edges of $I \times I$ [that is, $\alpha_i(t) = h(i, t)$ and $\beta_i(t) = h(t, i)$]. Then $(\alpha_0 * \beta_1) * (\alpha_1^{-1} * \beta_0^{-1})$ is a closed path in X at $h(0, 0)$ which represents the trivial element of $\pi(X, h(0, 0))$.

PROOF Let $\alpha'_0, \alpha'_1, \beta'_0$, and β'_1 be the paths in $I \times I$ defined by $\alpha'_i(t) = (i, t)$ and $\beta'_i(t) = (t, i)$. Then $(\alpha'_0 * \beta'_1) * (\alpha'_1^{-1} * \beta'_0^{-1})$ is a closed path in $I \times I$ at $(0, 0)$ and h maps this closed path into $(\alpha_0 * \beta_1) * (\alpha_1^{-1} * \beta_0^{-1})$. Since $I \times I$ is a convex subset of $\mathbb{R}^2$, it is contractible, and by lemma 5, it is simply connected. Therefore

$$(\alpha'_0 * \beta'_1) * (\alpha'_1^{-1} * \beta'_0^{-1}) \simeq \varepsilon_{(0,0)}$$

and

$$\begin{aligned} (\alpha_0 * \beta_1) * (\alpha_1^{-1} * \beta_0^{-1}) &= h \circ ((\alpha'_0 * \beta'_1) * (\alpha'_1^{-1} * \beta'_0^{-1})) \\ &\simeq h \circ \varepsilon_{(0,0)} = \varepsilon_{h(0,0)}. \quad \blacksquare \end{aligned}$$

7 THEOREM Let $f: (X, x_0) \rightarrow (Y, y_0)$ and $g: (X, x_0) \rightarrow (Y, y_1)$ be homotopic as maps of X to Y . Then there is a path ω in Y from y_0 to y_1 such that

$$f_{\#} = h_{[\omega]} \circ g_{\#}: \pi(X, x_0) \rightarrow \pi(Y, y_0)$$

PROOF Let $F: X \times I \rightarrow Y$ be a homotopy from f to g and let $\omega: I \rightarrow Y$ be defined by $\omega(t) = F(x_0, t)$. Then ω is a path in Y from y_0 to y_1 . If ω' is any closed path in X at x_0 , let $h: I \times I \rightarrow Y$ be defined by $h(t, t') = F(\omega'(t), t')$. Then $h(0, t') = F(x_0, t') = \omega(t')$, $h(t, 1) = g\omega'(t)$, $h(1, t') = \omega(t')$, and $h(t, 0) = f\omega'(t)$. By lemma 6 we have

$$(\omega * g\omega') * (\omega^{-1} * (f\omega')^{-1}) \simeq \varepsilon_{y_0}$$

This implies $[\omega] \circ g_{\#}[\omega'] \circ [\omega]^{-1} = f_{\#}[\omega']$, or $(h_{[\omega]} \circ g_{\#})[\omega'] = f_{\#}[\omega']$. Since $[\omega']$ is an arbitrary element of $\pi(X, x_0)$, $h_{[\omega]} \circ g_{\#} = f_{\#}$. ■

8 THEOREM Two path-connected spaces with the same homotopy type have isomorphic fundamental groups.

PROOF Let $f: X \rightarrow Y$ be a homotopy equivalence with homotopy inverse $g: Y \rightarrow X$. Let $x_0 \in X$ and set $y_0 = f(x_0)$, $x_1 = g(y_0)$, and $y_1 = f(x_1)$. Let $f_0: (X, x_0) \rightarrow (Y, y_0)$ and $f_1: (X, x_1) \rightarrow (Y, y_1)$ be maps defined by f (that is, f_0 and f_1 are both equal to f but are regarded as maps of pairs), and let $g': (Y, y_0) \rightarrow (X, x_1)$ be defined by g . Then $g' \circ f_0: (X, x_0) \rightarrow (X, x_1)$ is homotopic, as a map of X to X , to $1_{(X, x_0)}: (X, x_0) \subset (X, x_0)$ and $f_1 \circ g': (Y, y_0) \rightarrow (Y, y_1)$ is homotopic, as a map of Y to Y , to $1_{(Y, y_0)}: (Y, y_0) \subset (Y, y_0)$. It follows from theorem 7 that there are paths ω in X from x_1 to x_0 and ω' in Y from y_1 to y_0 such that

$$h_{[\omega]} = (g' \circ f_0)_{\#} = g'_{\#} \circ f_{0\#} \quad \text{and} \quad h_{[\omega']} = (f_1 \circ g')_{\#} = f_{1\#} \circ g'_{\#}$$

Therefore we have a commutative diagram

$$\begin{array}{ccc} \pi(X, x_0) & \xrightarrow{h_{1\omega}} & \pi(X, x_1) \\ f_{\omega} \downarrow & \nearrow g_{\omega}^* & \downarrow h_{1\omega} \\ \pi(Y, y_0) & \xrightarrow{h_{1\omega'}} & \pi(Y, y_1) \end{array}$$

g_{ω}^* is an epimorphism because $h_{1\omega}$ is, and it is a monomorphism because $h_{1\omega'}$ is. Therefore g_{ω}^* is an isomorphism. ■

We close with an example of a space with a nontrivial fundamental group. For this purpose we compute $\pi(S^1, p_0)$ following a method used by Tucker¹, where $S^1 = \{e^{i\theta}\}$ and $p_0 = 1$.

The exponential map $ex: \mathbf{R} \rightarrow S^1$ is defined by $ex(t) = e^{2\pi it}$. Then ex is continuous, $ex(t_1 + t_2) = ex(t_1)ex(t_2)$ (where the right-hand side is multiplication of complex numbers), and $ex(t_1) = ex(t_2)$ if and only if $t_1 - t_2$ is an integer. It follows that $ex|(-\frac{1}{2}, \frac{1}{2})$ is a homeomorphism of the open interval $(-\frac{1}{2}, \frac{1}{2})$ onto $S^1 - \{e^{\pi i}\}$. We let

$$lg: S^1 - \{e^{\pi i}\} \rightarrow (-\tfrac{1}{2}, \tfrac{1}{2})$$

be the inverse of $ex|(-\frac{1}{2}, \frac{1}{2})$.

A subset $X \subset \mathbf{R}^n$ will be called *starlike* from a point $x_0 \in X$ if, whenever $x \in X$, the closed line segment $[x_0, x]$ from x_0 to x lies in X .

9 LEMMA *Let X be compact and starlike from $x_0 \in X$. Given any continuous map $f: X \rightarrow S^1$ and any $t_0 \in \mathbf{R}$ such that $ex(t_0) = f(x_0)$, there exists a continuous map $f': X \rightarrow \mathbf{R}$ such that $f'(x_0) = t_0$ and $ex(f'(x)) = f(x)$ for all $x \in X$.*

PROOF Clearly, we can translate X so that it is starlike from the origin; hence there is no loss of generality in assuming $x_0 = 0$. Since X is compact, f is uniformly continuous and there exists $\epsilon > 0$ such that if $\|x - x'\| < \epsilon$, then $\|f(x) - f(x')\| < 2$ [that is, $f(x)$ and $f(x')$ are not antipodes in S^1]. Since X is bounded, there exists a positive integer n such that $\|x\|/n < \epsilon$ for all $x \in X$. Then for each $0 \leq j < n$ and all $x \in X$

$$\begin{aligned} \left\| \frac{(j+1)x}{n} - \frac{jx}{n} \right\| &= \frac{\|x\|}{n} < \epsilon \\ \left\| f\left(\frac{(j+1)x}{n}\right) - f\left(\frac{jx}{n}\right) \right\| &< 2 \end{aligned}$$

and so

It follows that the quotient $f((j+1)x/n)/f(jx/n)$ is a point of $S^1 - \{e^{\pi i}\}$. Let

$g_j: X \rightarrow S^1 - \{e^{\pi i}\}$ for $0 \leq j < n$ be the map defined by $g_j(x) =$

Therefore we have a commutative diagram

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PROOF Clearly, we can translate X so that it is starlike from the origin; hence there is no loss of generality in assuming $x_0 = 0$. Since X is compact, f is uniformly continuous and there exists $\epsilon > 0$ such that if $\|x - x'\| < \epsilon$, then $\|f(x) - f(x')\| < 2$ [that is, $f(x)$ and $f(x')$ are not antipodes in S^1]. Since X is bounded, there exists a positive integer n such that $\|x\|/n < \epsilon$ for all $x \in X$. Then for each $0 \leq j < n$ and all $x \in X$

$$\begin{aligned} \left\| \frac{(j+1)x}{n} - \frac{jx}{n} \right\| &= \frac{\|x\|}{n} < \epsilon \\ \left\| f\left(\frac{(j+1)x}{n}\right) - f\left(\frac{jx}{n}\right) \right\| &< 2 \end{aligned}$$

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g_{ω}^* is an epimorphism because $h_{1\omega}$ is, and it is a monomorphism because $h_{1\omega'}$ is. Therefore g_{ω}^* is an isomorphism. ■

We define $f': X \rightarrow \mathbf{R}$ by

$$f'(x) = t_0 + lg(g_0(x)) + lg(g_1(x)) + \cdots + lg(g_{n-1}(x))$$

f' is the sum of $n + 1$ continuous functions from X to $\mathbf{R}$, so it is continuous. Clearly, $f'(0) = t_0$ and $ex(f'(x)) = f(x)$. ■

10 LEMMA *Let X be a connected space and let $f', g': X \rightarrow \mathbf{R}$ be maps such that $ex \circ f' = ex \circ g'$ and $f'(x_0) = g'(x_0)$ for some $x_0 \in X$. Then $f' = g'$.*

PROOF Let $h = f' - g': X \rightarrow \mathbf{R}$. Since $ex \circ f' = ex \circ g'$, $ex \circ h$ is the constant map of X to p_0 . Therefore h is a continuous map of X to $\mathbf{R}$, taking only integral values. Because X is connected, h is constant, and since $h(x_0) = 0$, $h(x) = 0$ for all $x \in X$. ■

Let $\alpha: I \rightarrow S^1$ be a closed path at p_0 . Because I is starlike from 0 and $\alpha(0) = p_0 = ex(0)$, it follows from lemma 9 that there exists $\alpha': I \rightarrow \mathbf{R}$ such that $\alpha'(0) = 0$ and $ex \circ \alpha' = \alpha$. By lemma 10, α' is uniquely characterized by these properties. Because $ex(\alpha'(1)) = p_0$, it follows that $\alpha'(1)$ is an integer. We define the *degree* of α by $\deg \alpha = \alpha'(1)$.

11 LEMMA *Let α and β be homotopic closed paths in S^1 at p_0 . Then $\deg \alpha = \deg \beta$.*

PROOF Let $F: I \times I \rightarrow S^1$ be a homotopy relative to I from α to β . Because $I \times I$ is a starlike subset of $\mathbf{R}^2$ from $(0, 0)$, it follows from lemma 9 that there is a map $F': I \times I \rightarrow \mathbf{R}$ such that $F'(0, 0) = 0$ and $ex \circ F' = F$. Since F is a homotopy relative to I , $F(0, t') = F(1, t')$ for all $t \in I$. Therefore $F'(0, t')$ and $F'(1, t')$ take on only integral values for all $t' \in I$. It follows that $F'(0, t') = F'(1, t')$ must be constant and $F'(1, t')$ must be constant. Because $F'(0, 0) = 0$, $F'(0, t') = 0$ for all $t' \in I$. Define $\alpha', \beta': I \rightarrow \mathbf{R}$ by $\alpha'(t) = F'(t, 0)$ and $\beta'(t) = F'(t, 1)$. Then $\alpha'(0) = 0$ and $ex \circ \alpha' = \alpha$. Therefore $\deg \alpha = \alpha'(1) = F'(1, 0)$. Similarly, $\beta'(0) = 0$ and $ex \circ \beta' = \beta$, so $\deg \beta = \beta'(1) = F'(1, 1)$. Because $F'(1, t')$ is constant, $F'(1, 0) = F'(1, 1)$ and $\deg \alpha = \deg \beta$. ■

It follows that there is a well-defined function $\deg$ from $\pi(S^1, p_0)$ to $\mathbf{Z}$ defined by

$$\deg [\alpha] = \deg \alpha$$

where α is a closed path in S^1 at p_0 .

12 THEOREM *The function $\deg$ is an isomorphism*

$$\deg: \pi(S^1, p_0) \simeq \mathbf{Z}$$

PROOF To prove that $\deg$ is a homomorphism, let α and β be two closed paths in S^1 at p_0 and let $\alpha\beta$ be the closed path which is their pointwise

¹ See A. W. Tucker, Some topological properties of disk and sphere, *Proceedings of the Canadian Mathematical Congress*, 1945, pp. 285-309.

product in the group multiplication of S^1 . We know from theorem 4 that $[\alpha] * [\beta] = [\alpha\beta]$. Let $\alpha', \beta': I \rightarrow \mathbf{R}$ be such that $\alpha'(0) = 0$, $ex \circ \alpha' = \alpha$, $\beta'(0) = 0$, and $ex \circ \beta' = \beta$. Then $\alpha' + \beta': I \rightarrow \mathbf{R}$ is such that $(\alpha' + \beta')(0) = 0$ and $ex \circ (\alpha' + \beta') = \alpha\beta$. Therefore

$$\begin{aligned}deg([\alpha] * [\beta]) &= deg[\alpha\beta] = (\alpha' + \beta')(1) \\&= deg \alpha + deg \beta = deg [\alpha] + deg [\beta]\end{aligned}$$

showing that deg is a homomorphism.

The map deg is an epimorphism; for if n is an integer, there is a path α'_n in $\mathbf{R}$ defined by $\alpha'_n(t) = tn$. Let $\alpha_n = ex \circ \alpha'_n$. Then clearly, $deg[\alpha_n] = \alpha'_n(1) = n$.

The map deg is a monomorphism; for if $deg[\alpha] = 0$, there is a closed path α' in $\mathbf{R}$ at 0 such that $ex \circ \alpha' = \alpha$. Since $\mathbf{R}$ is simply connected (because it is contractible, and by lemma 5), $\alpha' \simeq \epsilon_0$. Then $ex \circ \alpha' \simeq \epsilon_{p_0}$. Therefore $\alpha \simeq \epsilon_{p_0}$ and $[\alpha]$ is the identity element of $\pi(S^1, p_0)$. ■

The method we have used to compute $\pi(S^1, p_0)$ will be generalized in Chapter 2 to give relations between the fundamental group of a space and the fundamental groups of its covering spaces.

It follows from theorem 2 that $\pi(S^1, p_0) \simeq [S^1, p_0; S^1, p_0]$. Because S^1 is a topological group, the set $[S^1, S^1]$ (with no base-point condition) is also a group under pointwise multiplication of maps, and there is an obvious homomorphism

$$\gamma: [S^1, p_0; S^1, p_0] \rightarrow [S^1, S^1]$$

1.3 LEMMA The homomorphism

$$\gamma: [S^1, p_0; S^1, p_0] \rightarrow [S^1, S^1]$$

is an isomorphism.

PROOF To show that γ is an epimorphism, let $f: S^1 \rightarrow S^1$ and let $f(p_0) = e^{i\theta}$ for some $0 \leq \theta < 2\pi$. Define a homotopy $F: S^1 \times I \rightarrow S^1$ by

$$F(z, t) = f(z)e^{-it\theta}$$

Then F is a homotopy from f to a map f' such that $f'(p_0) = p_0$. Therefore $\gamma[f']_{p_0} = [f']$.

To show that γ is a monomorphism, assume that $f: (S^1, p_0) \rightarrow (S^1, p_0)$ is such that $\gamma[f]_{p_0} = [f]$ is trivial. Then $f: S^1 \rightarrow S^1$ is null homotopic. By theorem 1.3.12, f is null homotopic relative to p_0 . Therefore $[f]_{p_0}$ is trivial. ■

It follows from theorem 12 and lemma 13 that $[S^1, S^1] \simeq \mathbf{Z}$. For each integer n the map $z \rightarrow z^n$ from S^1 to itself represents a homotopy class corresponding to n under the above isomorphism.

EXERCISES

A CONTRACTIBLE SPACES

1 The cone over a topological space X with vertex v is defined to be the mapping cylinder of the constant map $X \rightarrow v$. Prove that X is contractible if and only if it is a retract of any cone over X .

2 Prove that S^n is a retract of E^{n+1} if and only if S^n is contractible.

3 If CX is a cone over X , prove that (CX, X) has the homotopy extension property with respect to any space.

4 Prove that a space Y is contractible if and only if, given a pair (X, A) having the homotopy extension property with respect to Y , any map $A \rightarrow Y$ can be extended over X .

5 Let Y be the comb space of example 1.3.9 and let y_0 be the point $(0, 1) \in Y$. Let Y' be another copy of Y , with corresponding point y'_0 . Let X be the space obtained by forming the disjoint union of Y and Y' and identifying y_0 with y'_0 . Prove that X is n -connected for all n but not contractible. (Hint: Any deformation of X in itself must be a homotopy relative to y_0 .)

B ADJUNCTION SPACES

1 Let A be a subspace of a space X and let $f: A \rightarrow Y$ be a continuous map. The adjunction space Z of X to Y by f is defined to be the quotient space of the disjoint union of X and Y by the identifications $x \in A$ equals $f(x) \in Y$ for all $x \in A$. Prove that if X and Y are normal spaces and A is closed in X , then Z is a normal space.

2 A space X is said to be *binormal* if $X \times I$ is a normal space. If X is a binormal space, Y is a normal space, and $f: X \rightarrow Y$ is continuous, prove that the mapping cylinder of f is a normal space.

3 Given a continuous map $f: A \rightarrow Y$, where A is a subspace of a space X , prove that f can be extended over X if and only if Y is a retract of the adjunction space of X to Y by f .

4 Let Z be the adjunction space of X to Y by a map $f: A \rightarrow Y$. Prove that (Z, Y) has the homotopy extension property with respect to a space W if (X, A) has the homotopy extension property with respect to W .

C ABSOLUTE RETRACTS AND ABSOLUTE NEIGHBORHOOD RETRACTS

A space Y is said to be an *absolute retract* (or *absolute neighborhood retract*) if, given a normal space X , closed subset $A \subset X$, and a continuous map $f: A \rightarrow Y$, then f can be extended over X (or f can be extended over some neighborhood of A in X).

1 Prove that a normal space Y is an absolute retract (or absolute neighborhood retract) if and only if, whenever Y is imbedded as a closed subset of a normal space Z , then Y is a retract of Z (or a retract of some neighborhood of Y in Z).

2 Prove that the product of arbitrarily many absolute retracts (or finitely many absolute neighborhood retracts) is itself an absolute retract (or absolute neighborhood retract).

3 Prove that $\mathbf{R}^n$ is an absolute retract for all n .

- 4 Prove that a retract of an absolute retract is an absolute retract and that a retract of some open subset of an absolute neighborhood retract is an absolute neighborhood retract.
- 5 Prove that E^n is an absolute retract and S^n is an absolute neighborhood retract for all n .
- 6 Prove that a binormal absolute neighborhood retract is *locally contractible* (that is, every neighborhood U of a point x contains a neighborhood V of x deformable to x in U).
- 7 Prove that a binormal absolute neighborhood retract is an absolute retract if and only if it is contractible.

D HOMOTOPY EXTENSION PROPERTY

- 1 Let A be a closed subset of a normal space X , let $f: X \rightarrow Y$ be continuous (where Y is arbitrary), and let $G: A \times I \rightarrow Y$ be a homotopy of $f|_A$. If there exists a homotopy $G': U \times I \rightarrow Y$ of $f|_U$ which extends G , where U is an open neighborhood of A , show that there exists a homotopy $F: X \times I \rightarrow Y$ of f which extends G .
- 2 *Borsuk's homotopy extension theorem.* Let A be a closed subspace of a binormal space X . Then (X, A) has the homotopy extension property with respect to any absolute neighborhood retract Y .
- 3 Let A be a closed subset of a binormal space X and assume that the subspace $A \times I \cup X \times 0 \subset X \times I$ is an absolute neighborhood retract. Then (X, A) has the homotopy extension property with respect to any space Y .
- 4 Let A be a closed subset of X and B a subset of Y . Assume that (X, A) has the homotopy extension property with respect to B and that $(X \times I, X \times I \cup A \times I)$ has the homotopy extension property with respect to Y . Prove that if $f: (X, A) \rightarrow (Y, B)$ is homotopic (as a map of pairs) to a map which sends all of X to B , then it is homotopic relative to A to such a map.

E COFIBRATIONS

- 1 Prove that any cofibration is an injective function.
- 2 Prove that a composite of cofibrations is a cofibration.
- 3 For a closed subspace A of X prove that the inclusion map $A \subset X$ is a cofibration if and only if $X \times 0 \cup A \times I$ is a retract of $X \times I$.
- 4 If A is a subspace of a Hausdorff space X , prove that if $A \subset X$ is a cofibration, then A is closed in X .
- 5 Assume that X is the union of closed subsets X_1 and X_2 and let A be a subset of X such that $X_1 \cap X_2 \subset A$. Prove that if $A \cap X_1 \subset X_1$ and $A \cap X_2 \subset X_2$ are cofibrations, so is $A \subset X$.
- 6 Let A be a closed subspace of a space X . Prove that the following are equivalent:
 - (a) $A \subset X$ is a cofibration.
 - (b) There is a deformation $D: X \times I \rightarrow X$ rel A [that is, $D(x, 0) = x$ and $D(a, t) = a$ for $x \in X$, $a \in A$, and $t \in I$] and a map $q: X \rightarrow I$ such that $A = q^{-1}(1)$ and $D(q^{-1}(0), 1] \times 1) \subset A$.

¹ If X is normal, the equivalence of (a) and (c) is proved in G. S. Young, A condition for the absolute homotopy extension property, *American Mathematical Monthly*, vol. 71, pp. 896-897, 1964.

- (c) There is a neighborhood U of A deformable in X to A rel A [that is, there is a homotopy $H: U \times I \rightarrow X$ such that $H(x, 0) = x$, $H(a, t) = a$, and $H(x, 1) \in A$ for $x \in U$, $a \in A$, and $t \in I$] and a map $q: X \rightarrow I$ such that $A = q^{-1}(1)$ and $q(x) = 0$ if $x \in X - U$.
- 7 If $A \subset X$ and $B \subset Y$ are cofibrations with A and B closed in X and Y , respectively, prove that $A \times B \subset X \times B \cup A \times Y$ and $X \times B \cup A \times Y \subset X \times Y$ are cofibrations.

F LOCAL SYSTEMS¹

- 1 A *local system* on a space X is a covariant functor from the fundamental groupoid of X to some category. For any category $\mathcal{C}$ show that there is a category of local systems on X with values in $\mathcal{C}$. (Two local systems on X are said to be *equivalent* if they are equivalent objects in this category.)
- 2 Given a map $f: X \rightarrow Y$, show that f induces a covariant functor from the category of local systems on Y with values in $\mathcal{C}$ to the category of local systems on X with values in $\mathcal{C}$.
- 3 If A is an object of a category $\mathcal{C}$, let $\text{Aut } A$ be the group of self-equivalences of A in $\mathcal{C}$. If $q: A \xrightarrow{\sim} B$ is an equivalence in $\mathcal{C}$, then show that $\tilde{q}: \text{Aut } A \rightarrow \text{Aut } B$ defined by $\tilde{q}(\alpha) = q \circ \alpha \circ q^{-1}$ is an isomorphism of groups.
- 4 If Γ is a local system on X and $x_0 \in X$, show that Γ induces a homomorphism $\Gamma_{x_0}: \pi_1(X, x_0) \rightarrow \text{Aut } \Gamma(x_0)$.
- 5 If X is path connected, prove that two local systems Γ and Γ' on X with values in $\mathcal{C}$ are equivalent if and only if there is an equivalence $q: \Gamma(x_0) \xrightarrow{\sim} \Gamma'(x_0)$, such that $\tilde{q} \circ \Gamma_{x_0}$ is conjugate to Γ'_{x_0} in $\text{Aut } \Gamma'(x_0)$.
- 6 If X is path connected, given an object $A \in \mathcal{C}$ and a homomorphism $\alpha: \pi_1(X, x_0) \rightarrow \text{Aut } A$, prove that there is a local system Γ on X with values in $\mathcal{C}$ such that $\Gamma(x_0) = A$ and $\Gamma_{x_0} = \alpha$.

G THE FUNDAMENTAL GROUP

- 1 Prove that the fundamental group functor commutes with direct products.
- 2 If ω and ω' are paths in X from x_0 to x_1 , prove that $\omega \simeq \omega'$ if and only if $\omega * \omega'^{-1} \simeq \varepsilon_{x_0}$.
- 3 Let a space X be the union of two open simply connected subsets U and V such that $U \cap V$ is nonempty and path connected. Prove that X is simply connected.
- 4 Prove that S^n is simply connected for $n \geq 2$.
- 5 If there exists a space with a nonabelian fundamental group, prove that the "figure eight" (that is, the union of two circles with a point in common) has a nonabelian fundamental group (cf. exercise 2.B.4).
- 6 Let $f: I \rightarrow \mathbb{R}^2$ be a continuously differentiable simple closed curve in the plane with a nowhere-vanishing tangent vector (that is, $f(0) = f(1)$, $f'(0) = f'(1)$, and $f'(t) \neq 0$ for

¹ See N. E. Steenrod, Homology with local coefficients, *Annals of Mathematics*, vol. 44, pp. 610-627, 1943.

$0 \leq t \leq 1$. Let $\omega: I \rightarrow S^1$ be the closed path defined by $\omega(t) = f'(t)/\|f'(t)\|$. Prove that $[\omega]$ is a generator of $\pi(S^1)$.¹

7 In $\mathbf{R}^2$, let X be the space consisting of the union of the circles C_n , where C_n has center $(1/n, 0)$ and radius $1/n$ for all positive integers n . In $\mathbf{R}^3$ (with $\mathbf{R}^2$ imbedded as the plane $x_3 = 0$), let Y be the set of points on the closed line segments joining $(0, 0, 1)$ to X and let Y' be the reflection of Y through the origin of $\mathbf{R}^3$. Then Y and Y' are closed simply connected subsets of $\mathbf{R}^3$ such that $Y \cap Y'$ is a single point. Prove that $Y \cup Y'$ is not simply connected.²

SOME APPLICATIONS OF THE FUNDAMENTAL GROUP

- 1** Prove that S^1 is not a retract of E^2 .
- 2** Prove that S^1 and S^n for $n \geq 2$ are not of the same homotopy type.
- 3** Prove that $\mathbf{R}^2$ and $\mathbf{R}^n$ for $n \geq 2$ are not homeomorphic.
- 4** Let $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0$ be a polynomial of degree n , having complex coefficients and leading coefficient 1, and let $q(z) = z^n$. For $r > 0$ let $C_r = \{x \in \mathbf{R}^2 \mid \|x\| = r\}$. Prove that for r large enough, $p|C_r$ and $q|C_r$ are homotopic maps of C_r into $\mathbf{R}^2 - 0$.
- 5** *Fundamental theorem of algebra.* Prove that every complex polynomial has a root. (*Hint:* For any $r > 0$ the map $q|C_r; C_r \rightarrow \mathbf{R}^2 - 0$ is not null homotopic because it induces a nontrivial homomorphism of fundamental groups.)

¹ See H. Hopf, Über die Drehung der Tangenten und Schnen ebener Kurven, *Compositio Mathematica*, vol. 2, pp. 50-62, 1935. For generalizations see H. Whitney, On regular closed curves in the plane, *Compositio Mathematica*, vol. 4, pp. 276-284, 1937, and S. Smale, Regular curves on Riemannian manifolds, *Transactions of the American Mathematical Society*, vol. 87, pp. 495-512, 1958.

² See H. B. Griffiths, The Fundamental group of two spaces with a common point, *Quarterly Journal of Mathematics*, vol. 5, pp. 175-190, 1954.

CHAPTER TWO COVERING SPACES AND FIBRATIONS

The lifting theorem is applied in Sec. 2.5 to classify the covering spaces of a connected locally path-connected space by means of subgroups of its fundamental group. This entails the construction of a covering space starting with the base space and a subgroup of its fundamental group. In Sec. 2.6 a converse problem is considered. The base space is constructed, starting with a covering space and a suitable group of transformations on it.

In Sec. 2.7 fiber bundles are introduced as natural generalizations of covering spaces. The main result of the section is that local fibrations are fibrations. This implies that a fiber bundle with paracompact base space is a fibration. Section 2.8 considers properties of general fibrations and the concept of fiber homotopy equivalence. These will be important in our later study of homotopy theory.

I COVERING PROJECTIONS

A covering projection is a continuous map that is a uniform local homeomorphism. This and related concepts are introduced in this section, along with some examples and elementary properties.

Let $p: \tilde{X} \rightarrow X$ be a continuous map. An open subset $U \subset X$ is said to be *evenly covered* by p if $p^{-1}(U)$ is the disjoint union of open subsets of $\tilde{X}$ each of which is mapped homeomorphically onto U by p . If U is evenly covered by p , it is clear that any open subset of U is also evenly covered by p . A continuous map $p: \tilde{X} \rightarrow X$ is called a *covering projection* if each point $x \in X$ has an open neighborhood evenly covered by p . $\tilde{X}$ is called the *covering space* and X the *base space* of the covering projection.

The following are examples of covering projections.

- 1 Any homeomorphism is a covering projection.
- 2 If $\tilde{X}$ is the product of X with a discrete space, the projection $\tilde{X} \rightarrow X$ is a covering projection.
- 3 The map $ex: \mathbf{R} \rightarrow S^1$, defined by $ex(t) = e^{2\pi i t}$, (considered in Sec 1.8) is a covering projection.
- 4 For any positive integer n the map $p: S^1 \rightarrow S^1$, defined by $p(z) = z^n$, is a covering projection.
- 5 For any integer $n \geq 1$ the map $p: S^n \rightarrow P^n$, which identifies antipodal points, is a covering projection.
- 6 If G is a topological group, H is a discrete subgroup of G , and G/H is the space of left (or right) cosets, then the projection $G \rightarrow G/H$ is a covering projection.

A continuous map $f: Y \rightarrow X$ is called a *local homeomorphism* if each point $y \in Y$ has an open neighborhood mapped homeomorphically by f onto

THE THEORY OF COVERING SPACES IS IMPORTANT NOT ONLY IN TOPOLOGY, BUT also in differential geometry, complex analysis, and Lie groups. The theory is presented here because the fundamental group functor provides a faithful representation of covering-space problems in terms of algebraic ones. This justifies our special interest in the fundamental group functor.

This chapter contains the theory of covering spaces, as well as an introduction to the related concepts of fiber bundle and fibration. These concepts will be considered again later in other contexts. Here we adopt the view that certain fibrations, namely, those having the property of unique path lifting, are generalized covering spaces. Because of this, we shall consider these fibrations in some detail.

Covering spaces are defined in Sec. 2.1, and fibrations are defined in Sec. 2.2, where it is proved that every covering space is a fibration. Section 2.3 deals with relations between the fundamental groups of the total space and base space of a fibration with unique path lifting, and Sec. 2.4 contains a solution of the lifting problem for such fibrations in terms of the fundamental group functor.

an open subset of X . If this is so, each point of Y has arbitrarily small neighborhoods with this property, and we have the following lemmas.

7 LEMMA A local homeomorphism is an open map. ■

8 LEMMA A covering projection is a local homeomorphism.

PROOF Let $p: \tilde{X} \rightarrow X$ be a covering projection and let $\tilde{x} \in \tilde{X}$. Let U be an open neighborhood of $p(\tilde{x})$ evenly covered by p . Then $p^{-1}(U)$ is the disjoint union of open sets, each mapped homeomorphically onto U by p . Let $\tilde{U}$ be that one of these open sets which contains $\tilde{x}$. Then $\tilde{U}$ is an open neighborhood of $\tilde{x}$ such that $p|_{\tilde{U}}$ is a homeomorphism of $\tilde{U}$ onto the open subset U of X . ■

A local homeomorphism need not be a covering projection, as shown by the following example.

9 EXAMPLE Let $p: (0, 3) \rightarrow S^1$ be the restriction of the map $ex: \mathbf{R} \rightarrow S^1$ of example 3 to the open interval $(0, 3)$. Because p is the restriction of a local homeomorphism to an open subset, it is a local homeomorphism. It is also a surjection, but it is not a covering projection because the complex number $1 \in S^1$ has no neighborhood evenly covered by p .

The following is a consequence of lemmas 7 and 8 and the fact (immediate from the definition) that a covering projection is a surjection.

10 COROLLARY A covering projection exhibits its base space as a quotient space of its covering space. ■

For locally connected spaces there is the following reduction of covering projections to the components of the base space.

11 THEOREM If X is locally connected, a continuous map $p: \tilde{X} \rightarrow X$ is a covering projection if and only if for each component C of X the map

$$p|_{p^{-1}C}: p^{-1}C \rightarrow C$$

is a covering projection.

PROOF If p is a covering projection and C is a component of X , let $x \in C$ and let U be an open neighborhood of x evenly covered by p . Let V be the component of U containing x . Since X is locally connected, V is open in X , and hence open in C . Clearly, V is evenly covered by $p|_{p^{-1}C}$. Therefore $p|_{p^{-1}C}$ is a covering projection.

Conversely, assume that the map $p|_{p^{-1}C}: p^{-1}C \rightarrow C$ is a covering projection for each component C of X . Let $x \in C$ and let U be an open neighborhood of x in C evenly covered by $p|_{p^{-1}C}$. Since X is locally connected, C is open in X . Therefore U is also open in X and is clearly evenly covered by p . Hence p is a covering projection. ■

In general, the representation of the inverse image of an evenly covered open set as a disjoint union of open sets, each mapped homeomorphically, is

not unique (consider the case of an evenly covered discrete set); however, for connected evenly covered open sets there is the following characterization of these open subsets.

12 LEMMA Let U be an open connected subset of X which is evenly covered by a continuous map $p: \tilde{X} \rightarrow X$. Then p maps each component of $p^{-1}(U)$ homeomorphically onto U .

PROOF By assumption, $p^{-1}(U)$ is the disjoint union of open subsets, each mapped homeomorphically onto U by p . Since U is connected, each of these open subsets is also connected. Because they are open and disjoint, each is a component of $p^{-1}(U)$. ■

13 COROLLARY Consider a commutative triangle

$$\begin{array}{ccc} \tilde{X}_1 & \xrightarrow{p} & \tilde{X}_2 \\ p_1 \searrow & & \swarrow p_2 \\ & X & \end{array}$$

where X is locally connected and p_1 and p_2 are covering projections. If p is a surjection, it is a covering projection.

PROOF If U is a connected open subset of X which is evenly covered by p_1 and p_2 , it follows easily from lemma 12 that each component of $p_2^{-1}(U)$ is evenly covered by p . ■

14 THEOREM If $p: \tilde{X} \rightarrow X$ is a covering projection onto a locally connected base space, then for any component $\tilde{C}$ of $\tilde{X}$ the map

$$p|_{\tilde{C}}: \tilde{C} \rightarrow p(\tilde{C})$$

is a covering projection onto some component of X .

PROOF Let $\tilde{C}$ be a component of $\tilde{X}$. We show that $p(\tilde{C})$ is a component of X . $p(\tilde{C})$ is connected; to show that it is an open and closed subset of X , let x be in the closure of $p(\tilde{C})$ and let U be an open connected neighborhood of x evenly covered by p . Because U meets $p(\tilde{C})$, $p^{-1}(U)$ meets $\tilde{C}$. Therefore some component $\tilde{U}$ of $p^{-1}(U)$ meets $\tilde{C}$. Since $\tilde{C}$ is a component of $\tilde{X}$, $\tilde{U} \subset \tilde{C}$. Then, by lemma 12, $p(\tilde{C}) \supset p(\tilde{U}) = U$. Therefore the closure of $p(\tilde{C})$ is contained in the interior of $p(\tilde{C})$, which implies that $p(\tilde{C})$ is open and closed. The same argument shows that if $x \in p(\tilde{C})$ and U is an open connected neighborhood of x in X evenly covered by p , then $U \subset p(\tilde{C})$ and $(p|_{\tilde{C}})^{-1}(U)$ is the disjoint union of those components of $p^{-1}(U)$ that meet $\tilde{C}$. It follows from lemma 12 that U is evenly covered by $p|_{\tilde{C}}$. Therefore $p|_{\tilde{C}}: \tilde{C} \rightarrow p(\tilde{C})$ is a covering projection. ■

The following example shows that the converse of theorem 14 is false.

15 EXAMPLE Let $X = S^1 \times S^1 \times \cdots$ be a countable product of 1-spheres and for $n \geq 1$ let $\tilde{X}_n = \mathbf{R}^n \times S^1 \times S^1 \times \cdots$. Define $p_n: \tilde{X}_n \rightarrow X$ by

a continuous map $f': X \rightarrow E$ such that $f = p \circ f'$ —that is, whether the dotted arrow in the diagram

$$\begin{array}{ccc} E & & \\ \nearrow f' & \downarrow p & \\ X & \xrightarrow{f} & B \end{array}$$

corresponds to a continuous map making the diagram commutative. If there is such a map f' , then f can be *lifted* to E , and we call f' a *lifting*, or *lift*, of f . In order that the lifting problem be a problem in the homotopy category, we need an analogue of the homotopy extension property, called the homotopy lifting property, defined as follows. A map $p: E \rightarrow B$ is said to have the *homotopy lifting property with respect to a space* X if, given maps $f': X \rightarrow E$ and $F: X \times I \rightarrow B$ such that $F(x, 0) = pf'(x)$ for $x \in X$, there is a map $F': X \times I \rightarrow E$ such that $F'(x, 0) = f'(x)$ for $x \in X$ and $p \circ F' = F$. If f' is regarded as a map of $X \times 0$ to E , the existence of F' is equivalent to the existence of a map represented by the dotted arrow that makes the following diagram commutative:

$$\begin{array}{ccc} X \times 0 & \xrightarrow{f'} & E \\ \downarrow \cap & \nearrow & \downarrow p \\ X \times I & \xrightarrow{F'} & B \end{array}$$

If $p: E \rightarrow B$ has the homotopy lifting property with respect to X and $f_0, f_1: X \rightarrow B$ are homotopic, it is easy to see that f_0 can be lifted to E if and only if f_1 can be lifted to E . Hence, whether or not a map $X \rightarrow B$ can be lifted to E is a property of the homotopy class of the map. Thus the homotopy lifting property implies that the lifting problem for maps $X \rightarrow B$ is a problem in the homotopy category.

A map $p: E \rightarrow B$ is called a *fibration* (or *Hurewicz fiber space* in the literature) if p has the homotopy lifting property with respect to every space. E is called the *total space* and B the *base space* of the fibration. For $b \in B$, $p^{-1}(b)$ is called the *fiber over* b .

If $p: E \rightarrow B$ is a fibration, any path ω in B such that $\omega(0) \in p(E)$ can be lifted to a path in E . In fact, ω can be regarded as a homotopy $\omega: P \times I \rightarrow B$ where P is a one-point space, and a point $e_0 \in E$ such that $p(e_0) = \omega(0)$ corresponds to a map $f: P \rightarrow E$ such that $pf(P) = \omega(P, 0)$. It follows from the homotopy lifting property of p that there exists a path $\tilde{\omega}$ in E such that $\tilde{\omega}(0) = e_0$ and $p \circ \tilde{\omega} = \omega$. Then $\tilde{\omega}$ is a lifting of ω .

EXAMPLE Let F be any space and let $p: B \times F \rightarrow B$ be the projection to the first factor. Then p is a fibration, and for any $b \in B$ the fiber over b is homeomorphic to F .

To prove that a covering projection is a fibration, we first establish the following *unique-lifting property* of covering projections for connected spaces.

SEC. 2 THE HOMOTOPY LIFTING PROPERTY 65

$$p_n(t_1, \dots, t_n, z_1, z_2, \dots) = (ex(t_1), \dots, ex(t_n), z_1, z_2, \dots)$$

Let $\tilde{X} = \bigvee \tilde{X}_n$ and define $p: \tilde{X} \rightarrow X$ so that $p|_{\tilde{X}_n} = p_n$. The components of $\tilde{X}$ are the spaces $\tilde{X}_n$ and the map $p|_{\tilde{X}_n} = p_n: \tilde{X}_n \rightarrow X$ is a covering projection. However, p is not a covering projection, because no open subset of X is evenly covered by p .

For later purposes we should like to have the analogues of theorems 11 and 14, in which “component” is replaced by “path component.” For this we need the following definition: a topological space is said to be *locally path connected* if the path components of open subsets are open. The following are easy consequences of this definition.

16 Any open subset of a locally path-connected space is itself locally path connected. ■

17 A locally path-connected space is locally connected. ■

18 In a locally path-connected space the components and path components coincide. ■

19 A connected locally path-connected space is path connected. ■

From statements 17 and 18 we obtain the following extension of theorems 11 and 14.

20 THEOREM If X is locally path connected, a continuous map $p: \tilde{X} \rightarrow X$ is a covering projection if and only if for each path component A of X

$$p|_{p^{-1}A}: p^{-1}A \rightarrow A$$

is a covering projection. In this case, if $\tilde{A}$ is any path component of $\tilde{X}$, then $p|_{\tilde{A}}$ is a covering projection of $\tilde{A}$ onto some path component of X . ■

2 THE HOMOTOPY LIFTING PROPERTY

The homotopy lifting property is dual to the homotopy extension property. It leads to the concept of fibration, which is dual to that of cofibration introduced in Sec. 1.4. In this section we define the concept of fibration and prove that a covering projection is a special kind of fibration. This special class of fibrations will be regarded as generalized covering projections, and our subsequent study of covering projections will be based on a study of the more general concept. At the end of the chapter we return to the general consideration of fibrations.

We begin with an important problem of algebraic topology, called the lifting problem, which is dual to the extension problem. Let $p: E \rightarrow B$ and $f: X \rightarrow B$ be maps. The *lifting problem* for f is to determine whether there is

2 THEOREM Let $p: \tilde{X} \rightarrow X$ be a covering projection and let $f, g: Y \rightarrow \tilde{X}$ be liftings of the same map (that is, $p \circ f = p \circ g$). If Y is connected and f agrees with g for some point of Y , then $f = g$.

PROOF Let $Y_1 = \{y \in Y \mid f(y) = g(y)\}$. We show that Y_1 is open in Y . If $y \in Y_1$, let U be an open neighborhood of $pf(y)$ evenly covered by p and let $\tilde{U}$ be an open subset of $\tilde{X}$ containing $f(y)$ such that p maps $\tilde{U}$ homeomorphically onto U . Then $f^{-1}(\tilde{U}) \cap g^{-1}(\tilde{U})$ is an open subset of Y containing y and contained in Y_1 .

Let $Y_2 = \{y \in Y \mid f(y) \neq g(y)\}$. We show that Y_2 is also open in Y (if $\tilde{X}$ were assumed to be Hausdorff, this would follow from a general property of Hausdorff spaces). Let $y \in Y_2$ and let U be an open neighborhood of $pf(y)$ evenly covered by p . Since $f(y) \neq g(y)$, there are disjoint open subsets $\tilde{U}_1$ and $\tilde{U}_2$ of $\tilde{X}$ such that $f(y) \in \tilde{U}_1$ and $g(y) \in \tilde{U}_2$ and p maps each of the sets $\tilde{U}_1$ and $\tilde{U}_2$ homeomorphically onto U . Then $f^{-1}(\tilde{U}_1) \cap g^{-1}(\tilde{U}_2)$ is an open subset of Y containing y and contained in Y_2 .

Since $Y = Y_1 \cup Y_2$ and Y_1 and Y_2 are disjoint open sets, it follows from the connectedness of Y that either $Y_1 = \emptyset$ or $Y_1 = Y$. By hypothesis, $Y_1 \neq \emptyset$, so $Y = Y_1$ and $f = g$. ■

We are now ready to prove that a covering projection has the homotopy lifting property.

3 THEOREM A covering projection is a fibration.

PROOF Let $p: \tilde{X} \rightarrow X$ be a covering projection and let $f': Y \rightarrow \tilde{X}$ and $F: Y \times I \rightarrow \tilde{X}$ be maps such that $F(y, 0) = pf'(y)$ for $y \in Y$. We show that for each $y \in Y$ there is an open neighborhood N_y of y in Y and a map $F_y': N_y \times I \rightarrow \tilde{X}$ such that $F_y'(y', 0) = f'(y')$ for $y' \in N_y$ and $pf_y' = F|_{N_y \times I}$. Assume that we have such neighborhoods N_y and maps F_y' . If $y'' \in N_y \cap N_{y'}$, then $F_y|_{y'' \times I}$ and $F_{y'}|_{y'' \times I}$ are maps of the connected space $y'' \times I$ into $\tilde{X}$ such that for $t \in I$

$$p \circ (F_y|_{y'' \times I})(y'', t) = F(y'', t) = F(y'', 0) = p \circ (F_{y'}|_{y'' \times I})(y'', t)$$

Because $(F_y|_{y'' \times I})(y'', 0) = f'(y'') = (F_{y'}|_{y'' \times I})(y'', 0)$, it follows from theorem 2 that $F_y|_{y'' \times I} = F_{y'}|_{y'' \times I}$. Since this is true for all $y'' \in N_y \cap N_{y'}$, it follows that $F_y|_{(N_y \cap N_{y'}) \times I} = F_{y'}|_{(N_y \cap N_{y'}) \times I}$. Hence there is a continuous map $F': Y \times I \rightarrow \tilde{X}$ such that $F|_{N_y \times I} = F_y'$, and F' is a lifting of F such that $F'(y, 0) = f'(y)$ for $y \in Y$. Thus we have reduced the theorem to the construction of the open neighborhoods N_y and maps F_y' .

It follows from the fact that $p: \tilde{X} \rightarrow X$ is a covering projection (and the compactness of I) that for each $y \in Y$ there is an open neighborhood N_y of y and a sequence $0 = t_0 < t_1 < \dots < t_m = 1$ of points of I such that for $i = 1, \dots, m$, $F(N_y \times [t_{i-1}, t_i])$ is contained in some open subset of $\tilde{X}$ evenly covered by p . We show that there is a map $F_y': N_y \times I \rightarrow \tilde{X}$ with the desired properties. It suffices to define maps

$$G_i: N_y \times [t_{i-1}, t_i] \rightarrow \tilde{X} \quad i = 1, \dots, m$$

such that

$$\begin{aligned} p \circ G_i &= F|_{N_y \times [t_{i-1}, t_i]} \\ G_1(y', 0) &= f'(y') & y' \in N_y \\ G_{i-1}(y', t_{i-1}) &= G_i(y', t_{i-1}) & y' \in N_y, i = 2, \dots, m \end{aligned}$$

because, given such maps G_i , there is a map $F_y': N_y \times I \rightarrow \tilde{X}$ such that $F_y'|_{N_y \times [t_{i-1}, t_i]} = G_i$ for $i = 1, \dots, m$. Then F_y' has the desired properties.

The maps G_i are defined by induction on i . To define G_1 , let U be an open subset of X evenly covered by p such that $F(N_y \times [t_0, t_1]) \subset U$. Let $\{\tilde{U}_j\}$ be a collection of disjoint open subsets of $\tilde{X}$ such that $p^{-1}(U) = \bigcup \tilde{U}_j$ and p maps $\tilde{U}_j$ homeomorphically onto U for each j . Let $V_j = f^{-1}(\tilde{U}_j)$. Then $\{V_j\}$ is a collection of disjoint open sets covering N_y , and G_1 is defined to be the unique map such that for each j , G_1 maps $V_j \times [t_0, t_1]$ into $\tilde{U}_j$ to be a lifting of $F|_{V_j \times [t_0, t_1]}$. This defines G_1 .

Assume G_{i-1} defined for $1 < i \leq m$. Let U' be an open subset of X evenly covered by p such that $F(N_y \times [t_{i-1}, t_i]) \subset U'$. Let $\{\tilde{U}'_k\}$ be a collection of disjoint open subsets of $\tilde{X}$ such that $p^{-1}(U') = \bigcup \tilde{U}'_k$ and p maps $\tilde{U}'_k$ homeomorphically onto U' for each k . Let $V'_k = \{y' \in N_y \mid G_{i-1}(y', t_{i-1}) \in \tilde{U}'_k\}$. Then $\{V'_k\}$ is a collection of disjoint open sets covering N_y , and G_i is defined to be the unique map such that for each k , G_i maps $V'_k \times [t_{i-1}, t_i]$ into $\tilde{U}'_k$ to be a lifting of $F|_{V'_k \times [t_{i-1}, t_i]}$. This defines G_i . ■

A map $p: E \rightarrow B$ is said to have *unique path lifting* if, given paths ω and ω' in E such that $p \circ \omega = p \circ \omega'$ and $\omega(0) = \omega'(0)$, then $\omega = \omega'$. It follows from theorem 2 that a covering projection has unique path lifting.

4 LEMMA If a map has unique path lifting, it has the unique-lifting property for path-connected spaces.

PROOF Assume that $p: E \rightarrow B$ has unique path lifting. Let Y be path connected and suppose that $f, g: Y \rightarrow E$ are maps such that $p \circ f = p \circ g$ and $f(y_0) = g(y_0)$ for some $y_0 \in Y$. We must show $f = g$. Let $y \in Y$ and let ω be a path in Y from y_0 to y . Then $f \circ \omega$ and $g \circ \omega$ are paths in E that are liftings of the same path in B and have the same origin. Because p has unique path lifting, $f \circ \omega = g \circ \omega$. Therefore

$$f(y) = (f \circ \omega)(1) = (g \circ \omega)(1) = g(y)$$

The following theorem characterizes fibrations with unique path lifting.

5 THEOREM A fibration has unique path lifting if and only if every fiber has no nonconstant paths.

PROOF Assume that $p: E \rightarrow B$ is a fibration with unique path lifting. Let ω be a path in the fiber $p^{-1}(b)$ and let ω' be the constant path in $p^{-1}(b)$ such that $\omega'(0) = \omega(0)$. Then $p \circ \omega = p \circ \omega'$, which implies $\omega = \omega'$. Hence ω is a constant path.

Conversely, assume that $p: E \rightarrow B$ is a fibration such that every fiber has no nontrivial path and let ω and ω' be paths in E such that $p \circ \omega = p \circ \omega'$ and $\omega(0) = \omega'(0)$. For $t \in I$, let ω'_t be the path in E defined by

$$\omega'_t(t') = \begin{cases} \omega((1 - 2t')t) & 0 \leq t' \leq \frac{1}{2} \\ \omega'((2t' - 1)t) & \frac{1}{2} \leq t' \leq 1 \end{cases}$$

Then ω'_t is a path in E from $\omega(t)$ to $\omega'(t)$, and $p \circ \omega'_t$ is a closed path in B that is homotopic relative to $\bar{I}$ to the constant path at $p\omega(t)$. By the homotopy lifting property of p , there is a map $F: I \times I \rightarrow E$ such that $F(t', 0) = \omega'_t(t')$ and F maps $0 \times I \cup I \times 1 \cup 1 \times I$ to the fiber $p^{-1}(p\omega(t))$. Because $p^{-1}(p\omega(t))$ has no nonconstant paths, F maps $0 \times I, I \times 1$, and $1 \times I$ to a single point. It follows that $F(0, 0) = F(1, 0)$. Therefore $\omega'_t(0) = \omega'_t(1)$ and $\omega(t) = \omega'(t)$. ■

We have seen that a covering projection is a fibration with unique path lifting. It will be shown in Sec. 2.4 that if the base space satisfies some mild hypotheses, any fibration with unique path lifting is a covering projection. One reason for studying fibrations with unique path lifting as generalized covering projections is that the following two theorems are easily proved, but both are false for covering projections.

6 THEOREM *The composite of fibrations (with unique path lifting) is a fibration (with unique path lifting). ■*

7 THEOREM *The product of fibrations (with unique path lifting) is a fibration (with unique path lifting). ■*

An example shows that theorem 6 is false for covering projections.

8 EXAMPLE Let X and X_n , for $n \geq 1$, be a countable product of 1-spheres. Let $\tilde{X}_n = \mathbb{R}^n \times X$ and define $p_n: \tilde{X}_n \rightarrow X_n$ by

$$p_n(t_1, \dots, t_n, z_1, z_2, \dots) = (ex(t_1), \dots, ex(t_n), z_1, z_2, \dots)$$

Then p_n is a covering projection for $n \geq 1$. It follows from theorem 2.1.11 that $\bigvee p_n: \bigvee \tilde{X}_n \rightarrow \bigvee X_n$ is a covering projection. Since $\bigvee X_n$ is the product of X and the set of positive integers, there is a covering projection $\bigvee X_n \rightarrow X$ (see example 2.1.2). The composite

$$\bigvee \tilde{X}_n \rightarrow \bigvee X_n \rightarrow X$$

is not a covering projection (cf. example 2.1.15).

Similarly, theorem 7 is false for covering projections.

9 EXAMPLE For $n \geq 1$, let $p_n: \tilde{X}_n \rightarrow X_n$ be the covering projection $ex: \mathbb{R} \rightarrow S^1$. Then

$$\times p_n: \times \tilde{X}_n \rightarrow \times X_n$$

is not a covering projection.

It follows from theorem 6 that there is a category whose objects are topological spaces and whose morphisms are fibrations with unique path

lifting. From theorem 7, this category has products. It is also easy to verify that it has sums. We shall now describe a category, depending on a given base space, which is of more use in studying covering projections or fibrations. For a given space X there is a category whose objects are maps $p: \tilde{X} \rightarrow X$, which are fibrations with unique path lifting, and whose morphisms are commutative triangles

$$\begin{array}{ccc} X_1 & \xrightarrow{f} & X_2 \\ p_1 \searrow & & \swarrow p_2 \\ & X & \end{array}$$

If $p: \tilde{X}_j \rightarrow X$ is an indexed family of objects in this category, let $p: \bigvee \tilde{X}_j \rightarrow X$ be the map such that $p|_{\tilde{X}_j} = p_j$. Then p is also an object in the category and is the sum of the collection $\{p_j\}$ in the category.

To show that this category also has products, given maps $p_j: \tilde{X}_j \rightarrow X$, let $\tilde{X} = \{(x_j) \in \times \tilde{X}_j \mid p_j(x_j) = p_j(x_j') \text{ for all } j, j'\}$

and define $p: \tilde{X} \rightarrow X$ by $p(x_j) = p_j(x_j)$. If each p_j is a fibration, so is p , and if each p_j has unique path lifting, so does p . Hence p is a product of $\{p_j\}$ in the category of fibrations with unique path lifting. This map p is called the *fibred product* of the maps $\{p_j\}$. We consider it in more detail in Sec. 2.8.

There is a similar category whose objects are covering projections with base space X and whose morphisms are commutative triangles. This category has finite sums and finite products, but neither arbitrary sums nor arbitrary products. In fact, for each n let

$$p_n: \mathbb{R}^n \times S^1 \times S^1 \times \dots \rightarrow S^1 \times S^1 \times \dots$$

be defined by $p_n(t_1, \dots, t_n, z_1, z_2, \dots) = (e^{2\pi i t_1}, \dots, e^{2\pi i t_n}, z_1, z_2, \dots)$, as in example 8. Then the collection $\{p_n\}$ has neither a sum nor a product in the category of covering projections with base space X .

3 RELATIONS WITH THE FUNDAMENTAL GROUP

In a fibration with unique path lifting the fundamental group of the total space is isomorphic to a subgroup of the fundamental group of the base space. The corresponding subgroup of the fundamental group will lead to a classification of fibrations with unique path lifting. In fact, we shall see in the next section that the fundamental group functor solves the lifting problem for fibrations with unique path lifting. The present section is devoted to consideration of the relation between the fundamental groups of the total space and the base space of a fibration with unique path lifting.

We begin with a localization property for fibrations which is an analogue of theorem 2.1.14.

LEMMA Let $p: E \rightarrow B$ be a fibration. If A is any path component of E , then pA is a path component of B and $p|_A: A \rightarrow pA$ is a fibration.

PROOF Since pA is the continuous image of a path-connected space, it is path connected. It is a maximal path-connected subset of B , for if ω is a path in B that begins in pA , there is a lifting $\tilde{\omega}$ of ω that begins in A . Since A is a path component of E , $\tilde{\omega}$ is a path in A . Therefore $\omega = p \circ \tilde{\omega}$ is a path in pA . Hence pA is a maximal path-connected subset of B and, by theorem 1.7.9, a path component of B .

To show that $p|_A: A \rightarrow pA$ has the homotopy lifting property, let $f': Y \rightarrow A$ and $F: Y \times I \rightarrow pA$ be maps such that $F(y, 0) = pf'(y)$. Because p is a fibration, there is a map $F': Y \times I \rightarrow E$ such that $p \circ F' = F$ and $F'(y, 0) = f'(y)$. For any $y \in Y$, F' must map $y \times I$ into the path component of E containing $F'(y, 0)$. Therefore $F'(y \times I) \subset A$ for all y , and $F': Y \times I \rightarrow A$ is a lifting of F such that $F'(y, 0) = f'(y)$. ■

For locally path-connected spaces we have the following analogue of theorem 2.1.20, which reduces the study of fibrations to the study of fibrations with total space and base space path connected.

THEOREM Let $p: E \rightarrow B$ be a map. If E is locally path connected, p is a fibration if and only if for each path component A of E , pA is a path component of B and $p|_A: A \rightarrow pA$ is a fibration.

PROOF If $p: E \rightarrow B$ is a fibration and A is a path component of E , it follows from lemma 1 that pA is a path component of B and $p|_A: A \rightarrow pA$ is a fibration.

To prove the converse, let $f': Y \rightarrow E$ and $F: Y \times I \rightarrow B$ be such that $F(y, 0) = f'(y)$. Let $\{A_j\}$ be the path components of E . Then $\{A_j\}$ are disjoint open subsets of E . Let $V_j = f'^{-1}(A_j)$. The collection $\{V_j\}$ is a disjoint open covering of Y . Therefore, to construct a map $F': Y \times I \rightarrow E$ such that $p \circ F' = F$ and $F'(y, 0) = f'(y)$, it suffices to construct maps $F'_j: V_j \times I \rightarrow E$ for all j such that $p \circ F'_j = F|_{V_j \times I}$ and $F'_j(y, 0) = f'(y, 0)$.

Because $F(y \times I)$ is contained in the path component of B containing $F(y, 0) = pf'(y)$, it follows from the fact that pA_j is a path component of B that $F(V_j \times I) \subset pA_j$ for all j . Because $p|_{A_j}: A_j \rightarrow pA_j$ is a fibration, there is a map $F'_j: V_j \times I \rightarrow A_j$ such that $pF'_j = F|_{V_j \times I}$ and $F'_j(y, 0) = f'(y)$ for $y \in V_j$. Therefore p has the homotopy lifting property. ■

Since every path in a topological space lies in some path component of the space, it is clear that theorem 2 remains valid if the term "fibration" is replaced throughout by "fibration with unique path lifting."

The main result on fibrations with unique path lifting is embodied in the following statement.

LEMMA Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. If ω and ω' are paths in $\tilde{X}$ such that $\omega(0) = \omega'(0)$ and $p \circ \omega \simeq p \circ \omega'$, then $\omega \simeq \omega'$.

PROOF Let $F: I \times I \rightarrow X$ be a homotopy relative to $\tilde{I}$ from $p \circ \omega$ to $p \circ \omega'$ [that is, $F(t, 0) = p\omega(t)$ and $F(t, 1) = p\omega'(t)$, and $F(0, t) = p\omega(0)$ and $F(1, t) = p\omega(1)$]. By the homotopy lifting property of fibrations, there is a map $F': I \times I \rightarrow \tilde{X}$ such that $F'(t, 0) = \omega(t)$ and $p \circ F' = F$. Then $F'(0 \times I)$ and $F'(1 \times I)$ are contained in $p^{-1}(p\omega(0))$ and $p^{-1}(p\omega(1))$, respectively. By theorem 2.2.5, $F'(0 \times I)$ and $F'(1 \times I)$ are single points. Hence F' is a homotopy relative to $\tilde{I}$ from ω to some path ω'' such that $\omega''(0) = \omega(0)$ and $p \circ \omega'' = p \circ \omega'$. Since $\omega''(0) = \omega(0)$, it follows from the unique-path-lifting property of p that $\omega' = \omega''$ and $F': \omega \simeq \omega'$ rel $\tilde{I}$. ■

It follows from lemma 3 that if $p: \tilde{X} \rightarrow X$ is a fibration with unique path lifting, then for any two objects $\tilde{x}_0$ and $\tilde{x}_1$ in the fundamental groupoid of $\tilde{X}$, $p_{\#}$ maps $\text{hom}(\tilde{x}_0, \tilde{x}_1)$ injectively into $\text{hom}(p(\tilde{x}_0), p(\tilde{x}_1))$. In particular, if $\tilde{x}_0 = \tilde{x}_1$, we obtain the following theorem.

THEOREM Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. For any $\tilde{x}_0 \in \tilde{X}$ the homomorphism

$$p_{\#}: \pi(\tilde{X}, \tilde{x}_0) \rightarrow \pi(X, x_0)$$

is a monomorphism. ■

This last result provides the basis for the reduction of problems concerning fibrations with unique path lifting to problems about the fundamental group. In order that the fundamental group be really representative of the space in question, we assume that the spaces involved are path connected. It follows from theorem 2 that this is no loss of generality for locally path-connected spaces.

LEMMA Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting and assume that $\tilde{X}$ is a nonempty path-connected space. If $\tilde{x}_0, \tilde{x}_1 \in \tilde{X}$, there is a path ω in X from $p(\tilde{x}_0)$ to $p(\tilde{x}_1)$ such that

$$p_{\#}\pi(\tilde{X}, \tilde{x}_0) = h_{(p, \omega)}p_{\#}\pi(\tilde{X}, \tilde{x}_1)$$

Conversely, given a path ω in X from $p(\tilde{x}_0)$ to x_1 , there is a point $\tilde{x}_1 \in p^{-1}(x_1)$ such that

$$h_{(p, \omega)}p_{\#}\pi(\tilde{X}, \tilde{x}_1) = p_{\#}\pi(\tilde{X}, \tilde{x}_0)$$

PROOF For the first part, let $\tilde{\omega}$ be a path in $\tilde{X}$ from $\tilde{x}_0$ to $\tilde{x}_1$. Then $\pi(\tilde{X}, \tilde{x}_0) = h_{(p, \tilde{\omega})}\pi(\tilde{X}, \tilde{x}_1)$. Therefore

$$p_{\#}\pi(\tilde{X}, \tilde{x}_0) = h_{(p, \tilde{\omega})}p_{\#}\pi(\tilde{X}, \tilde{x}_1)$$

and so $p \circ \tilde{\omega}$ will do as the path from $p(\tilde{x}_0)$ to $p(\tilde{x}_1)$.

Conversely, given a path ω in X from $p(\tilde{x}_0)$ to x_1 , let $\tilde{\omega}$ be a path in $\tilde{X}$ such that $\tilde{\omega}(0) = \tilde{x}_0$ and $p\tilde{\omega} = \omega$. If $\tilde{x}_1 = \tilde{\omega}(1)$, then

$$h_{(p, \tilde{\omega})}p_{\#}\pi(\tilde{X}, \tilde{x}_1) = p_{\#}(h_{(p, \tilde{\omega})}\pi(\tilde{X}, \tilde{x}_1)) = p_{\#}\pi(\tilde{X}, \tilde{x}_0)$$

This easily implies the following result.

6 THEOREM Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting and assume that $\tilde{X}$ is a nonempty path-connected space. For $x_0 \in p\tilde{X}$ the collection $\{p_\# \pi(\tilde{X}, \tilde{x}_0) \mid \tilde{x}_0 \in p^{-1}(x_0)\}$ is a conjugacy class in $\pi(X, x_0)$. If ω is a path in $p\tilde{X}$ from x_0 to x_1 , then $h_{[\omega]}$ maps the conjugacy class in $\pi(X, x_1)$ to the conjugacy class in $\pi(X, x_0)$. ■

Let $p: \tilde{X} \rightarrow X$ be a fibration and let ω be a path in X beginning at x_0 . Define a map $F_\omega: p^{-1}(x_0) \times I \rightarrow X$ by $F_\omega(\tilde{x}, t) = \omega(t)$ and let $i: p^{-1}(x_0) \subset \tilde{X}$. Then $p(i\tilde{x}) = F_\omega(\tilde{x}, 0)$ for $\tilde{x} \in p^{-1}(\tilde{x}_0)$. It follows from the homotopy lifting property of p that there exists a map $G_\omega: p^{-1}(x_0) \times I \rightarrow \tilde{X}$ such that $G_\omega(\tilde{x}, 0) = i(\tilde{x}) = \tilde{x}$ and $p \circ G_\omega = F_\omega$.

Suppose now that p has unique path lifting. We prove that the map $\tilde{x} \rightarrow G_\omega(\tilde{x}, 1)$ of $p^{-1}(x_0)$ to $p^{-1}(\omega(1))$ depends only on the path class of ω . If $\omega' \simeq \omega$ and $G'_\omega: p^{-1}(x_0) \times I \rightarrow \tilde{X}$ is a map such that $G'_\omega(\tilde{x}, 0) = \tilde{x}$ and $p \circ G'_\omega = F_\omega$, then for any $\tilde{x} \in p^{-1}(x_0)$, let $\tilde{\omega}$ and $\tilde{\omega}'$ be the paths in $\tilde{X}$ defined by $\tilde{\omega}(t) = G_\omega(\tilde{x}, t)$ and $\tilde{\omega}'(t) = G'_\omega(\tilde{x}, t)$. Then $\tilde{\omega}$ and $\tilde{\omega}'$ begin at $\tilde{x}$ and

$$p \circ \tilde{\omega} = \omega \simeq \omega' = p \circ \tilde{\omega}'$$

It follows from lemma 3 that $\tilde{\omega} \simeq \tilde{\omega}'$. Then $G_\omega(\tilde{x}, 1) = G'_\omega(\tilde{x}, 1)$ for every $\tilde{x} \in p^{-1}(x_0)$. Therefore there is a well-defined continuous map

$$f_{[\omega]}: p^{-1}(\omega(0)) \rightarrow p^{-1}(\omega(1))$$

defined by $f_{[\omega]}(\tilde{x}) = G_\omega(\tilde{x}, 1)$, where G_ω is as above. It is clear that if $\omega(1) = \omega'(0)$, then $f_{[\omega]} \circ f_{[\omega']} = f_{[\omega]}$.

7 THEOREM Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. There is a contravariant functor from the fundamental groupoid of X to the category of topological spaces and maps which assigns to $x \in X$ the fiber over x and to $[\omega]$ the function $f_{[\omega]}$. ■

The fact that $f_{[\omega]}$ is a homeomorphism for every $[\omega]$ leads to the following corollary.

8 COROLLARY If $p: \tilde{X} \rightarrow X$ is a fibration with unique path lifting and X is path connected, then any two fibers are homeomorphic. ■

If X is path connected and $p: \tilde{X} \rightarrow X$ is a fibration with unique path lifting, the number of sheets of p (or the multiplicity of p) is defined to be the cardinal number of $p^{-1}(x)$ (which is independent of $x \in X$, by corollary 8). For a path-connected total space, the multiplicity is determined by the conjugacy class as follows.

9 THEOREM Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting and assume $\tilde{X}$ and X to be nonempty path-connected spaces. If $\tilde{x}_0 \in \tilde{X}$, the multiplicity of p is the index of $p_\# \pi(\tilde{X}, \tilde{x}_0)$ in $\pi(X, p(\tilde{x}_0))$.

PROOF By theorem 7, $\pi(X, p(\tilde{x}_0))$ acts as a group of transformations on the right on $p^{-1}(p(\tilde{x}_0))$ by $\tilde{x} \circ [\omega] = f_{[\omega]}(\tilde{x})$ for $\tilde{x} \in p^{-1}(p(\tilde{x}_0))$. If $\tilde{x}_1, \tilde{x}_2 \in p^{-1}(p(\tilde{x}_0))$, let $\tilde{\omega}$ be a path in $\tilde{X}$ from $\tilde{x}_1$ to $\tilde{x}_2$. Then $[p \circ \tilde{\omega}] \in \pi(X, p(\tilde{x}_0))$ and $\tilde{x}_1 \circ [p\tilde{\omega}] = \tilde{x}_2$.

Therefore $\pi(X, p(\tilde{x}_0))$ acts transitively on $p^{-1}(p(\tilde{x}_0))$. The isotropy group of $\tilde{x}_0$ [that is, the subgroup of $\pi(X, p(\tilde{x}_0))$ leaving $\tilde{x}_0$ fixed] is clearly equal to $p_\# \pi(\tilde{X}, \tilde{x}_0)$. From general considerations¹ there is a bijection between the set of right cosets of $p_\# \pi(\tilde{X}, \tilde{x}_0)$ in $\pi(X, p(\tilde{x}_0))$ and $p^{-1}(p(\tilde{x}_0))$. ■

10 EXAMPLE For $n \geq 2$ the covering $p: S^n \rightarrow P^n$ of example 2.1.5 has multiplicity 2. Because S^n is simply connected, $\pi(P^n) \simeq \mathbb{Z}_2$ for $n \geq 2$.

A fibration $p: \tilde{X} \rightarrow X$ with unique path lifting is said to be *regular* if, given any closed path ω in X , either every lifting of ω is closed or none is closed.

11 THEOREM Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting, p is regular if and only if $p_\# \pi(\tilde{X}, \tilde{x}_0) = p_\# \pi(\tilde{X}, \tilde{x}_1)$ whenever $p(\tilde{x}_0) = p(\tilde{x}_1)$.

PROOF Assume that p is regular and let $\tilde{\omega}$ be a closed path in $\tilde{X}$ at $\tilde{x}_0$. Then $\tilde{\omega}$ is a closed lifting of $p\tilde{\omega}$. Therefore there is a closed lifting $\tilde{\omega}_1$ of $p\tilde{\omega}$ at $\tilde{x}_1$. It follows that $p_\#[\tilde{\omega}] = [p\tilde{\omega}] = [p\tilde{\omega}_1] = p_\#[\tilde{\omega}_1]$. Therefore $p_\# \pi(\tilde{X}, \tilde{x}_0) \subset p_\# \pi(\tilde{X}, \tilde{x}_1)$. Since the roles of $\tilde{x}_0$ and $\tilde{x}_1$ can be interchanged, it follows that $p_\# \pi(\tilde{X}, \tilde{x}_0) = p_\# \pi(\tilde{X}, \tilde{x}_1)$.

Conversely, if $p_\# \pi(\tilde{X}, \tilde{x}_0) = p_\# \pi(\tilde{X}, \tilde{x}_1)$ whenever $p(\tilde{x}_0) = p(\tilde{x}_1)$, let ω be a closed path in X at $p(\tilde{x}_0)$ having a closed lifting $\tilde{\omega}$ at $\tilde{x}_0$. Then

$$[\omega] = p_\#[\tilde{\omega}] \in p_\# \pi(\tilde{X}, \tilde{x}_0) = p_\# \pi(\tilde{X}, \tilde{x}_1)$$

Therefore there is a closed path $\tilde{\omega}_1$ in $\tilde{X}$ at $\tilde{x}_1$ such that $p\tilde{\omega}_1 \simeq \omega$. If $\tilde{\omega}_1$ is a lifting of ω such that $\tilde{\omega}_1(0) = \tilde{x}_1$, then by the unique-path-lifting property of p , $\tilde{\omega}_1 = \tilde{\omega}$. Therefore $\tilde{\omega}_1$ is a closed lifting of ω at $\tilde{x}_1$ and p is regular. ■

In case $\tilde{X}$ is a nonempty path-connected space, theorems 6 and 11 give the following result.

12 THEOREM Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting and assume that $\tilde{X}$ is a nonempty path-connected space. Then p is regular if and only if for some $\tilde{x}_0 \in \tilde{X}$, $p_\# \pi(\tilde{X}, \tilde{x}_0)$ is a normal subgroup of $\pi(X, p(\tilde{x}_0))$. ■

4 THE LIFTING PROBLEM

In this section we show that the fundamental group functor solves the lifting problem for fibrations with unique path lifting. As a consequence of this, the fundamental group functor provides a classification of covering projections, which is discussed in the next section.

Our first result is that any map of a contractible space to the base space of a fibration can be lifted.

1 LEMMA Let $p: E \rightarrow B$ be a fibration. Any map of a contractible space to B whose image is contained in $p(E)$ can be lifted to E .

¹ Whenever a group G acts transitively on the right on a set S there is induced a bijection between the set of right cosets of the isotropy group (of any $s \in S$) in G and the set S .

PROOF Let Y be contractible and let $f: Y \rightarrow B$ be a map such that $f(Y) \subset p(E)$. Because Y is contractible, f is homotopic to a constant map of Y to some point of $f(Y)$. $f(Y) \subset p(E)$, so this constant map can be lifted to E . The homotopy lifting property then implies that f can be lifted to E . ■

Because we use the fundamental group functor, it will prove technically simpler to consider the lifting problem for spaces with base points.

2 LEMMA Let $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a fibration with unique path lifting. If y_0 is a strong deformation retract of Y , any map $(Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ can be lifted to a map $(Y, y_0) \rightarrow (X, x_0)$.

PROOF Let $f: (Y, y_0) \rightarrow (X, x_0)$ be a map. f is homotopic relative to y_0 to the constant map $Y \rightarrow x_0$. The constant map can be lifted to the constant map $Y \rightarrow \tilde{x}_0$. By the homotopy lifting property, f can be lifted to a map $f': Y \rightarrow \tilde{X}$ such that f' is homotopic to the constant map $Y \rightarrow \tilde{x}_0$ by a homotopy which maps $y_0 \times I$ to $p^{-1}(x_0)$. Because $p^{-1}(x_0)$ has no nonconstant path by theorem 2.2.5, $f'(y_0) = \tilde{x}_0$. ■

We shall apply lemma 2 to a contractible space in order to lift certain quotient spaces of the contractible space. The usual way to represent a space as the quotient space of a contractible space is to show it is a quotient space of its path space. Given $y_0 \in Y$, the path space $P(Y, y_0)$ is the space of continuous maps $\omega: (I, 0) \rightarrow (Y, y_0)$ topologized by the compact-open topology. There is a function $q: P(Y, y_0) \rightarrow Y$ defined by $q(\omega) = \omega(1)$. If U is an open set in Y ,

$$q^{-1}(U) = \langle 1; U \rangle = \{ \omega \in P(Y, y_0) \mid \omega(1) \in U \}$$

is an open set in $P(Y, y_0)$. Therefore q is continuous.

3 LEMMA The constant path at y_0 is a strong deformation retract of the path space $P(Y, y_0)$.

PROOF A strong deformation retraction $F: P(Y, y_0) \times I \rightarrow P(Y, y_0)$ to the constant path at y_0 is defined by

$$F(\omega, t)(t') = \omega((1 - t)t') \quad \omega \in P(Y, y_0); t, t' \in I$$

We have shown that q is a continuous map of the contractible path space $P(Y, y_0)$ to Y . If Y is path connected, q is clearly surjective. If Y is also locally path connected, the following theorem shows that q is a quotient projection.

4 THEOREM A connected locally path-connected space Y is the quotient space of its path space $P(Y, y_0)$ by the map q .

PROOF We know that q is continuous, and because a connected locally path-connected space is path connected, it is surjective. To complete the proof it suffices to show that q is an open map. Let $\omega \in P(Y, y_0)$ and let $W = \cap_{1 \leq i \leq n} \langle K_i, U_i \rangle$ be a neighborhood of ω , where K_i is compact in I and U_i is open in Y . We enumerate the K_i 's so that for some $0 \leq k \leq n$, $1 \in K_1 \cap \dots \cap K_k$ and

$1 \notin K_{k+1} \cup \dots \cup K_n$. Because $\omega(1) \in U_1 \cap \dots \cap U_k$, there is a path-connected neighborhood V of $\omega(1)$ contained in $U_1 \cap \dots \cap U_k$. Choose $0 < t' < 1$ such that $\{t', 1\} \cap (K_{k+1} \cup \dots \cup K_n) = \emptyset$ and $\omega(\{t', 1\}) \subset V$.

To prove that $q(W) \supset V$, which completes the proof, let $y' \in V$ and let ω' be a path in V from $\omega(t')$ to y' . Define $\tilde{\omega}: I \rightarrow Y$ by

$$\tilde{\omega}(t) = \begin{cases} \omega(t) & 0 \leq t \leq t' \\ \omega' \left(\frac{t - t'}{1 - t'} \right) & t' \leq t \leq 1 \end{cases}$$

For $i > k$, $\tilde{\omega}(K_i) = \omega(K_i) \subset U_i$. For $i \leq k$,

$$\tilde{\omega}(K_i) = \tilde{\omega}(K_i \cap [0, t']) \cup \tilde{\omega}(K_i \cap [t', 1]) \subset \omega(K_i) \cup \omega'(I) \subset U_i \cup V = U_i$$

Therefore $\tilde{\omega} \in W$ and $q(\tilde{\omega}) = y'$. Hence $q(W) \supset V$. ■

We can put these results together to obtain the following result, called the lifting theorem.

5 THEOREM Let $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a fibration with unique path lifting. Let Y be a connected locally path-connected space. A necessary and sufficient condition that a map $f: (Y, y_0) \rightarrow (X, x_0)$ have a lifting $(Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ is that in $\pi(X, x_0)$

$$f_{\#}\pi(Y, y_0) \subset p_{\#}\pi(\tilde{X}, \tilde{x}_0)$$

PROOF If $f': (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ is a lifting of f , then $f = p \circ f'$ and

$$f_{\#}\pi(Y, y_0) = p_{\#}f'_{\#}\pi(Y, y_0) \subset p_{\#}\pi(\tilde{X}, \tilde{x}_0)$$

which shows that the condition is necessary.

We now prove that the condition is sufficient. It follows from lemmas 3 and 2 that if ω_0 is the constant path at y_0 , the composite

$$(P(Y, y_0), \omega_0) \xrightarrow{q} (Y, y_0) \xrightarrow{f} (X, x_0)$$

can be lifted to a map $\tilde{f}: (P(Y, y_0), \omega_0) \rightarrow (\tilde{X}, \tilde{x}_0)$. We show that if $f_{\#}\pi(Y, y_0) \subset p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ and if $\omega, \omega' \in P(Y, y_0)$ are such that $q(\omega) = q(\omega')$, then $\tilde{f}(\omega) = \tilde{f}(\omega')$. Let $\tilde{\omega}$ and $\tilde{\omega}'$ be the paths in $P(Y, y_0)$ from ω_0 to ω and ω' , respectively, defined by $\tilde{\omega}(t)(t') = \omega(t)(t')$ and $\tilde{\omega}'(t)(t') = \omega'(t)(t')$. Then $\tilde{f} \circ \tilde{\omega}$ and $\tilde{f} \circ \tilde{\omega}'$ are paths in $\tilde{X}$ from $\tilde{x}_0$ to $\tilde{f}(\omega)$ and $\tilde{f}(\omega')$, respectively, such that

$$p \circ \tilde{f} \circ \tilde{\omega} = f \circ q \circ \tilde{\omega} = f \circ \omega \quad \text{and} \quad p \circ \tilde{f} \circ \tilde{\omega}' = f \circ \omega'$$

Because $\omega \circ \omega'^{-1}$ is a closed path in Y at y_0 and $f_{\#}\pi(Y, y_0) \subset p_{\#}\pi(\tilde{X}, \tilde{x}_0)$, there is a closed path $\tilde{\omega}$ in $\tilde{X}$ at $\tilde{x}_0$ such that $(f \circ \omega) \circ (f \circ \omega')^{-1} \simeq p \circ \tilde{\omega}$. Then

$$p \circ (\tilde{f} \circ \tilde{\omega}) = f \circ \omega \simeq (p \circ \tilde{\omega}) \circ (f \circ \omega') = p \circ (\tilde{\omega} \circ (\tilde{f} \circ \tilde{\omega}'))$$

By lemma 2.3.3, $\tilde{f} \circ \tilde{\omega} \simeq \tilde{\omega} \circ (\tilde{f} \circ \tilde{\omega}')$. In particular, the endpoint of $\tilde{f} \circ \tilde{\omega}$, which is $\tilde{f}(\omega)$, equals the endpoint of $\tilde{f} \circ \tilde{\omega}'$, which is $\tilde{f}(\omega')$.

It follows that there is a function $f': (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ such that $f' \circ q = \tilde{f}$,

and using theorem 4, we see that f' is continuous. Because

$$p \circ f' \circ \varphi = p \circ \tilde{f} = f \circ \varphi$$

and φ is surjective, $p \circ f' = f$. Therefore f' is a lifting of f . ■

Let $p: E \rightarrow B$ be a fibration. A section of p is a map $s: B \rightarrow E$ such that $p \circ s = 1_B$ (thus a section is a right inverse of p). It follows easily from the homotopy lifting property that there is a section of p if and only if $[p]$ has a right inverse in the homotopy category. Because a section is a lifting of the identity map $B \subset B$, the following is an immediate consequence of theorem 5.

6 COROLLARY Let $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a fibration with unique path lifting. If $\tilde{X}$ is a connected locally path-connected space, there is a section $(X, x_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ of p if and only if $p_{\#}\pi(\tilde{X}, \tilde{x}_0) = \pi(X, x_0)$. ■

7 COROLLARY Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. If $\tilde{X}$ is a nonempty path-connected space and X is connected and locally path connected, then p is a homeomorphism if and only if for some $\tilde{x}_0 \in \tilde{X}$, $p_{\#}\pi(\tilde{X}, \tilde{x}_0) = \pi(X, p(\tilde{x}_0))$.

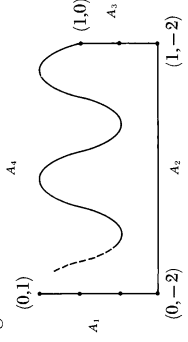
PROOF If p is a homeomorphism, $p_{\#}\pi(\tilde{X}, \tilde{x}_0) = \pi(X, p(\tilde{x}_0))$. Conversely, if $p_{\#}\pi(\tilde{X}, \tilde{x}_0) = \pi(X, p(\tilde{x}_0))$, then by theorem 2.3.9, p is a bijection. By corollary 6, it has a continuous right inverse. Therefore p is a homeomorphism. ■

If $p: \tilde{X} \rightarrow X$ is a covering projection and $\tilde{X}$ is path connected, a necessary and sufficient condition that p be a homeomorphism is that $p_{\#}\pi(\tilde{X}, \tilde{x}_0) = \pi(X, p(\tilde{x}_0))$ for some $\tilde{x}_0 \in \tilde{X}$. This condition on the fundamental groups implies that p is a bijection, and by lemmas 2.1.8 and 2.1.7, p is open; hence for covering projections corollary 7 is valid without the assumption that X be locally path connected. This is definitely false for fibrations with unique path lifting if X is not locally path connected, because p need not be open. The following example shows this.

8 EXAMPLE Let X be the subspace of $\mathbf{R}^2$ defined to be the union of the four sets

$$\begin{aligned} A_1 &= \{(x, y) \mid x = 0, -2 \leq y \leq 1\} \\ A_2 &= \{(x, y) \mid 0 \leq x \leq 1, y = -2\} \\ A_3 &= \{(x, y) \mid x = 1, -2 \leq y \leq 0\} \\ A_4 &= \{(x, y) \mid 0 < x \leq 1, y = \sin 2\pi/x\} \end{aligned}$$

illustrated in the diagram



Let $\tilde{X}$ be the half-open interval $[0, 4)$ and define $p: \tilde{X} \rightarrow X$ to map $[0, 1]$ linearly onto A_1 , $[1, 2]$ linearly onto A_2 , $[2, 3]$ linearly onto A_3 , and $[3, 4)$ homeomorphically onto A_4 by the map $t \rightarrow (t - 3, \sin(2\pi/(t - 3)))$. Then $\tilde{X}$ and X are path connected and $p: \tilde{X} \rightarrow X$ is a fibration with unique path lifting. However, p is not a homeomorphism, although $\tilde{X}$ and X are both simply connected.

For locally path-connected spaces the lifting theorem provides the following criterion for determining whether an open path-connected subset of the base space is evenly covered by a fibration.

9 LEMMA Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. Assume that $\tilde{X}$ and X are locally path connected and let U be an open connected subset of X . Then U is evenly covered by p if and only if every lifting to $\tilde{X}$ of a closed path in U is a closed path.

PROOF If U is evenly covered by p and $\tilde{\omega}$ is a path in $p^{-1}(U)$, then $\tilde{\omega}$ is a path in some component U of $p^{-1}(U)$. By lemma 2.1.12, p maps U homeomorphically onto U . Therefore, if $p \circ \tilde{\omega}$ is a closed path in U , $\tilde{\omega}$ is a closed path in U . Hence the condition is necessary.

It is also sufficient, because if $x_0 \in U$ and $\tilde{x}_0 \in p^{-1}(x_0)$, the hypothesis that every lifting of a closed path in U at x_0 is a closed path in $\tilde{X}$ implies that in $\pi(X, x_0)$

$$i_{\#}\pi(U, x_0) \subset p_{\#}\pi(\tilde{X}, \tilde{x}_0) \quad \text{where } i: (U, x_0) \subset (X, x_0)$$

By theorem 5, there is a lifting $i_{\tilde{x}_0}^*: (U, x_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ of i . The collection $\{i_{\tilde{x}_0}^*(U) \mid \tilde{x}_0 \in p^{-1}(x_0)\}$ consists of path-connected sets which, by lemma 2.2.4, are disjoint. We show that their union equals $p^{-1}(U)$. If $\tilde{x} \in p^{-1}(U)$, let ω be a path in U from $p(\tilde{x})$ to x_0 and let $\tilde{\omega}$ be a lifting of ω such that $\tilde{\omega}(0) = \tilde{x}$. Then $\tilde{\omega}(1) \in p^{-1}(x_0)$, and therefore $\tilde{\omega}$ is a path in $i_{\tilde{x}_0}^*(U)$. Hence $\tilde{x} \in i_{\tilde{x}_0}^*(U)$ and $\{i_{\tilde{x}_0}^*(U) \mid \tilde{x}_0 \in p^{-1}(x_0)\}$ is a partition of $p^{-1}(U)$ into path-connected sets. Since $p^{-1}(U)$ is open and $\tilde{X}$ is locally path connected, $i_{\tilde{x}_0}^*(U)$ is open in $\tilde{X}$ for each $\tilde{x}_0 \in p^{-1}(x_0)$. Clearly, p is a homeomorphism of $i_{\tilde{x}_0}^*(U)$ onto U for each $\tilde{x}_0 \in p^{-1}(x_0)$, and U is evenly covered by p . ■

A space X is said to be *semilocally 1-connected* if every point $x_0 \in X$ has a neighborhood N such that $\pi(N, x_0) \rightarrow \pi(X, x_0)$ is trivial.

10 THEOREM Every fibration with unique path lifting whose base space is locally path connected and semilocally 1-connected and whose total space is locally path connected is a covering projection.

PROOF It follows from lemma 9 and the definition of semilocally 1-connected space that each point of the base space has an open neighborhood evenly covered by the fibration. ■

5 THE CLASSIFICATION OF COVERING PROJECTIONS

This section contains a classification of covering projections over a connected locally path-connected base space. It is based on the lifting theorem and reduces the problem of equivalence of covering projections to conjugacy of their corresponding subgroups of the fundamental group of the base space. A large part of the section is devoted to constructing a covering projection corresponding to a given subgroup of the fundamental group of the base space.

Let X be a connected space. The category of connected covering spaces of X has objects which are covering projections $p: \tilde{X} \rightarrow X$, where $\tilde{X}$ is connected, and morphisms which are commutative triangles

$$\begin{array}{ccc} \tilde{X}_1 & \xrightarrow{f} & \tilde{X}_2 \\ p_1 \searrow & & \swarrow p_2 \\ & X & \end{array}$$

If X is locally path connected and $p: \tilde{X} \rightarrow X$ is an object of this category, then, by lemma 2.1.8, p is a local homeomorphism and $\tilde{X}$ is also locally path connected. We show that in this case every morphism in this category is a covering projection.

1 LEMMA *In the category of connected covering spaces of a connected locally path-connected space every morphism is itself a covering projection.*

PROOF Consider a commutative triangle

$$\begin{array}{ccc} \tilde{X}_1 & \xrightarrow{f} & \tilde{X}_2 \\ p_1 \searrow & & \swarrow p_2 \\ & X & \end{array}$$

where p_1 and p_2 are covering projections and X is locally path connected. It follows from corollary 2.1.13 that f is a covering projection if it is surjective.

Because $\tilde{X}_2$ is connected and locally path connected, it is path connected. Let $\tilde{x}_1 \in \tilde{X}_1$ and $\tilde{x}_2 \in \tilde{X}_2$ be arbitrary and let $\tilde{\omega}_2$ be a path in $\tilde{X}_2$ from $f(\tilde{x}_1)$ to $\tilde{x}_2$. Because p_1 is a fibration, there is a path $\tilde{\omega}_1$ in $\tilde{X}_1$ beginning at $\tilde{x}_1$ such that $p_1 \circ \tilde{\omega}_1 = p_2 \circ \tilde{\omega}_2$. By the unique path lifting of $p_2, f \circ \tilde{\omega}_1 = \tilde{\omega}_2$. Therefore

$$f(\tilde{\omega}_1(1)) = \tilde{\omega}_2(1) = \tilde{x}_2$$

proving that f is surjective. ■

The next result determines when there is a morphism from one object to another in the category of connected covering spaces of X .

2 THEOREM *Let $p_1: \tilde{X}_1 \rightarrow X$ and $p_2: \tilde{X}_2 \rightarrow X$ be objects in the category*

of connected covering spaces of a connected locally path-connected space X . The following are equivalent:

- There is a covering projection $f: \tilde{X}_1 \rightarrow \tilde{X}_2$ such that $p_2 \circ f = p_1$.
- For all $\tilde{x}_1 \in \tilde{X}_1$ and $\tilde{x}_2 \in \tilde{X}_2$ such that $p_1(\tilde{x}_1) = p_2(\tilde{x}_2)$, $p_{1\#}\pi(\tilde{X}_1, \tilde{x}_1)$ is conjugate in $\pi(X, p_1(\tilde{x}_1))$ to a subgroup of $p_{2\#}\pi(\tilde{X}_2, \tilde{x}_2)$.
- There exist $x_1 \in \tilde{X}_1$ and $x_2 \in \tilde{X}_2$ such that $p_1(x_1) = p_2(x_2)$ and $p_{1\#}\pi(\tilde{X}_1, \tilde{x}_1)$ is conjugate in $\pi(X, p_1(\tilde{x}_1))$ to a subgroup of $p_{2\#}\pi(\tilde{X}_2, \tilde{x}_2)$.

PROOF (a) $\Rightarrow$ (b) Given $f: \tilde{X}_1 \rightarrow \tilde{X}_2$ such that $p_2 \circ f = p_1$, if $\tilde{x}_1 \in \tilde{X}_1$ and $\tilde{x}_2 \in \tilde{X}_2$ are such that $p_1(\tilde{x}_1) = p_2(\tilde{x}_2)$, then

$$p_{1\#}\pi(\tilde{X}_1, \tilde{x}_1) = p_{2\#} \circ f_{\#}\pi(\tilde{X}_1, \tilde{x}_1) \subset p_{2\#}\pi(\tilde{X}_2, f(\tilde{x}_1))$$

Because $f(\tilde{x}_1)$ and $\tilde{x}_2$ lie in the same fiber of $p_2: \tilde{X}_2 \rightarrow X$, it follows from theorem 2.3.6 that $p_{2\#}\pi(\tilde{X}_2, f(\tilde{x}_1))$ and $p_{2\#}\pi(\tilde{X}_2, \tilde{x}_2)$ are conjugate in $\pi(X, p_1(\tilde{x}_1))$.

(b) $\Rightarrow$ (c) The proof is trivial.

(c) $\Rightarrow$ (a) Assume that $\tilde{x}_1 \in \tilde{X}_1$ and $\tilde{x}_2 \in \tilde{X}_2$ are such that $p_1(\tilde{x}_1) = p_2(\tilde{x}_2)$ and that $p_{1\#}\pi(\tilde{X}_1, \tilde{x}_1)$ is conjugate in $\pi(X, p_1(\tilde{x}_1))$ to a subgroup of $p_{2\#}\pi(\tilde{X}_2, \tilde{x}_2)$. By theorem 2.3.6, there is a point $\tilde{x}'_2 \in \tilde{X}_2$ such that $p_2(\tilde{x}'_2) = p_2(\tilde{x}_2)$ and such that

$$p_{1\#}\pi(\tilde{X}_1, \tilde{x}_1) \subset p_{2\#}\pi(\tilde{X}_2, \tilde{x}'_2)$$

Because $\tilde{X}_1$ is a connected locally path-connected space, the lifting theorem implies the existence of a map $f: (\tilde{X}_1, \tilde{x}_1) \rightarrow (\tilde{X}_2, \tilde{x}'_2)$ such that $p_2 \circ f = p_1$. ■

3 COROLLARY *Two objects in the category of connected covering spaces of a connected locally path-connected space X are equivalent if and only if their fundamental groups (at some two points over the same point of X) map to conjugate subgroups of the fundamental group of X (at this point).* ■

We give two examples.

- Because every nontrivial subgroup of $\pi(S^1) \simeq \mathbf{Z}$ is infinite cyclic, by corollary 3 every connected covering space $\tilde{X} \rightarrow S^1$ is equivalent to $\text{ex: } \mathbf{R} \rightarrow S^1$ or to the map $S^1 \rightarrow S^1$ sending z to z^n for some positive integer n .
- For $n \geq 2$, $\pi(P^n) \simeq \mathbf{Z}_2$, and every connected covering space $\tilde{X} \rightarrow P^n$ is equivalent to the double covering $S^n \rightarrow P^n$ or to the trivial covering $P^n \subset P^n$.

A universal covering space of a connected space X is an object $p: \tilde{X} \rightarrow X$ of the category of connected covering spaces of X such that for any object $p': \tilde{X}' \rightarrow X$ of this category there is a morphism

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{f} & \tilde{X}' \\ p \searrow & & \swarrow p' \\ & X & \end{array}$$

in the category. The next result follows easily from theorem 2 and corollary 3.

6 COROLLARY *Two universal covering spaces of a connected locally path-connected space are equivalent.* ■

Another result also follows from theorem 2.

7 COROLLARY *A simply connected covering space of a connected locally path-connected space X is a universal covering space of X .* ■

Having reduced the comparison of connected covering spaces of X to a comparison of their corresponding subgroups of the fundamental group of X , we shall determine which subgroups of the fundamental group correspond to covering spaces. This necessitates the construction of covering spaces. Let X be a space and let $\mathfrak{q}_l$ be an open covering of X . If $x_0 \in X$, let $\pi(\mathfrak{q}_l, x_0)$ be the subgroup of $\pi(X, x_0)$ generated by homotopy classes of closed paths having a representative of the form $(\omega * \omega') * \omega^{-1}$, where ω' is a closed path lying in some element of $\mathfrak{q}_l$ and ω is a path from x_0 to $\omega'(0)$. The following statements are easily verified.

- 8** If $\mathfrak{q}_l$ is an open covering of X that refines $\mathfrak{q}_l$, then $\pi(\mathfrak{q}_l, x_0) \subset \pi(\mathfrak{q}_l, x_0)$. ■
- 9** $\pi(\mathfrak{q}_l, x_0)$ is a normal subgroup of $\pi(X, x_0)$. ■
- 10** If ω is a path in X , then $h_{[\omega]}\pi(\mathfrak{q}_l, \omega(1)) = \pi(\mathfrak{q}_l, \omega(0))$. ■

The connection of the groups $\pi(\mathfrak{q}_l, x_0)$ with covering projections is explained by the following result.

11 LEMMA *Let $p: \tilde{X} \rightarrow X$ be a covering projection and let $\mathfrak{q}_l$ be a covering of X by open sets each evenly covered by p . For any $\tilde{x}_0 \in \tilde{X}$*

$$\pi(\mathfrak{q}_l, p(\tilde{x}_0)) \subset p_\# \pi(\tilde{X}, \tilde{x}_0)$$

PROOF If ω' is a closed path lying in some element of $\mathfrak{q}_l$, then, by lemma 2.4.9, any lifting of ω' is a closed path in $\tilde{X}$. Hence any path of the form $(\omega * \omega') * \omega^{-1}$, where ω' is a closed path lying in some element of $\mathfrak{q}_l$, can be lifted to a closed path [namely, to $(\tilde{\omega} * \tilde{\omega}') * \tilde{\omega}^{-1}$, where $\tilde{\omega}$ and $\tilde{\omega}'$ are suitable liftings of ω and ω' , respectively]. Hence any element of $\pi(\mathfrak{q}_l, p(x_0))$ has a representative which can be lifted to a closed path at $\tilde{x}_0$. ■

The following theorem characterizes those fibrations with unique path lifting which are covering projections.

12 THEOREM *Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting, where X and $\tilde{X}$ are connected locally path-connected spaces. Then p is a covering projection if and only if there is an open covering $\mathfrak{q}_l$ of X and a point $\tilde{x}_0 \in \tilde{X}$ such that*

$$\pi(\mathfrak{q}_l, p(\tilde{x}_0)) \subset p_\# \pi(\tilde{X}, \tilde{x}_0)$$

PROOF If p is a covering projection, the desired result follows from lemma 11. Conversely, if there is such an open covering $\mathfrak{q}_l$ and point $\tilde{x}_0 \in \tilde{X}$, it follows from statements 9 and 10 that for any point $\tilde{x}'_0 \in \tilde{X}$, $\pi(\mathfrak{q}_l, p(\tilde{x}'_0)) \subset p_\# \pi(\tilde{X}, \tilde{x}'_0)$. Using lemma 2.4.9, it follows that every element of $\mathfrak{q}_l$ is evenly covered by p . ■

Lemma 11 gives a necessary condition for a subgroup of $\pi(X, x_0)$ to correspond to a covering space. The next result proves that this necessary condition is also sufficient.

13 THEOREM *Let X be a connected locally path-connected space and let $x_0 \in X$. Let H be a subgroup of $\pi(X, x_0)$ and assume that there is an open covering $\mathfrak{q}_l$ of X such that $\pi(\mathfrak{q}_l, x_0) \subset H$. Then there is a covering projection $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ such that $p_\# \pi(\tilde{X}, \tilde{x}_0) = H$.*

PROOF Suppose such a covering projection exists, and suppose, moreover, that the space $\tilde{X}$ is path connected. The projection $q: (P(X, x_0), \omega_0) \rightarrow (X, x_0)$ of the path space of (X, x_0) can then be lifted to a map $q': (P(X, x_0), \omega_0) \rightarrow (\tilde{X}, \tilde{x}_0)$, which is surjective. If ω and ω' are elements of $P(X, x_0)$, then $q'(\omega) = q'(\omega')$ if and only if $q(\omega) = q(\omega')$ and $[\omega * \omega'^{-1}] \in p_\# \pi(\tilde{X}, \tilde{x}_0) = H$. Therefore, for path connected $\tilde{X}$ there is a one-to-one correspondence between the points of $\tilde{X}$ and equivalence classes of $P(X, x_0)$ identifying ω with ω' if $\omega(1) = \omega'(1)$ and $[\omega * \omega'^{-1}] \in H$ (the group properties of H imply that this is an equivalence relation). Hence it is natural to try to construct $\tilde{X}$ by suitably topologizing these equivalence classes of $P(X, x_0)$. We could start with the compact-open topology on $P(X, x_0)$ and use the quotient topology on the set of equivalence classes, but it seems no simpler than merely topologizing the set of equivalence classes directly, as is done below.

We consider the set of all paths in X beginning at x_0 . If ω and ω' are two such paths, set $\omega \sim \omega'$ if $\omega(1) = \omega'(1)$ and $[\omega * \omega'^{-1}] \in H$. This is an equivalence relation, and the equivalence class of ω will be denoted by $\langle \omega \rangle$. Let $\tilde{X}$ be the set of equivalence classes. There is a function $p: \tilde{X} \rightarrow X$ such that $p(\langle \omega \rangle) = \omega(1)$. If U is an open subset of X and ω is a path beginning at x_0 and ending in U , $\langle \omega, U \rangle$ will denote the subset of $\tilde{X}$ consisting of all the equivalence classes having a representative of the form $\omega * \omega'$, where ω' is a path in U beginning at $\omega(1)$.

We prove that the collection $\{\langle \omega, U \rangle\}$ is a base for a topology on $\tilde{X}$. If $\langle \omega' \rangle \in \langle \omega, U \rangle$, then $\omega' \sim \omega * \omega''$ for some path ω'' lying in U . If $\tilde{\omega}$ is any path in U beginning at $\omega(1)$, then

$$\omega' * \tilde{\omega} \sim (\omega * \omega') * \tilde{\omega} \sim \omega'' * (\omega' * \tilde{\omega})$$

showing that $\langle \omega', U \rangle \subset \langle \omega, U \rangle$. Since $\omega \sim \omega' * \omega'^{-1}$, $\langle \omega \rangle \in \langle \omega', U \rangle$. The same argument shows that $\langle \omega, U \rangle \subset \langle \omega', U \rangle$, and so $\langle \omega, U \rangle = \langle \omega', U \rangle$. Therefore, if $\omega'' \in \langle \omega, U \rangle \cap \langle \omega', U' \rangle$, then $\langle \omega'', U \cap U' \rangle \subset \langle \omega, U \rangle \cap \langle \omega', U' \rangle$, and so the collection $\{\langle \omega, U \rangle\}$ is a base for a topology on $\tilde{X}$.

Let $\tilde{X}$ be topologized by the topology having $\{\langle \omega, U \rangle\}$ as a base. Then p is continuous; for if $p(\langle \omega \rangle) \in U$, then $p(\langle \omega, U \rangle) \subset U$. p is also open, because $p(\langle \omega, U \rangle)$ clearly equals the path component of U containing $\omega(1)$, and this is open because X is locally path connected.

Let $\mathfrak{q}_l$ be an open covering of X such that $\pi(\mathfrak{q}_l, x_0) \subset H$ and let V be an open path-connected subset of X contained in some element of $\mathfrak{q}_l$. We show that V is evenly covered by p , which will imply that p is a covering projection.

If $\langle \omega \rangle \in p^{-1}(V)$, then $\langle \omega, V \rangle \subset p^{-1}(V)$. The sets $\{\langle \omega, V \rangle \mid \langle \omega \rangle \in p^{-1}(V)\}$ are open and their union equals $p^{-1}(V)$. If $\langle \omega, V \rangle \cap \langle \omega', V \rangle \neq \emptyset$, let $\langle \omega'' \rangle \in \langle \omega, V \rangle \cap \langle \omega', V \rangle$. Then $\langle \omega'', V \rangle = \langle \omega, V \rangle$ and $\langle \omega'', V \rangle = \langle \omega', V \rangle$. Hence the sets $\{\langle \omega, V \rangle \mid \langle \omega \rangle \in p^{-1}(V)\}$ are either identical or disjoint. To prove that V is evenly covered by p , it suffices to show that p maps each set $\langle \omega, V \rangle$ bijectively to V (because p has already been shown to be continuous and open). If $x \in V$, let ω' be a path in V from $\omega(1)$ to x . Then $\langle \omega * \omega' \rangle \in \langle \omega, V \rangle$ and $p(\langle \omega * \omega' \rangle) = x$, showing that p is surjective. Assume $p\langle \omega * \omega_1 \rangle = p\langle \omega * \omega_2 \rangle$. Then $\omega_1(1) = \omega_2(1)$, so $(\omega * \omega_1) * (\omega * \omega_2)^{-1}$ is a closed path in X at x_0 . Also,

$$[(\omega * \omega_1) * (\omega * \omega_2)^{-1}] = [(\omega * (\omega_1 * \omega_2^{-1})) * \omega^{-1}]$$

Since $\omega_1 * \omega_2^{-1}$ is a path in V and V is contained in some element of U , $[(\omega * (\omega_1 * \omega_2^{-1})) * \omega^{-1}] \in \pi(\mathfrak{Q}_1, x_0) \subset H$. Therefore $\omega * \omega_1 \sim \omega * \omega_2$ and $\langle \omega * \omega_1 \rangle = \langle \omega * \omega_2 \rangle$, showing that p is injective.

We have shown that $p: \tilde{X} \rightarrow X$ is a covering projection. Let $\tilde{x}_0 = \langle \omega_0 \rangle$, where ω_0 is the constant path in X at x_0 . It remains only to verify that $p_{\#}\pi(\tilde{X}, \tilde{x}_0) = H$. For this we need an explicit expression for the lift of a path in X that begins at x_0 . Let ω be a path in X beginning at x_0 , and for $t \in I$, define a path ω_t in X beginning at x_0 by $\omega_t(t') = \omega(t't)$. Let $\tilde{\omega}: I \rightarrow \tilde{X}$ be defined by $\tilde{\omega}(t) = \langle \omega_t \rangle$. We prove that $\tilde{\omega}$ is continuous. If $\tilde{\omega}(t_0) \in \langle \omega', U \rangle$, then $p\tilde{\omega}(t_0) = \omega(t_0) \in U$ and $\langle \omega', U \rangle = \langle \tilde{\omega}(t_0), U \rangle$. Let N be any open interval in I containing t_0 such that $\omega(N) \subset U$. If $t \in N$, then $\omega_t \sim \omega_{t_0} * \omega_{t_0,t}$, where $\omega_{t_0,t}(t') = \omega(t_0 + t'(t - t_0))$. Therefore, for $t \in N$

$$\tilde{\omega}(t) = \langle \omega_t \rangle = \langle \tilde{\omega}(t_0) * \omega_{t_0,t} \rangle \in \langle \tilde{\omega}(t_0), U \rangle = \langle \omega', U \rangle$$

and so $\tilde{\omega}$ is continuous. Furthermore, $p\tilde{\omega}(t) = \omega_t(1) = \omega(t)$. Hence $\tilde{\omega}$ is a lift of ω beginning at $\tilde{\omega}(0) = \tilde{x}_0$ and ending at $\tilde{\omega}(1) = \langle \omega \rangle$.

If $[\omega] \in H$, then $\omega \sim \omega_0$ and $\langle \omega \rangle = \tilde{x}_0$. Therefore the lift $\tilde{\omega}$ of ω constructed above is a closed path in $\tilde{X}$ at $\tilde{x}_0$, proving that $H \subset p_{\#}\pi(\tilde{X}, \tilde{x}_0)$. On the other hand, if $\tilde{\omega}'$ is a closed path in $\tilde{X}$ at $\tilde{x}_0$ and $p\tilde{\omega}' = \omega$, let $\tilde{\omega}$ be the path in $\tilde{X}$ constructed above. Since $\tilde{\omega}$ is a lift of ω beginning at $\tilde{x}_0$, it follows from the unique path lifting of p that $\tilde{\omega} = \tilde{\omega}'$. Therefore $\tilde{\omega}(1) = \tilde{\omega}'(1) = \tilde{x}_0$. Since $\tilde{\omega}(1) = \langle \omega \rangle$, $\omega \sim \omega_0$, showing that $p_{\#}\pi(\tilde{X}, \tilde{x}_0) \subset H$. ■

A space X is semilocally 1-connected (defined in Sec. 2.4) if and only if there is an open covering $\mathfrak{Q}$ of X such that $\pi(\mathfrak{Q}_1, x_0) = 0$. Hence we have the following result.

1.4 COROLLARY A connected locally path-connected space X has a simply connected covering space if and only if X is semilocally 1-connected. ■

From corollaries 1.4 and 6 and theorem 2 we obtain the next result.

1.5 COROLLARY Any universal covering space of a connected locally path-connected semilocally 1-connected space is simply connected. ■

Not every connected locally path-connected space has a universal covering space. We give two examples.

1.6 An infinite product of 1-spheres has no universal covering space.

1.7 Let X be the subspace of $\mathbf{R}^2$ equal to the union of the circumferences of circles C_n with $n \geq 1$, where C_n has center at $(1/n, 0)$ and radius $1/n$. Then X is connected and locally path connected but has no universal covering space.

It is possible for a connected locally path-connected space to have a universal covering space that is not simply connected. We present an example.

1.8 EXAMPLE Let Y_1 be the cone with base X equal to the space of example 1.7. Y_1 can be visualized as the set of line segments in $\mathbf{R}^3$ joining the points of X to the point $(0, 0, 1)$ and let y_1 be the point at which all the circles of X are tangent. Let (Y_2, y_2) be another copy of (Y_1, y_1) . Let $Z = Y_1 \vee Y_2$. Then Z is connected and locally path connected but not simply connected (cf. exercise 1.6.7, a closed path oscillating back and forth from Y_1 to Y_2 around the decreasing circles C_n is not null homotopic). However, Y_1 and Y_2 are each closed contractible subsets of Z . By the lifting theorem, each of them can be lifted to any covering space of Z , so that y_1 is lifted arbitrarily and y_2 is lifted arbitrarily. Therefore any covering projection with base Z has a section. It follows that any connected covering space of Z is homeomorphic to Z .

In the category of fibrations with unique path lifting over a fixed path-connected base space (and with path-connected total spaces) there is always a universal object (that is, an object which has morphisms to any other object in the category). We sketch a proof of this fact. Let X be a path-connected space and let $\mathfrak{Q}(X)$ be the collection of topological spaces whose underlying sets are cartesian products of X and the set of right cosets of some subgroup of the fundamental group of X . It follows from theorem 2.3.9 that any fibration whose base space is X and total space is path connected is equivalent to a fibration $\tilde{X} \rightarrow X$, where $\tilde{X} \in \mathfrak{Q}(X)$. Since $\mathfrak{Q}(X)$ is a set, those fibrations $\tilde{X} \rightarrow X$ with unique path lifting, where $\tilde{X}$ is a path-connected space in $\mathfrak{Q}(X)$, constitute a set. We may form the fibered product of this set (as in Sec. 2.2). This fibered product is then the desired universal fibration with unique path lifting.

If X is a connected locally path-connected space, it follows from theorem 1.3 that for any open covering $\mathfrak{Q}$ of X there is a path-connected covering space of X whose fundamental group is isomorphic to $\pi(\mathfrak{Q}_1, x_0)$. This implies that if $\tilde{X}$ is a universal object in the category of path-connected fibrations over X with unique path lifting, then $\pi(\tilde{X}, \tilde{x}_0)$ is isomorphic to a subgroup of $\pi_{\mathfrak{Q}}(\pi(\mathfrak{Q}_1, x_0))$. In particular, if $\bigcap_{\mathfrak{Q}} \pi(\mathfrak{Q}_1, x_0) = 0$, then X has a simply connected fibration with unique path lifting that is a universal object in the category. Thus the spaces in examples 1.6 and 1.7 both have universal fibrations with unique path lifting that are simply connected. The space Z of example 1.8 is its own universal fibration with unique path lifting.

If $\tilde{\omega}$ is a path in $\tilde{X}$ from $\tilde{x}_0$ to $f(\tilde{x}_0)$, there is a commutative square

$$\begin{array}{ccc} \pi(\tilde{X}, \tilde{x}_0) & \xleftarrow{h_{\omega}} & \pi(\tilde{X}, f(\tilde{x}_0)) \\ p_{\#} \downarrow & & \downarrow p_{\#} \end{array}$$

$$\pi(X, p(\tilde{x}_0)) \xleftarrow{h_{p \circ \tilde{\omega}}} \pi(X, p(f(\tilde{x}_0)))$$

Since $f: (\tilde{X}, \tilde{x}_0) \rightarrow (\tilde{X}, f(\tilde{x}_0))$ is a homeomorphism,

$$f_{\#}\pi(\tilde{X}, \tilde{x}_0) = \pi(\tilde{X}, f(\tilde{x}_0))$$

and since $p_{\#}f_{\#} = p_{\#}$,

$$\begin{aligned} h_{(p \circ \tilde{\omega})\pi(\tilde{X}, f(\tilde{x}_0))} &= h_{(p \circ \tilde{\omega})\pi(\tilde{X}, \tilde{x}_0)} \circ f_{\#}\pi(\tilde{X}, \tilde{x}_0) = h_{(p \circ \tilde{\omega})\pi(\tilde{X}, \tilde{x}_0)} \circ p_{\#}\pi(\tilde{X}, \tilde{x}_0) \\ &= p_{\#}h_{(\tilde{\omega})\pi(\tilde{X}, f(\tilde{x}_0))} = p_{\#}\pi(\tilde{X}, \tilde{x}_0) \end{aligned}$$

Hence $[p \circ \tilde{\omega}] \in N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$. Because $\psi(f)$ is equal to the right coset $(p_{\#}\pi(\tilde{X}, \tilde{x}_0))[p \circ \tilde{\omega}]$, ψ is an injection of $G(\tilde{X} | X)$ into the set of right cosets of $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ by elements of $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$.

We now verify that ψ is an homomorphism. If $f_1, f_2 \in G(\tilde{X} | X)$ let $\tilde{\omega}_1$ and $\tilde{\omega}_2$ be paths in $\tilde{X}$ from $\tilde{x}_0$ to $f_1(\tilde{x}_0)$ and $f_2(\tilde{x}_0)$, respectively. Then $f_1 \circ \tilde{\omega}_2$ is a path from $f_1(\tilde{x}_0)$ to $f_1 f_2(\tilde{x}_0)$, and $\tilde{\omega}_1 * (f_1 \circ \tilde{\omega}_2)$ is a path from $\tilde{x}_0$ to $f_1 f_2(\tilde{x}_0)$. Therefore $\psi(f_1 f_2)$ is the right coset

$$(p_{\#}\pi(\tilde{X}, \tilde{x}_0))[(p \circ \tilde{\omega}_1) * (p \circ f_1 \circ \tilde{\omega}_2)] = (p_{\#}\pi(\tilde{X}, \tilde{x}_0))[p \circ \tilde{\omega}_1] * [p \circ \tilde{\omega}_2]$$

and this equals $\psi(f_1)\psi(f_2)$.

Finally, we show that if $\tilde{X}$ is locally path connected, ψ is an epimorphism to the set of right cosets of $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ in $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$. Assume that $[\omega] \in \pi(X, p(\tilde{x}_0))$ belongs to $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$. Let $\tilde{\omega}$ be a lifting of ω ending at $\tilde{x}_0$ and let $\tilde{x} = \tilde{\omega}(0)$. Then

$$p_{\#}\pi(\tilde{X}, \tilde{x}_0) = h_{(\omega)}(p_{\#}\pi(\tilde{X}, \tilde{x}_0)) = p_{\#}(h_{[\tilde{\omega}]} \pi(\tilde{X}, \tilde{x}_0)) = p_{\#}\pi(\tilde{X}, \tilde{x})$$

Because X is connected and locally path connected, the lifting theorem implies the existence of maps $f: (\tilde{X}, \tilde{x}_0) \rightarrow (\tilde{X}, \tilde{x})$ and $g: (\tilde{X}, \tilde{x}) \rightarrow (\tilde{X}, \tilde{x}_0)$ such that $p \circ f = p$ and $p \circ g = p$. From the unique-lifting property (lemma 2.2.4), it follows that $f \circ g = 1_{\tilde{x}}$ and $g \circ f = 1_{\tilde{x}}$. Therefore $f \in G(\tilde{X} | X)$ and $\psi(f)$ equals the right coset $(p_{\#}\pi(\tilde{X}, \tilde{x}_0))[\omega]^{-1}$. ■

Combining theorem 2 with theorem 2.3.12, we have the following corollary.

3 COROLLARY Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. If $\tilde{X}$ is connected and locally path connected and $\tilde{x}_0 \in \tilde{X}$, then p is regular if and only if $G(\tilde{X} | X)$ is transitive on each fiber of p , in which case

$$\psi: G(\tilde{X} | X) \simeq \pi(X, p(\tilde{x}_0)) / p_{\#}\pi(\tilde{X}, \tilde{x}_0) \quad \blacksquare$$

In this section we consider a problem inverse to the one of the last section, in which we constructed covering projections with given base space; we ask for covering projections with given covering space. On any regular covering space we prove that there is a group of covering transformations. The covering projection is then equivalent to the projection of the covering space onto the space of orbits of the group of covering transformations.

Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. It is clear that there is a group of self-equivalences of this fibration (a self-equivalence is a homeomorphism $f: \tilde{X} \rightarrow \tilde{X}$ such that $p \circ f = p$). We denote this group by $G(\tilde{X} | X)$. In case $p: \tilde{X} \rightarrow X$ is a covering projection, $G(\tilde{X} | X)$ is also called the group of covering transformations of p . In general, there is a close analogy of $G(\tilde{X} | X)$ with the group of automorphisms of an extension field leaving a subfield pointwise fixed.

If $\tilde{X}$ is path connected, it follows from lemma 2.2.4 that two self-equivalences of $p: \tilde{X} \rightarrow X$ that agree at one point are identical. Hence we have the following lemma.

1 LEMMA Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. If $\tilde{X}$ is path connected and $\tilde{x}_0 \in \tilde{X}$, then the function $f \rightarrow f(\tilde{x}_0)$ is an injection of $G(\tilde{X} | X)$ into the fiber of p over $p(\tilde{x}_0)$. ■

Theorem 2.3.9 established a bijection from the set of right cosets of $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ in $\pi(X, p(\tilde{x}_0))$ to the fiber of p over $p(\tilde{x}_0)$. Combining the inverse of this bijection with the function of lemma 1 yields an injection ψ from $G(\tilde{X} | X)$ to the set of right cosets of $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ in $\pi(X, p(\tilde{x}_0))$. ψ is defined explicitly as follows. For any $f \in G(\tilde{X} | X)$ let $\tilde{\omega}$ be a path in $\tilde{X}$ from $\tilde{x}_0$ to $f(\tilde{x}_0)$. Then $p \circ \tilde{\omega}$ is a closed path in X at $p(\tilde{x}_0)$, and the right coset $(p_{\#}\pi(\tilde{X}, \tilde{x}_0))[p \circ \tilde{\omega}]$ is independent of the choice of $\tilde{\omega}$. The function ψ assigns to f this right coset.

Given $\tilde{x}_0 \in \tilde{X}$, let $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$ be the normalizer of $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ in $\pi(X, p(\tilde{x}_0))$. Thus $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$ is the subgroup of $\pi(X, p(\tilde{x}_0))$ consisting of elements $[\omega] \in \pi(X, p(\tilde{x}_0))$ such that $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ is invariant under conjugation by $[\omega]$. $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$ is the largest subgroup of $\pi(X, p(\tilde{x}_0))$ containing $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ as a normal subgroup.

2 THEOREM Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting. Let $\tilde{X}$ be path connected and let $\tilde{x}_0 \in \tilde{X}$. Then ψ is a monomorphism of $G(\tilde{X} | X)$ to the quotient group $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0)) / p_{\#}\pi(\tilde{X}, \tilde{x}_0)$. If $\tilde{X}$ is also locally path connected, ψ is an isomorphism.

PROOF We already know that ψ is an injection. We show that ψ is a function from $G(\tilde{X} | X)$ to the set of right cosets of $p_{\#}\pi(\tilde{X}, \tilde{x}_0)$ by elements of $N(p_{\#}\pi(\tilde{X}, \tilde{x}_0))$.

If $\tilde{X}$ is simply connected, any fibration $p: \tilde{X} \rightarrow X$ is regular, and we also have the next result.

4 COROLLARY Let $p: \tilde{X} \rightarrow X$ be a fibration with unique path lifting, where $\tilde{X}$ is simply connected, locally path connected, and nonempty. Then the group of self-equivalences of p is isomorphic to the fundamental group of X . ■

If $p: \tilde{X} \rightarrow X$ is a regular covering projection and $\tilde{X}$ is connected and locally path connected, then X is homeomorphic to the space of orbits of $G(\tilde{X}|X)$ (an orbit of a group of transformations G acting on a set S is an equivalence class of S with respect to the equivalence relation $s_1 \sim s_2$ if there is $g \in G$ such that $gs_1 = s_2$). We are interested in the converse problem—that is, in knowing what conditions on a group G of homeomorphisms of a topological space Y will ensure that the projection of Y onto the space of orbits Y/G is a regular covering projection whose group of covering transformations is equal to G .

A group G of homeomorphisms of a topological space Y is said to be *discontinuous* if the orbits of G in Y are discrete subsets of Y . G is *properly discontinuous* if for $y \in Y$ there is an open neighborhood U of y in Y such that if $g, g' \in G$ and gU meets $g'U$, then $g = g'$. G acts *without fixed points* if the only element of G having fixed points is the identity element. The following are clear.

- 5** A properly discontinuous group of homeomorphisms is discontinuous and acts without fixed points. ■
- 6** A finite group of homeomorphisms acting without fixed points on a Hausdorff space is properly discontinuous. ■

If G is the group of covering transformations of a covering projection, then a simple verification shows that G is properly discontinuous. We now show that any properly discontinuous group of homeomorphisms defines a covering projection.

7 THEOREM Let G be a properly discontinuous group of homeomorphisms of a space Y . Then the projection of Y to the orbit space Y/G is a covering projection. If Y is connected, this covering projection is regular and G is its group of covering transformations.

PROOF Let $p: Y \rightarrow Y/G$ be the projection. Then p is continuous. It is an open map, for if U is an open set in Y , then $p^{-1}(p(U)) = \bigcup \{gU \mid g \in G\}$ is open in Y , and therefore pU is open in Y/G . Let U be an open subset of Y such that whenever gU meets $g'U$, then $g = g'$. We show that $p(U)$ is evenly covered by p . The hypothesis on U ensures that $\{gU \mid g \in G\}$ is a disjoint collection of open sets whose union is $p^{-1}(p(U))$. It suffices to prove that $p|_{gU}$ is a bijection from gU to $p(U)$. If $y \in U$, then $p(gy) = p(y)$, so $p(gU) = p(U)$. If $p(gy_1) = p(gy_2)$, with $y_1, y_2 \in U$, there is $g' \in G$ such that $gy_1 = g'gy_2$.

Therefore $g'U$ meets $g'gU$, and $g = g'g$. Hence $g' = 1_Y$ and $gy_1 = gy_2$. We have proved that p is a homeomorphism of gU onto $p(U)$. Since G is properly discontinuous, the sets $p(U)$ evenly covered by p constitute an open covering of Y/G .

Because $p(gy) = p(y)$, we see that G is contained in the group of covering transformations of p . Since G is transitive on the fibers of p , it follows from theorem 2.2.2 that if Y is connected, G equals the group of covering transformations. Since the group of covering transformations is transitive on each fiber, the covering projection is regular. ■

8 COROLLARY Let G be a properly discontinuous group of homeomorphisms of a simply connected space Y . Then the fundamental group of the orbit space Y/G is isomorphic to G .

PROOF By theorem 7, G is the group of covering transformations of the regular covering projection $p: Y \rightarrow Y/G$. By theorem 2, ψ is a monomorphism of G into the fundamental group of Y/G . Because G is transitive on the fibers of p , ψ is an isomorphism. ■

9 EXAMPLE Let $S^3 = \{(z_0, z_1) \in \mathbb{C}^2 \mid |z_0|^2 + |z_1|^2 = 1\}$ and let p and q be relatively prime integers. Define $h: S^3 \rightarrow S^3$ by

$$h(z_0, z_1) = (e^{2\pi i/p} z_0, e^{2\pi i/q} z_1)$$

Then h is a homeomorphism of S^3 with period p (that is, $h^p = 1$), and Z_p acts on S^3 by

$$n(z_0, z_1) = h^n(z_0, z_1)$$

where n denotes the residue class of the integer n modulo p . In this way Z_p acts without fixed points on S^3 . The orbit space of this action of Z_p on S^3 is called a *lens space* and is denoted by $L(p, q)$. By statement 6 and corollary 8, the fundamental group of $L(p, q)$ is isomorphic to Z_p .

10 EXAMPLE Let $S^{2n+1} = \{(z_0, z_1, \dots, z_n) \in \mathbb{C}^{n+1} \mid \sum |z_i|^2 = 1\}$ and let $q_1, \dots, q_n$ be integers relatively prime to p . Define $h: S^{2n+1} \rightarrow S^{2n+1}$ by

$$h(z_0, z_1, \dots, z_n) = (e^{2\pi i/p} z_0, e^{2\pi i q_1/p} z_1, \dots, e^{2\pi i q_n/p} z_n)$$

Then, as in example 9, h determines an action of Z_p on S^{2n+1} without fixed points; the orbit space is called a *generalized lens space* and is denoted by $L(p, q_1, \dots, q_n)$. Its fundamental group is isomorphic to Z_p .

It is possible to use theorem 7 to show that the projection $Y \rightarrow Y/G$ is a regular fibration with unique path lifting even when it may not be a covering projection. Note that if G acts on Y without fixed points, so does any subgroup of G , and if G' is a normal subgroup of Y , then G'/G' acts without fixed points on Y/G' .

11 THEOREM Let G be a group of homeomorphisms acting without fixed points on a path-connected space Y and assume that there is a decreasing sequence of subgroups

$$G = G_0 \supset G_1 \supset \dots \supset G_n \supset G_{n+1} \supset \dots$$

such that

- (a) $G_n \cap G_n = \{1_Y\}$
- (b) G_{n+1} is a normal subgroup of G_n for $n \geq 0$
- (c) G_n/G_{n+1} is a properly discontinuous group of homeomorphisms on Y/G_{n+1} and the projection $Y \rightarrow Y/G_n$ is a closed map for $n \geq 0$
- (d) Any orbit of Y under G_n for $n \geq 0$ is compact

Then the projection $p: Y \rightarrow Y/G$ is a regular fibration with unique path lifting whose group of self-equivalences is G .

PROOF Since $Y/G_n = (Y/G_{n+1})/(G_n/G_{n+1})$, it follows from (c) and theorem 7 that the projection

$$p_{n+1}: Y/G_{n+1} \rightarrow Y/G_n$$

is a regular covering projection for $n \geq 0$. Let

$$\tilde{Y} = \{(y_n) \in \times (Y/G_n) \mid p_{n+1}(y_{n+1}) = y_n \text{ for } n \geq 0\}$$

and define $\tilde{p}: \tilde{Y} \rightarrow Y/G$ by $\tilde{p}((y_n)) = y_0$. It is easy to verify that $\tilde{p}$ is a fibration with unique path lifting (it is the fibered product of the maps $\{p_1 \circ \dots \circ p_l\}$).

For $n \geq 0$ there is a continuous closed projection map $q_n: Y \rightarrow Y/G_n$ such that $p_{n+1} \circ q_{n+1} = q_n$. Therefore there is a continuous closed map $q: Y \rightarrow \tilde{Y}$ defined by $q(y) = (q_n(y))$ and such that $\tilde{p} \circ q = p$. To prove that q is a homeomorphism, it suffices to show that it is a bijection. If $q(y) = q(y')$, then for $n \geq 0$ there is $g_n \in G_n$ such that $y = g_n y'$. Then $g_n y' = g_n y'$ for all m and n , and because G acts without fixed points, $g_m = g_n$ for all m and n . Therefore $g_n \in G_m$ for all m , and by (d), $g_n = 1_Y$. It follows that $y = y'$, and hence that q is injective.

If $(y_n) \in \tilde{Y}$, then $q_n^{-1}y_n$ is an orbit of Y under G_n . By (d), $q_n^{-1}y_n$ is compact. Since

$$q_n^{-1}y_n = q_{n+1}^{-1}p_{n+1}^{-1}y_n \supset q_{n+1}^{-1}y_{n+1}$$

the collection $\{q_n^{-1}y_n\}$ consists of compact sets having the finite-intersection property. Therefore $\bigcap q_n^{-1}y_n \neq \emptyset$. If $y \in \bigcap q_n^{-1}y_n$, then $q(y) = (y_n)$, showing that q is surjective.

We have shown that $q: Y \rightarrow \tilde{Y}$ is a homeomorphism. Therefore $p: Y \rightarrow Y/G$ is a fibration with unique path lifting. Since each element of G is a self-equivalence of p , the group of self-equivalences of p is transitive on each fiber. By corollary 3, p is a regular fibration and G is the group of self-equivalences of p . ■

7 FIBER BUNDLES

A covering space is locally the product of its base space and a discrete space. This is generalized by the concept of fiber bundle, defined in this section, because the total space of a fiber bundle is locally the product of its base

space and its fiber. The main result is that the bundle projection of a fiber bundle is a fibration.¹

A fiber bundle $\xi = (E, B, F, p)$ consists of a total space E , a base space B , a fiber F , and a bundle projection $p: E \rightarrow B$ such that there exists an open covering $\{U\}$ of B and, for each $U \in \{U\}$, a homeomorphism $q_U: U \times F \rightarrow p^{-1}(U)$ such that the composite

$$U \times F \xrightarrow{q_U} p^{-1}(U) \xrightarrow{p} U$$

is the projection to the first factor. Thus the bundle projection $p: E \rightarrow B$ and the projection $B \times F \rightarrow B$ are locally equivalent. The fiber over $b \in B$ is defined to equal $p^{-1}(b)$, and we note that F is homeomorphic to $p^{-1}(b)$ for every $b \in B$. Usually there is also given a structure group G for the bundle consisting of homeomorphisms of F , and we define this concept next.

Let G be a group of homeomorphisms of F . Given a space F' and a collection $\Phi = \{q\}$ of homeomorphisms $q: F \rightarrow F'$, define $qg: F \rightarrow F'$ for $q \in \Phi$ and $g \in G$ by $qg(y) = q(gy)$ for $y \in F$. The collection Φ is called a G structure on F' if

- (a) Given $q \in \Phi$ and $g \in G$, then $qg \in \Phi$
- (b) Given $q_1, q_2 \in \Phi$, there is $g \in G$ such that $q_1 = q_2g$

Condition (a) implies that G acts on the right on Φ , and condition (b) implies that this action of G is transitive on Φ . A fiber bundle (E, B, F, p) is said to have structure group G if each fiber $p^{-1}(b)$ has a G structure $\Phi(b)$ such that there exists an open covering $\{U\}$ of B and, for each $U \in \{U\}$, a homeomorphism $q_U: U \times F \rightarrow p^{-1}(U)$ such that for $b \in U$, the map $F \rightarrow p^{-1}(b)$ sending x to $q_U(b, x)$ is in $\Phi(b)$. It is clear that a given fiber bundle can always be given the structure of a fiber bundle with structure group the group of all homeomorphisms of F . It is also clear that a given fiber bundle can sometimes be given the structure of a fiber bundle with two different structure groups of homeomorphisms of F .

An n -plane bundle, or real vector bundle, is a fiber bundle whose fiber is $\mathbb{R}^n$ and whose structure group is the general linear group $GL(\mathbb{R}^n)$, which consists of all linear automorphisms of $\mathbb{R}^n$. A complex n -plane bundle, or complex vector bundle, is a fiber bundle whose fiber is $\mathbb{C}^n$ and whose structure group is $GL(\mathbb{C}^n)$.

We give some examples.

1 For spaces B and F the product bundle is the fiber bundle $(B \times F, B, F, p)$, where $p: B \times F \rightarrow B$ is the projection to the first factor (it has the trivial group as structure group).

2 Given that $p: \tilde{X} \rightarrow X$ is a covering projection and X is a connected space, if $x_0 \in X$, then $(\tilde{X}, X, p^{-1}(x_0), p)$ is a fiber bundle (and if X is path connected, it can be given the structure of a fiber bundle with structure group

¹ For the general theory of fiber bundles see N. E. Steenrod, *The Topology of Fibre Bundles*, Princeton University Press, Princeton, N.J., 1951.

$\pi(X, x_0)$, where $\pi(X, x_0)$ acts on $p^{-1}(x_0)$ by $[\omega]\bar{x} = \bar{x}[\omega]^{-1}$, with the right-hand side as in the proof of theorem 2.3.9).

3 Given that M is a differentiable n -manifold and $T(M)$ is the set of all tangent vectors to M , there is a fiber bundle $(T(M), M, R^n, p)$, where $p: T(M) \rightarrow M$ assigns to each tangent vector its origin. This is called the *tangent bundle* and is denoted by $\tau(M)$. Because it can be given the structure group $GL(\mathbb{R}^n)$, it is an n -plane bundle, and if M is a complex manifold of complex dimension m , then $\tau(M)$ is a complex m -plane bundle.

4 Given that H is a closed subgroup of a Lie group G and that G/H is the quotient space of left cosets and $p: G \rightarrow G/H$ the projection, then $(G, G/H, H, p)$ is a fiber bundle (having structure group H acting on itself by left translation).

5 Represent S^n as the union of closed hemispheres E_-^n and E_+^n with intersection S^{n-1} and let G be a group of homeomorphisms of a space F . Given a map $q: S^{n-1} \rightarrow G$ such that the map $S^{n-1} \times F \rightarrow F$ sending (x, y) to $q(x)y$ is continuous, let E_q be the space obtained from $(E_-^n \times F) \vee (E_+^n \times F)$ by identifying $(x, y) \in E_-^n \times F$ with $(x, q(x)y) \in E_+^n \times F$ for $x \in S^{n-1}$ and $y \in F$. These identifications are compatible with the projections $E_-^n \times F \rightarrow E_-^n$ and $E_+^n \times F \rightarrow E_+^n$. Therefore there is a map $p_q: E_q \rightarrow S^n$ such that each of the composites

$$E_-^n \times F \rightarrow E_q \xrightarrow{p_q} S^n \quad \text{and} \quad E_+^n \times F \rightarrow E_q \xrightarrow{p_q} S^n$$

is projection to the first factor. Then (E_q, S^n, F, p_q) is a fiber bundle (having structure group G) which is said to be defined by the *characteristic map* q .

6 Let $P_n(\mathbb{C})$ be the n -dimensional complex projective space coordinatized by homogeneous coordinates. If $z_0, z_1, \dots, z_n \in \mathbb{C}$ are not all zero, let $[z_0, z_1, \dots, z_n] \in P_n(\mathbb{C})$ be that point of $P_n(\mathbb{C})$ having homogeneous coordinates $z_0, z_1, \dots, z_n$. Regard S^{2n+1} as the set $\{(z_0, z_1, \dots, z_n) \in \mathbb{C}^{n+1} \mid \sum |z_i|^2 = 1\}$ and define $p: S^{2n+1} \rightarrow P_n(\mathbb{C})$ by $p(z_0, z_1, \dots, z_n) = [z_0, z_1, \dots, z_n]$. If $U_i \subset P_n(\mathbb{C})$ is the subset of points having a nonzero i th homogeneous coordinate, it is easy to see that $p^{-1}(U_i)$ is homeomorphic to $U_i \times S^1$. Therefore there is a fiber bundle $(S^{2n+1}, P_n(\mathbb{C}), S^1, p)$ (having structure group S^1 acting on itself by left translation), and this is called the *Hopf bundle*.

7 If $\mathbf{Q}$ is the division ring of quaternions, there is a similar map $p: S^{4n+3} \rightarrow P_n(\mathbf{Q})$ and a *quaternionic Hopf bundle* $(S^{4n+3}, P_n(\mathbf{Q}), S^3, p)$ (having structure group S^3 acting on itself by left translation).

The structure group will not be important for our purposes. Thus we define an *n -sphere bundle* to be a fiber bundle whose fiber is S^n [usually it is also required that it have as structure group the *orthogonal group* $O(n+1)$ of all isometries in $GL(\mathbb{R}^{n+1})$]. If ξ is an n -sphere bundle, we shall denote its total space by $\tilde{E}_\xi$. The mapping cylinder of the bundle projection $\tilde{E}_\xi \rightarrow B$ is the total space E_ξ of a fiber bundle $(E_\xi, B, F^{n+1}, p_\xi)$, where $p_\xi: \tilde{E}_\xi \rightarrow B$ is the retraction of the mapping cylinder to B (and $p_\xi|_{\tilde{E}_\xi}: \tilde{E}_\xi \rightarrow B$ is the original bundle projection).

If $\xi = (E, B, R^{n+1}, p)$ is an $(n+1)$ -plane bundle having structure group $O(n+1)$, it is possible to introduce a norm in each fiber $p^{-1}(b)$. The subset $E' \subset E$ of all elements in E having unit norm is the total space of an n -sphere bundle $(E', B, S^n, p|_{E'})$ called the *unit n -sphere bundle* of ξ . If the base space B of an $(n+1)$ -plane bundle is a paracompact Hausdorff space, the bundle can always be given $O(n+1)$ as structure group. In particular, there is a *unit tangent bundle* of a paracompact differentiable manifold.

Two fiber bundles (E_1, B, F, p_1) and (E_2, B, F, p_2) with the same fiber and same base are said to be *equivalent* if there is a homeomorphism $h: E_1 \rightarrow E_2$ such that $p_2 \circ h = p_1$. If they both have structure group G , they are *equivalent over G* if there is a homeomorphism h as above, with the additional property that if $q \in \Phi_1(b)$, then $h \circ q \in \Phi_2(b)$ for $b \in B$. A fiber bundle is said to be *trivial* if it is equivalent to the product bundle of example 1 (or, equivalently, if it can be given the trivial group as structure group).

In view of example 2, fiber bundles are related to covering spaces in much the same way that fibrations are related to fibrations with unique path lifting. The rest of this section is devoted to a proof of the fact that in a fiber bundle (E, B, F, p) whose base space B is a paracompact Hausdorff space the map p is a fibration.

A map $p: E \rightarrow B$ is called a *local fibration* if there is an open covering $\{U\}$ of B such that $p|_{p^{-1}(U)}: p^{-1}(U) \rightarrow U$ is a fibration for every $U \in \{U\}$. It is clear that a fibration is a local fibration¹ and that any bundle projection is a local fibration.

Given a map $p: E \rightarrow B$, we define a subspace $\tilde{B} \subset E \times B'$ by

$$\tilde{B} = \{(e, \omega) \in E \times B' \mid \omega(0) = p(e)\}$$

There is a map $\tilde{p}: \tilde{B} \rightarrow \tilde{B}$ defined by $\tilde{p}(\tilde{\omega}) = (\tilde{\omega}(0), p \circ \tilde{\omega})$ for $\tilde{\omega}: I \rightarrow \tilde{B}$. A *lifting function* for p is a map

$$\lambda: \tilde{B} \rightarrow E'$$

which is a right inverse of $\tilde{p}$. Thus a lifting function assigns to each point $e \in E$ and path ω in B starting at $p(e)$ a path $\lambda(e, \omega)$ in E starting at e that is a lift of ω . The relation between lifting functions and fibrations is contained in the following theorem.

THEOREM A map $p: E \rightarrow B$ is a fibration if and only if there exists a lifting function for p .

PROOF The proof involves repeated use of theorem 2.8 in the Introduction. If p is a fibration, let $f': B \rightarrow E$ and $F: B \times I \rightarrow B$ be defined by $f'(e, \omega) = e$

¹Our proof of the converse for paracompact Hausdorff spaces B can be found in W. Hurewicz, On the concept of fibre space, *Proceedings of the National Academy of Sciences, U.S.A.*, vol. 41, pp. 956-961 (1955). Another proof can be found in W. Huebsch, On the covering homotopy theorem, *Annals of Mathematics*, vol. 61, pp. 555-563 (1955). Generalizations and related questions are treated in A. Dold, Partitions of unity in the theory of fibrations, *Annals of Mathematics*, vol. 78, pp. 223-255 (1963).

and $F((e, \omega), t) = \omega(t)$. Then

$$F((e, \omega), 0) = \omega(0) = p(e) = (p \circ f')(e, \omega)$$

By the homotopy lifting property of p , there is a map $F': \bar{B} \times I \rightarrow E$ such that $F'((e, \omega), 0) = f'(e, \omega) = e$ and $p \circ F' = F$. F' defines a lifting function λ for p by $\lambda(e, \omega)(t) = F'((e, \omega), t)$.

Conversely, if λ is a lifting function for p , let $f': X \rightarrow E$ and $F: X \times I \rightarrow B$ be such that $F(x, 0) = pf'(x)$. Let $g: X \rightarrow B'$ be defined by $g(x)(t) = F(x, t)$. There is a map $F': X \times I \rightarrow E$ such that $F'(x, t) = \lambda(f'(x), g(x))(t)$. Because $F'(x, 0) = f'(x)$ and $p \circ F' = F$, p has the homotopy lifting property. ■

Let $p: E \rightarrow B$ and let W be a subset of B' . Let $\tilde{W}$ be defined by

$$\tilde{W} = \{(e, \omega, s) \in E \times W \times I \mid \omega(s) = p(e)\}$$

An *extended lifting function* over W is a map

$$\Lambda: \tilde{W} \rightarrow E'$$

such that $p(\Lambda(e, \omega, s)(t)) = \omega(t)$ and $\Lambda(e, \omega, s)(s) = e$. Thus an extended lifting function is a function which lifts paths to paths that pass through a given point of E at a given parameter value. It is reasonable to expect the following relation between the existence of lifting functions and extended lifting functions.

9 LEMMA *A map $p: E \rightarrow B$ has a lifting function if and only if there is an extended lifting function over B' .*

PROOF If Λ is an extended lifting function over B' , a lifting function λ for p is defined by $\lambda(e, \omega) = \Lambda(e, \omega, 0)$.

To prove the converse, given a path ω in B , let ω_s and ω^s be the paths in B defined by

$$\begin{aligned} \omega_s(t) &= \begin{cases} \omega(s-t) & 0 \leq t \leq s \\ \omega(0) & s \leq t \leq 1 \end{cases} \\ \omega^s(t) &= \begin{cases} \omega(s+t) & 0 \leq t \leq 1-s \\ \omega(1) & 1-s \leq t \leq 1 \end{cases} \end{aligned}$$

The maps $(\omega, s) \rightarrow \omega_s$ and $(\omega, s) \rightarrow \omega^s$ are continuous maps $B' \times I \rightarrow B'$. Given a lifting function $\lambda: \bar{B} \rightarrow E'$ for p , we define an extended lifting function Λ over B' by

$$\Lambda(e, \omega, s)(t) = \begin{cases} \lambda(e, \omega_s)(s-t) & 0 \leq t \leq s \\ \lambda(e, \omega^s)(t-s) & s \leq t \leq 1 \end{cases} \quad \blacksquare$$

The main step in proving that a local fibration is a fibration is the fitting together of extended lifting functions over various open subsets of B' . For this we need an additional concept. A covering $\{W\}$ of a space X is said to be *numerable* if it is locally finite and if for each W there is a function $f_W: X \rightarrow [0, 1]$ such that $W = \{x \in X \mid f_W(x) \neq 0\}$.

10 LEMMA *Let $p: E \rightarrow B$ be a map. If there is a numerable covering $\{W_i\}$ of B' such that for each i there is an extended lifting function over W_i , then there is a lifting function for p .*

PROOF Let the indexing set be $J = \{j\}$ and for each j let $f_j: B' \rightarrow I$ be a map such that $W_j = \{\omega \in B' \mid f_j(\omega) \neq 0\}$. For any subset $\alpha \subset J$ let $W_\alpha = \bigcup_{j \in \alpha} W_j$ and define $f_\alpha: B' \rightarrow \mathbf{R}$ by

$$f_\alpha(\omega) = \sum_{j \in \alpha} f_j(\omega)$$

(this is a finite sum and is continuous because $\{W_j\}$ is locally finite). Then $f_\alpha(\omega) \geq 0$ for $\omega \in B'$ and

$$W_\alpha = \{\omega \in B' \mid f_\alpha(\omega) \neq 0\}$$

We define $\bar{B}_\alpha = \{(e, \omega) \in \bar{B} \mid \omega \in W_\alpha\}$.

Consider the set of pairs (α, λ_α) , where $\alpha \subset J$ and $\lambda_\alpha: \bar{B}_\alpha \rightarrow E'$ is a lifting function over $\bar{B}_\alpha$ [that is, $\lambda_\alpha(e, \omega)(0) = e$ and $p\lambda_\alpha(e, \omega)(t) = \omega(t)$]. We define a partial order $\leq$ in this set by $(\alpha, \lambda_\alpha) \leq (\beta, \lambda_\beta)$ if $\alpha \subset \beta$ and $\lambda_\alpha(e, \omega) = \lambda_\beta(e, \omega)$ whenever $(e, \omega) \in \bar{B}_\alpha$ and $f_\alpha(\omega) = f_\beta(\omega)$ [so if $(e, \omega) \in \bar{B}_\alpha$ and $\lambda_\alpha(e, \omega) \neq \lambda_\beta(e, \omega)$, then $\omega \in W_j$ for some $j \in \beta - \alpha$].

To prove that every simply ordered subset $\{(\alpha_i, \lambda_{\alpha_i})\}$ has an upper bound, let $\beta = \bigcup \alpha_i$. We shall define $\lambda_\beta: \bar{B}_\beta \rightarrow E'$ so that $(\alpha_i, \lambda_{\alpha_i}) \leq (\beta, \lambda_\beta)$ for all i . Let U be any open subset of W_β meeting only finitely many W_j with $j \in \beta$. Say $W_{j_1}, \dots, W_{j_r}$ (W_β can be covered by such sets U). Choose i so that $j_1, \dots, j_r$ all belong to α_i . Then if $\alpha_i \subset \alpha_k$, $f_{\alpha_i} \mid U = f_{\alpha_k} \mid U$. Because $(\alpha_i, \lambda_{\alpha_i}) \leq (\alpha_k, \lambda_{\alpha_k})$, it follows that $\lambda_{\alpha_i}(e, \omega) = \lambda_{\alpha_k}(e, \omega)$ for $(e, \omega) \in \bar{B}_{\alpha_i}$, with $\omega \in U$. Therefore there exists a map $\lambda_\beta: \bar{B}_\beta \rightarrow E'$ such that $\lambda_\beta(e, \omega) = \lambda_{\alpha_i}(e, \omega)$ for α_i sufficiently large. We now show that $(\alpha_i, \lambda_{\alpha_i}) \leq (\beta, \lambda_\beta)$. If $(e, \omega) \in \bar{B}_{\alpha_i}$ and $\lambda_{\alpha_i}(e, \omega) \neq \lambda_\beta(e, \omega)$, there exists α_k such that $(\alpha_i, \lambda_{\alpha_i}) \leq (\alpha_k, \lambda_{\alpha_k})$ and $\lambda_{\alpha_i}(e, \omega) \neq \lambda_{\alpha_k}(e, \omega)$. This implies $\omega \in W_j$ for some $j \in \alpha_k - \alpha_i$. Therefore $\omega \in W_j$ for some $j \in \beta - \alpha_i$, hence $(\alpha_i, \lambda_{\alpha_i}) \leq (\beta, \lambda_\beta)$.

By Zorn's lemma, there is a maximal element (α, λ_α) . To complete the proof we need only show that $\alpha = J$. If $\alpha \neq J$, let $j_0 \in J - \alpha$ and let $\beta = \alpha \cup \{j_0\}$. Define $g: W_\beta \rightarrow \mathbf{R}$ by $g(\omega) = f_\alpha(\omega)/f_\beta(\omega)$. Then $0 \leq g(\omega) \leq 1$, $g(\omega) \neq 0 \Leftrightarrow \omega \in W_\alpha$, and $g(\omega) \neq 1 \Leftrightarrow \omega \in W_{j_0}$. Define $\mu: \bar{B}_\beta \rightarrow E$ by

$$\mu(e, \omega) = \begin{cases} \lambda_\alpha(e, \omega)(g(\omega)) & g(\omega) \neq 0 \\ e & g(\omega) = 0 \end{cases}$$

Then μ is continuous. Let Λ be an extended lifting function over W_{j_0} and define $\lambda_\beta: \bar{B}_\beta \rightarrow E'$ by

$$\lambda_\beta(e, \omega)(t) = \begin{cases} \lambda_\alpha(e, \omega)(t) & 0 \leq t \leq g(\omega), g(\omega) \neq 0 \\ \Lambda(\mu(e, \omega), \omega, g(\omega))(t) & g(\omega) \leq t \leq 1, g(\omega) \neq 1 \end{cases}$$

Then λ_β is a well-defined lifting function over W_{j_0} . Moreover, for $(e, \omega) \in \bar{B}_\alpha$, if $\lambda_\alpha(e, \omega) \neq \lambda_\beta(e, \omega)$, then $g(\omega) \neq 1$ and $\omega \in W_{j_0}$. Since $j_0 \in \beta - \alpha$, this means that $(\alpha, \lambda_\alpha) < (\beta, \lambda_\beta)$, contradicting the maximality of (α, λ_α) . ■

In case p has unique path lifting, lemma 10 would hold for any open covering $\{W_j\}$ of B' such that there is a lifting function over W_j for each j (because the uniqueness of lifted paths enables the extended liftings to be amalgamated to a lifting for p). This was used in the proof of the theorem that a covering projection is a fibration (theorem 2.2.3), which was valid without any assumption on the base space.

11 LEMMA Given a map $p: E \rightarrow B$ and subsets $U_1, \dots, U_k$ of B such that there is an extended lifting function over $U_1^t, U_2^t, \dots, U_k^t$, let W be the subset of B' defined by

$$W = \left\{ \omega \in B' \mid \omega \left(\left[\frac{i-1}{k}, \frac{i}{k} \right] \right) \subset U_i \text{ for } i = 1, \dots, k \right\}$$

Then there is an extended lifting function over W .

PROOF Let Λ_i be an extended lifting function over U_i^t for $i = 1, \dots, k$. Given a path $\omega \in W$, let ω_i be the path equal to ω on $[(i-1)/k, i/k]$ and constant on $[0, (i-1)/k]$ and on $[i/k, 1]$. Given $(e, \omega, s) \in W$ such that $(n-1)/k \leq s \leq n/k$, define $e_i \in E$ for $i = 1, \dots, k$ inductively so that

$$e_{n-1} = \Lambda_n(e, \omega_{n,s}) \left(\frac{n-1}{k} \right)$$

$$e_n = \Lambda_n(e, \omega_{n,s}) \left(\frac{n}{k} \right)$$

$$\text{and } e_{i-1} = \Lambda_{i-1} \left(e_i, \omega_{i-1, \frac{i}{k}} \right) \left(\frac{i-1}{k} \right) \quad 0 \leq i < n-1$$

$$e_{i+1} = \Lambda_{i+1} \left(e_i, \omega_{i+1, \frac{i}{k}} \right) \left(\frac{i+1}{k} \right) \quad n < i+1 \leq k$$

An extended lifting function Λ over W is defined by

$$\Lambda(e, \omega, s)(t) = \Lambda_i \left(e_{i-1}, \omega_{i-1, \frac{i}{k}} \right) \left(t \right) \quad \frac{i-1}{k} \leq t \leq \frac{i}{k}, i = 1, \dots, k \quad \blacksquare$$

We are now ready for the main result on the passage from a local fibration to a fibration.

12 THEOREM Given a map $p: E \rightarrow B$ and a numerable covering $\mathfrak{U}$ of B such that for $U \in \mathfrak{U}$, $p|_{p^{-1}(U)}: p^{-1}(U) \rightarrow U$ is a fibration, then p is a fibration.

PROOF Let $\mathfrak{U} = \{U_j\}$ and for $k \geq 1$, given a set of indices $i_1, \dots, i_k$, let $W_{j_1 i_1} \dots j_k i_k$ be the subset of B' defined by

$$W_{j_1 i_1} \dots j_k i_k = \left\{ \omega \in B' \mid \omega \left(\left[\frac{i-1}{k}, \frac{i}{k} \right] \right) \subset U_{j_i}, i = 1, \dots, k \right\}$$

It is then clear that the collection $\{W_{j_1 i_1} \dots j_k i_k\}$ (with k varying) is an open covering of B' , and by lemma 11, each set $W_{j_1 i_1} \dots j_k i_k$ has an extended lifting function. For k fixed the collection $\{W_{j_1 i_1} \dots j_k i_k\}$ is locally finite. In fact, if

$\omega \in B'$, for each $i = 1, \dots, k$ there is a neighborhood V_i of $\omega((i-1)/k, i/k)$ meeting only finitely many U_j . Then $\bigcap_{1 \leq i \leq k} \langle [(i-1)/k, i/k], V_i \rangle$ is a neighborhood of ω meeting only finitely many $\{W_{j_1 i_1} \dots j_k i_k\}$.

For each j let $f_j: B \rightarrow I$ be a continuous map such that $f_j(b) \neq 0$ if and only if $b \in U_j$. Define $f_{j_1 \dots j_k}: B' \rightarrow I$ by

$$\bar{f}_{j_1 \dots j_k}(\omega) = \inf \left\{ f_{j_i}(\omega(t)) \mid \frac{i-1}{k} \leq t \leq \frac{i}{k}, i = 1, \dots, k \right\}$$

Then $\bar{f}_{j_1 \dots j_k}(\omega) \neq 0$ if and only if $\omega \in W_{j_1 \dots j_k}$.

Unfortunately, the collection $\{W_{j_1 i_1} \dots j_k i_k\}$ (all k) is not locally finite, otherwise the proof would be complete by lemma 10. This difficulty is circumvented by modifying the sets $W_{j_1 i_1} \dots j_k i_k$. Since for fixed m the collection $\{W_{j_1 \dots j_k}\}$ with $k < m$ is locally finite, the sum of the functions $\bar{f}_{j_1 \dots j_k}$ with $k < m$ is a continuous real-valued function g_m on B' . Define

$$f_{j_1 \dots j_m}(\omega) = \inf(\sup(0, \bar{f}_{j_1 \dots j_m} - mg_m), 1)$$

Then $f_{j_1 \dots j_m}: B' \rightarrow I$ and we define $W_{j_1 \dots j_m} = \{\omega \in B' \mid f_{j_1 \dots j_m}(\omega) \neq 0\}$. Clearly, $W_{j_1 \dots j_m} \subset W_{j_1 \dots j_m}$; therefore there is an extended lifting function over $W_{j_1 \dots j_m}$. To complete the proof, it follows from lemma 10 that we need only verify that $\{W_{j_1 \dots j_k}\}$ (with k varying) is a locally finite covering of B' .

For $\omega \in B'$, let m be the smallest integer such that $f_{j_1 \dots j_m}(\omega) \neq 0$ for some $j_1, \dots, j_m$. Then $g_m(\omega) = 0$ and $f_{j_1 \dots j_m}(\omega) = f_{j_1 \dots j_m}(\omega) \neq 0$. Therefore $\omega \in W_{j_1 \dots j_m}$, proving that $\{W_{j_1 \dots j_m}\}$ is a covering of B' . To show that it is locally finite, assume N chosen so that $N > m$ and $f_{j_1 \dots j_m}(\omega) > 1/N$. Then $g_N(\omega) > 1/N$ and $N g_N(\omega) > 1$. Hence $N g_N(\omega') > 1$ for all ω' in some neighborhood V of ω . Therefore all functions $f_{j_1 \dots j_k}$ with $k \geq N$ vanish on V . But this means that the corresponding set $W_{j_1 \dots j_k}$ is disjoint from V . Since the collection $\{W_{j_1 \dots j_k}\}$ with $k < N$ is locally finite, the collection $\{W_{j_1 \dots j_k}\}$ (all k) is locally finite. $\blacksquare$

The fact that any open covering of a paracompact Hausdorff space has a numerable refinement, leads to our next theorem.

13 THEOREM If B is a paracompact Hausdorff space, a map $p: E \rightarrow B$ is a fibration if and only if it is a local fibration. $\blacksquare$

A bundle projection is a local fibration. Therefore, we have the following corollary.

14 COROLLARY If (E, B, F, p) is a fiber bundle with base space B paracompact and Hausdorff, then p is a fibration. $\blacksquare$

FIBRATIONS

This section contains a general discussion of fibrations. We establish a relation between cofibrations and fibrations which allows the construction of fibrations from cofibrations by means of function spaces. We also prove that every map is equivalent, up to homotopy, to a map that is a fibration (this dualizes a

Hausdorff spaces, so is $\bar{X}$, and by theorem 2.9 in the introduction, $Y^{X \times I} \simeq (Y^X)^I$ and

$$Y^{\bar{X}} \simeq \{(g, G) \in Y^X \times (Y^{X \times I}) \mid g \circ f = G(0)\}$$

Therefore ρ' corresponds to a lifting function for $p: Y^X \rightarrow Y^X$, and by theorem 2.7.8, p is a fibration. ■

3 COROLLARY For any space Y let $p: Y^I \rightarrow Y \times Y$ be the map $p(\omega) = (\omega(0), \omega(1))$ for $\omega: I \rightarrow Y$. Then p is a fibration.

PROOF Because $I \times I \cup I \times 0$ is a retract of $I \times I$, the inclusion map $I \subset I$ is a cofibration [equivalently, the pair (I, I) has the homotopy extension property with respect to any space]. The result follows from theorem 2 and the observation that Y^I is homeomorphic to $Y \times Y$ under the map $g \rightarrow (g(0), g(1))$ for $g: I \rightarrow Y$. ■

Let $f: B' \rightarrow B$ and $p: E \rightarrow B$ be maps and let E' be the subset of $B' \times E$ defined by

$$E' = \{(b', e) \in B' \times E \mid f(b') = p(e)\}$$

E' is called the *fibred product* of B' and E (more precisely, the fibred product of f and p ; cf. Sec. 2.2). Note that there are maps $p': E' \rightarrow B'$ and $f': E' \rightarrow E$ defined by $p'(b', e) = b'$ and $f'(b', e) = e$. E' and the maps p' and f' are characterized as the product of $f: B' \rightarrow B$ and $p: E \rightarrow B$ in the category whose objects are continuous maps with range B and whose morphisms are commutative triangles

$$\begin{array}{ccc} X_1 & \xrightarrow{h} & X_2 \\ g_1 \searrow & & \searrow g_2 \\ & B & \end{array}$$

The following properties are easily verified.

- 4** If p is injective (or surjective), so is p' . ■
- 5** If $p: B \times F \rightarrow B$ is the trivial fibration, then $p': E' \rightarrow B'$ is equivalent to the trivial fibration $B' \times F \rightarrow B'$. ■
- 6** If p is a fibration (with unique path lifting), so is p' . ■
- 7** If p is a fibration, f can be lifted to E if and only if p' has a section. ■

Note that since the fibred product is symmetric in B and E (or rather, in f and p), there is a similar set of statements where p and p' are replaced by f and f' .

If $p: E \rightarrow B$ is a fibration (or covering projection) and $f: B' \rightarrow B$ is a map, then, by property 6 (or property 5), $p': E' \rightarrow B'$ is a fibration (or covering projection) and is called the *fibration induced* from p by f (or *covering projection induced* from p by f). If $\xi = (E, B, F, p)$ is a fiber bundle and $f: B' \rightarrow B$ is a map, it follows from property 5 that there is a fiber bundle (E', B', F', p') . This is called the *fiber bundle induced* from ξ by f and is denoted by $f^* \xi$. In the case of an inclusion map $i: B' \subset B$ we use $E|B'$ to denote the fibred

similar result concerning cofibrations). The section contains definitions of the concepts of fiber homotopy type and induced fibration and a proof of the result that homotopic maps induce fiber-homotopy-equivalent fibrations.

We begin with an analogue of theorem 2.7.8 for cofibrations. Given a map $f: X' \rightarrow X$, let $\bar{X}$ be the quotient space of the sum $(X' \times I) \vee (X \times 0)$, obtained by identifying $(x', 0) \in X' \times I$ with $(f(x'), 0) \in X \times 0$ for all $x' \in X'$. We use $[x', t]$ and $[x, 0]$ to denote the points of $\bar{X}$ corresponding to $(x', t) \in X' \times I$ and $(x, 0) \in X \times 0$, respectively. Then $[x', 0] = [f(x'), 0]$. There is a map

$$\bar{i}: \bar{X} \rightarrow X \times I$$

defined by

$$\begin{aligned} \bar{i}[x', t] &= (f(x'), t) & x' \in X', t \in I \\ \bar{i}[x, 0] &= (x, 0) & x \in X \end{aligned}$$

A *retracting function* for f is a map

$$\rho: X \times I \rightarrow \bar{X}$$

which is a left inverse of $\bar{i}$. In case f is an inclusion map, so is $\bar{i}$, and a retracting function for f is a retraction of $X \times I$ to the subspace $X' \times I \cup X \times 0$.

1 THEOREM A map $f: X' \rightarrow X$ is a cofibration if and only if there exists a retracting function for f .

PROOF If f is a cofibration, let $g: X \rightarrow \bar{X}$ and $G: X' \times I \rightarrow \bar{X}$ be the maps defined by $g(x) = [x, 0]$ and $G(x', t) = [x', t]$. Because

$$G(x', 0) = [x', 0] = [f(x'), 0] = g f'(x')$$

it follows from the fact that f is a cofibration that there exists a map $\rho: X \times I \rightarrow \bar{X}$ such that $\rho(x, 0) = g(x)$ and $\rho(f(x'), t) = G(x', t)$. Then ρ is a retracting function for f .

Conversely, given maps $g: X \rightarrow Y$ and $G: X' \times I \rightarrow Y$ such that $G(x', 0) = g f'(x')$ for $x' \in X'$, define

$$\bar{G}: \bar{X} \rightarrow Y$$

by $\bar{G}[x', t] = G(x', t)$ and $\bar{G}[x, 0] = g(x)$. If $\rho: X \times I \rightarrow \bar{X}$ is a retracting function for f , the map $F = \bar{G} \circ \rho: X \times I \rightarrow Y$ has the properties $F(x, 0) = g(x)$ and $F(f(x'), t) = G(x', t)$, showing that f is a cofibration. ■

This leads to the following construction of fibrations from cofibrations.

2 THEOREM Let $f: X' \rightarrow X$ be a cofibration, where X' and X are locally compact Hausdorff spaces, and let Y be any space. Then the map $p: Y^{X \times I} \rightarrow Y^{X' \times I}$ defined by $p(\xi) = g \circ f$ is a fibration.

PROOF Let $\rho: X \times I \rightarrow \bar{X}$ be a retracting function for f (which exists by theorem 1). Then ρ defines a map

$$\rho': Y^{\bar{X}} \rightarrow Y^{X \times I}$$

such that $\rho'(\bar{g}) = g \circ \rho$ for $g: \bar{X} \rightarrow Y$. Because X' and X are locally compact

product of B' and E , and if ξ is a fiber bundle with base space B , $\xi|B'$ will denote the fiber bundle with base space B' induced by i . Observe that $\xi|B'$ is equivalent to $(p^{-1}(B'), B', F, p|p^{-1}(B'))$.

8 COROLLARY For any space Y and point $y_0 \in Y$, let $p: P(Y, y_0) \rightarrow Y$ be the map sending each path starting at y_0 to its endpoint. Then p is a fibration whose fiber over y_0 is the loop space ΩY .

PROOF Let $f: Y \rightarrow Y \times Y$ be defined by $f(y) = (y, y)$ and let $\tilde{p}: Y' \rightarrow Y \times Y$ be the fibration of corollary 3. The fibration induced by f is equivalent to the map $p: P(Y, y_0) \rightarrow Y$, where $p(\omega) = \omega(1)$, and $p^{-1}(y_0)$ the fiber over y_0 , is by definition, the loop space ΩY . ■

It follows from corollary 3 that the map $p': Y' \rightarrow Y$ defined by $p'(\omega) = \omega(0)$ [or by $p'(\omega) = \omega(1)$] is a fibration, because it is the composite of fibrations $Y' \rightarrow Y \times Y \rightarrow Y$. If $p: E \rightarrow B$ is any map and $p': B' \rightarrow B$ is the fibration defined by $p'(\omega) = \omega(0)$, then the fibered product of E and B' is just the space $\tilde{B}$ used to define the concept of lifting function for p .

These remarks about fibered products and induced fibrations have analogues for cofibrations. Given maps $f_1: X \rightarrow X_1$ and $f_2: X \rightarrow X_2$, the cofibered sum of X_1 and X_2 is the quotient space X' of $X_1 \vee X_2$ obtained by identifying $f_1(x)$ with $f_2(x)$ for all $x \in X$. There are maps $i_1: X_1 \rightarrow X'$ and $i_2: X_2 \rightarrow X'$, and these characterize X' as the sum of f_1 and f_2 in the category whose objects are maps with domain X and whose morphisms are commutative triangles. If $f_1: X \rightarrow X_1$ is a cofibration, so is $i_2: X_2 \rightarrow X'$, and this is called the *cofibration induced from f_1 by f_2* .

The map $h_0: X' \rightarrow X' \times I$ defined by $h_0(x') = (x', 0)$ is a cofibration for any space X' , and if $f: X' \rightarrow X$ is any map, the cofibered sum of $X' \times I$ and X is just the space $\tilde{X}$ used to define the concept of retracting function for f .

Let $p: E \rightarrow B$ be a fibration. Maps $f_0, f_1: X \rightarrow E$ are said to be *fiber homotopic*, denoted by $f_0 \approx f_1$, if there is a homotopy $F: f_0 \simeq f_1$ such that $pF(x, t) = pf_0(x)$ for $x \in X$ and $t \in I$ (in which case $p \circ f_0 = p \circ f_1$). This is an equivalence relation in the set of maps $X \rightarrow E$. The equivalence classes are denoted by $[X; E]_p$, and if $f: X \rightarrow E$, $[f]_p$ denotes its fiber homotopy class. The concept of fiber homotopy is dual to the concept of relative homotopy.

We use induced fibrations to prove that any map is, up to homotopy equivalence, a fibration. Let $f: X \rightarrow Y$ and let $p': Y' \rightarrow Y$ be the fibration defined by $p'(\omega) = \omega(0)$. Let $p: P_I \rightarrow X$ be the fibration induced from p' by f . It is called the *mapping path fibration* of f and is dual to the mapping cylinder. There is a section $s: X \rightarrow P_I$ of p defined by $s(x) = (x, \omega_{(x)})$, where $\omega_{(x)}$ is the constant path in Y at $f(x)$. There is also a map $p'': P_I \rightarrow Y$ defined by $p''(x, \omega) = \omega(1)$. We then have the following dual of theorem 1.4.12.

9 THEOREM Given a map $f: X \rightarrow Y$, there is a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{s} & P_I \\ f \searrow & & \swarrow p'' \\ & Y & \end{array}$$

such that

- (a) $1_{P_I} \approx s \circ p$
- (b) p'' is a fibration

PROOF The triangle is commutative by the definition of the maps involved. (a) Define $F: P_I \times I \rightarrow P_I$ by $F(x, \omega), t) = (x, \omega_{1-t})$, where $\omega_{1-t}(t') = \omega((1-t)t')$. Then F is a fiber homotopy from 1_{P_I} to $s \circ p$.

(b) Let $g: W \rightarrow P_I$ and $G: W \times I \rightarrow Y$ be such that $G(w, 0) = p''g(w)$ for $w \in W$. Then there exist maps $g': W \rightarrow X$ and $g'': W \rightarrow Y'$ such that $g'(w)(0) = fg'(w)$ and $g(w) = (g'(w), g''(w))$ for $w \in W$. We define a lifting $G': W \times I \rightarrow P_I$ of G beginning with g by $G'(w, t) = (g'(w), \tilde{g}(w, t))$, where $\tilde{g}(w, t) \in Y'$ is defined by

$$\tilde{g}(w, t)(t') = \begin{cases} g''(w)(2t'/(2-t)) & 0 \leq 2t' \leq 2-t \leq 2, w \in W \\ G(w, 2t' + t - 2) & 1 \leq 2-t \leq 2t' \leq 2, w \in W \end{cases}$$

Since p'' has the homotopy lifting property, it is a fibration. ■

It follows that the fibration $p'': P_I \rightarrow Y$ is equivalent (by means of $s: X \rightarrow P_I$ and $p: P_I \rightarrow X$) in the homotopy category of maps with range Y to the original map $f: X \rightarrow Y$. In replacing f by an equivalent fibration, we replaced X by a space P_I of the same homotopy type, whereas in Sec. 1.4, when f was replaced by an equivalent cofibration, the space Y was replaced by a space Z_I of the same homotopy type.

Two fibrations $p_1: E_1 \rightarrow B$ and $p_2: E_2 \rightarrow B$ are said to be *fiber homotopy equivalent* (or to have the *same fiber homotopy type*) if there exist maps $f: E_1 \rightarrow E_2$ and $g: E_2 \rightarrow E_1$ such that $g \circ f \approx 1_{E_1}$, and $f \circ g \approx 1_{E_2}$ (in which case f and g both preserve fibers in the sense that $p_2 \circ f = p_1$ and $p_1 \circ g = p_2$). Each of the maps f and g is called a *fiber homotopy equivalence*. The rest of this section is concerned with fiber homotopy equivalence.

We begin with the following result concerning liftings of homotopic maps.

10 THEOREM Let $p: E \rightarrow B$ be a fibration and let $F_0, F_1: X \times I \rightarrow E$ be maps. Given homotopies $H: p \circ F_0 \simeq p \circ F_1$ and $G: F_0|X \times 0 \simeq F_1|X \times 0$ such that $H(x, 0, t) = pG(x, 0, t)$, there is a lifting $H': X \times I \times I \rightarrow E$ of H which is a homotopy from F_0 to F_1 and is an extension of G .

PROOF Let $A = (I \times 0) \cup (0 \times I) \cup (I \times I) \subset I \times I$ and define $f: X \times A \rightarrow E$ by

$$\begin{aligned} f(x, t, 0) &= F_0(x, t) \\ f(x, 0, t) &= G(x, 0, t) \\ f(x, t, 1) &= F_1(x, t) \end{aligned}$$

Then $H|X \times A = p \circ f$. Because there is a homeomorphism of $I \times I$ with itself taking A onto $I \times 0$, there is a homeomorphism of $X \times I \times I$ with itself taking $X \times A$ onto $X \times I \times 0$. It follows from the homotopy lifting property of p that there is a lifting $H': X \times I \times I \rightarrow E$ of H such that $H'|X \times A = f$. ■

Taking H and G to be constant homotopies, we obtain the following corollary.

11 COROLLARY *Let $p: E \rightarrow B$ be a fibration and let $F_0, F_1: X \times I \rightarrow E$ be liftings of the same map such that $F_0|_X \times 0 = F_1|_X \times 0$. Then $F_0 \simeq_{\overline{p}} F_1 \text{ rel } X \times 0$. ■*

Let $p: E \rightarrow B$ be a fibration and let $\omega: I \rightarrow B$ be a path in its base space. By the homotopy lifting property of p , there exists a map $F: p^{-1}(\omega(0)) \times I \rightarrow E$ such that $pF(x, t) = \omega(t)$ and $F(x, 0) = x$ for $x \in p^{-1}(\omega(0))$ and $t \in I$. Let $f: p^{-1}(\omega(0)) \rightarrow p^{-1}(\omega(1))$ be the map $f(x) = F(x, 1)$. It follows from theorem 10 that if $\omega \simeq \omega'$ are homotopic paths in B and if $F, F': p^{-1}(\omega(0)) \times I \rightarrow E$ are such that $pF(x, t) = \omega(t)$, $pF'(x, t) = \omega'(t)$, and $F(x, 0) = x = F'(x, 0)$ for $x \in p^{-1}(\omega(0))$ and $t \in I$, then the maps $f, f': p^{-1}(\omega(0)) \rightarrow p^{-1}(\omega(1))$ defined by $f(x) = F(x, 1)$ and $f'(x) = F'(x, 1)$ are homotopic. Hence there is a well-defined homotopy class $[f] \in [p^{-1}(\omega(0)); p^{-1}(\omega(1))]$ corresponding to a path class $[\omega]$ in B . We let $h[\omega] = [f]$.

The following is the form theorem 2.3.7 takes for an arbitrary fibration.

12 THEOREM *Let $p: E \rightarrow B$ be a fibration. There is a contravariant functor from the fundamental groupoid of B to the homotopy category which assigns to $b \in B$ the fiber over b and to a path class $[\omega]$ the homotopy class $h[\omega]$.*

PROOF If ω_b is the constant path at b , let $F: p^{-1}(b) \times I \rightarrow E$ be the map $F(x, t) = x$. The corresponding map $f: p^{-1}(b) \rightarrow p^{-1}(b)$ defined by $f(x) = F(x, 1)$ is the identity map. Hence

$$h[\omega_b] = [1_{p^{-1}(b)}]$$

showing that h preserves identities.

Let ω and ω' be paths in B such that $\omega(1) = \omega'(0)$. Given a map $F: p^{-1}(\omega(0)) \times I \rightarrow E$ such that $F(x, 0) = x$ and $pF(x, t) = \omega(t)$ for $x \in p^{-1}(\omega(0))$ and $t \in I$, and given $F': p^{-1}(\omega(1)) \times I \rightarrow E$ such that $F'(x', 0) = x'$ and $pF'(x', t) = \omega'(t)$ for $x' \in p^{-1}(\omega'(0))$ and $t \in I$, let $f: p^{-1}(\omega(0)) \rightarrow p^{-1}(\omega'(0))$ be defined by $f(x) = F(x, 1)$ and let $F'': p^{-1}(\omega(0)) \times I \rightarrow E$ be defined by

$$F''(x, t) = \begin{cases} F(x, 2t) & 0 \leq t \leq \frac{1}{2}, x \in p^{-1}(\omega(0)) \\ F'(f(x), 2t - 1) & \frac{1}{2} \leq t \leq 1, x \in p^{-1}(\omega(0)) \end{cases}$$

Then $pF''(x, t) = (\omega * \omega')(t)$ and $F''(x, 0) = x$ for $x \in p^{-1}(\omega(0))$ and $t \in I$. Let $f': p^{-1}(\omega'(0)) \rightarrow p^{-1}(\omega(1))$ be defined by $f'(x') = F'(x', 1)$. Then $F''(x, 1) = f'(f(x))$ for $x \in p^{-1}(\omega(0))$, which shows that

$$h[\omega * \omega'] = h[\omega'] * h[\omega]$$

Therefore h is a contravariant functor. ■

This yields the following analogue of corollary 2.3.8 for an arbitrary fibration.

13 COROLLARY *If $p: E \rightarrow B$ is a fibration with a path-connected base space, any two fibers have the same homotopy type. ■*

The following result asserts that homotopic maps induce fiber-homotopy-equivalent fibrations.

14 THEOREM *Let $p: E \rightarrow B$ be a fibration and let $f_0, f_1: X \rightarrow B$ be homotopic. The fibrations induced from p by f_0 and by f_1 are fiber homotopy equivalent.*

PROOF Let $p_0: E_0 \rightarrow X$ and $p_1: E_1 \rightarrow X$ be the fibrations induced from p by f_0 and f_1 , respectively, and let $f'_0: E_0 \rightarrow E$ and $f'_1: E_1 \rightarrow E$ be the corresponding maps such that $p \circ f'_0 = f_0 \circ p_0$ and $p \circ f'_1 = f_1 \circ p_1$. Given a homotopy $F: X \times I \rightarrow B$ from f_0 to f_1 , there are maps $F'_0: E_0 \times I \rightarrow E$ and $F'_1: E_1 \times I \rightarrow E$ such that $p \circ F'_0 = F \circ (p_0 \times 1_I)$ and $p \circ F'_1 = F \circ (p_1 \times 1_I)$ and $F'_0|_{E_0 \times 0} = f'_0$ and $F'_1|_{E_1 \times 0} = f'_1$. Let $g_0: E_0 \rightarrow E_1$ and $g_1: E_1 \rightarrow E_0$ be the maps defined by the property $F'_0(x, 1) = f'_1(g_0(x))$ for $x \in E_0$ and $F'_1(y, 0) = f'_0(g_1(y))$ for $y \in E_1$. Then

$$p \circ F'_0 \circ (g_1 \times 1_I) = F \circ (p_0 \times 1_I) \circ (g_1 \times 1_I) = F \circ (p_1 \times 1_I)$$

and $F'_0 \circ (g_1 \times 1_I)|_{E_1 \times 0} = F'_1|_{E_1 \times 0}$

It follows from theorem 10 that $F'_1 \simeq_{\overline{p}} F'_0 \circ (g_1 \times 1_I)$. In a similar fashion $F'_0 \simeq_{\overline{p}} F'_1 \circ (g_0 \times 1_I)$. This implies that $g_0 g_1 \simeq_{\overline{p}} 1_{E_1}$ and $g_1 g_0 \simeq_{\overline{p}} 1_{E_0}$. ■

Clearly, a constant map induces a trivial fibration, and we have the following result.

15 COROLLARY *If $p: E \rightarrow B$ is a fibration and B is contractible, then p is fiber homotopy equivalent to the trivial fibration $B \times p^{-1}(b_0) \rightarrow B$ for any $b_0 \in B$. ■*

Let B be a space which is the join of some space Y with S^0 . Then $B = C_-Y \cup C_+Y$, where C_-Y and C_+Y are cones over Y and $C_-Y \cap C_+Y = Y$. Let $y_0 \in Y$ and let $p: E \rightarrow B$ be a fibration with fiber $F_0 = p^{-1}(y_0)$. It follows from corollary 15 that there are fiber homotopy equivalences $f_-: C_-Y \times F_0 \rightarrow p^{-1}(C_-Y)$ and $g_+: p^{-1}(C_+Y) \rightarrow C_+Y \times F_0$. A clutching function $\mu: Y \times F_0 \rightarrow F_0$ for p is a function μ defined by the equation

$$g_+ f_-(y, z) = (y, \mu(y, z)) \quad y \in Y, z \in F_0$$

where $f_-: C_-Y \times F_0 \rightarrow p^{-1}(C_-Y)$ and $g_+: p^{-1}(C_+Y) \rightarrow C_+Y \times F_0$ are fiber homotopy equivalences. If C_-Y and C_+Y are contractible to y_0 relative to y_0 , it follows from theorem 10 that f_- and g_+ can be chosen so that $z \rightarrow f_-(y_0, z)$ is homotopic to the map $F_0 \subset p^{-1}(C_-Y)$ and $z \rightarrow g_+(z)$ is homotopic to the map $z \rightarrow (y_0, z)$ of F_0 to $C_+Y \times F_0$. In this case the clutching function μ corresponding to f_- and g_+ has the property that the map $z \rightarrow \mu(y_0, z)$ is homotopic to the identity map $F_0 \subset F_0$.

Let E_ϕ be the fiber bundle over S^n defined by a characteristic map $\phi: S^{n-1} \rightarrow G$, as in example 2.7.5 (where G is a group of homeomorphisms of the fiber F). Then $E_\phi^- = C_-S^{n-1}$ and $E_\phi^+ = C_+S^{n-1}$, and it is easy to verify that f_- and g_+ can be chosen so that the corresponding clutching function $\mu: S^{n-1} \times F \rightarrow F$ is the map $\mu(x, z) = \phi(x)z$.

EXERCISES

A LOCAL CONNECTEDNESS

- 1 Prove that a space X is locally path connected if and only if for any neighborhood U of x in X there exists a neighborhood V of x such that every pair of points in V can be joined by a path in U .
- 2 If X is a space, let $\tilde{X}$ denote the set X retopologized by the topology generated by path components of open sets of X . Prove that $\tilde{X}$ is locally path connected and that the identity map of X is a continuous function $j: \tilde{X} \rightarrow X$ having the property that for any locally path-connected space Y a function $f: Y \rightarrow \tilde{X}$ is continuous if and only if $j \circ f: Y \rightarrow X$ is continuous.
- 3 For any space X let $\tilde{X}$ and $j: \tilde{X} \rightarrow X$ be as in exercise 2. Prove that $j_{\#}: \pi(\tilde{X}, x_0) \simeq \pi(X, x_0)$.

B COVERING SPACES

- 1 Let X be the union of two closed simply connected and locally path-connected subsets A and B such that $A \cap B$ consists of a single point. Prove that if $p: \tilde{X} \rightarrow X$ is a non-empty path-connected fibration with unique path lifting, then p is a homeomorphism.
- 2 Let $\tilde{X} = \{(x, y) \in \mathbb{R}^2 \mid x \text{ or } y \text{ an integer}\}$ and let

$$X = S^1 \vee S^1 = \{(x_1, x_2) \in S^1 \times S^1 \mid x_1 = 1 \text{ or } x_2 = 1\}$$

Prove that the map $p: \tilde{X} \rightarrow X$ such that $p(x, y) = (e^{2\pi i x}, e^{2\pi i y})$ is a covering projection.

- 3 With $p: \tilde{X} \rightarrow X$ as in exercise 2 above, let $Y \subset \tilde{X}$ be defined by

$$Y = \{(x, y) \in \tilde{X} \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$$

Prove that Y is a retract of $\tilde{X}$ and that $(p|_Y)_*$ maps a generator of $\pi(Y)$ to the commutator of the two elements of $\pi(X)$ corresponding to the two circles of X .

- 4 Prove that $\pi(S^1 \vee S^1)$ is nonabelian.

C THE COVERING SPACE $ex: \mathbb{R} \rightarrow S^1$

- 1 For an arbitrary space X , prove that a map $f: X \rightarrow S^1$ can be lifted to a map $\tilde{f}: X \rightarrow \mathbb{R}$ such that $f = ex \circ \tilde{f}$ if and only if f is null homotopic.
- 2 Let X be a connected locally path-connected space with base point $x_0 \in X$. Prove that the map

$$[X, x_0, S^1, 1] \rightarrow \text{Hom}(\pi(X, x_0), \pi(S^1, 1))$$

which assigns to $[f]$ the homomorphism

$$f_{\#}: \pi(X, x_0) \rightarrow \pi(S^1, 1)$$

is a monomorphism (the set of homotopy classes being a group by virtue of the group structure on S^1).

- 3 Prove that any two maps from a simply connected locally path-connected space to S^1 are homotopic.
- 4 Prove that any map of the real projective space P^n for $n \geq 2$ to S^1 is null homotopic.

- 5 Prove that there is no map $f: S^n \rightarrow S^1$ for $n \geq 2$ such that $f(-x) = -f(x)$.
- 6 *Borsuk-Ulam theorem.* Prove that if $f: S^2 \rightarrow \mathbb{R}^2$ is a map such that $f(-x) = -f(x)$, then there exists a point $x_0 \in S^2$ such that $f(x_0) = 0$.

D COVERING SPACES OF TOPOLOGICAL GROUPS

- 1 Let H be a subgroup of a topological group and let G/H be the homogeneous space of right cosets. Prove that the projection $G \rightarrow G/H$ is a covering projection if and only if H is discrete.
- 2 Prove that a connected locally path-connected covering space of a topological group can be given a group structure that makes it a topological group and makes the projection map a homomorphism.

A *local homomorphism* φ from one topological group G to another G' is a continuous map from some neighborhood U of e in G to G' such that if $g_1, g_2, g_1 g_2 \in U$, then $\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$. A *local isomorphism* from G to G' is a homeomorphism φ from some neighborhood U of e to some neighborhood U' of e' such that φ and φ^{-1} are both local homomorphisms (in which case G and G' are said to be *locally isomorphic*).

- 3 Prove that a continuous homomorphism $\varphi: G \rightarrow G'$ between connected topological groups is a covering projection if and only if there exists a neighborhood U of e in G such that $\varphi|_U$ is a local isomorphism from G to G' .
- 4 Let φ be a local homomorphism from a connected topological group G to a topological group G' defined on a connected neighborhood U of e in G . Let $\tilde{G}$ be the subgroup of $G \times G'$ generated by the graph of φ (that is, generated by $\{(g, g') \in G \times G' \mid g' = \varphi(g), g \in U\}$). $\tilde{G}$ is topologized by taking as a base for neighborhoods of (e, e') the graph of $\varphi|_N$ as N varies over neighborhoods of e in U . Prove that $\tilde{G}$ is a connected topological group, the projection $p_1: \tilde{G} \rightarrow G$ is a covering projection, and the projection $p_2: \tilde{G} \rightarrow G'$ is continuous.

- 5 Prove that two connected locally path-connected topological groups are locally isomorphic if and only if there is a topological group which is a covering space of each of them.

- 6 If G is a simply connected locally path-connected topological group and φ is a local homomorphism from G to a topological group G' , prove that there is a continuous homomorphism $\varphi': G \rightarrow G'$ which agrees with φ on some neighborhood of e in G .

E FIBRATIONS

- 1 If $p: E \rightarrow B$ is a fibration, prove that $p(E)$ is a union of path components of B .
- 2 If a fibration has path-connected base and some fiber is path connected, prove that its total space is also path connected.
- 3 Let $p: E \rightarrow B$ be a fibration and let X be a locally compact Hausdorff space. Define $p': E^X \rightarrow B^X$ by $p'(g) = p \circ g$ for $g: X \rightarrow E$. Prove that p' is a fibration.
- 4 Let $p: E \rightarrow B$ be a fibration and let $b_0 \in p(E)$, $F = p^{-1}(b_0)$. Let X be a space regarded as a subset of some cone CX . Prove that the map

$$p_{\#}: [CX, X; E, F] \rightarrow [CX, X; B, b_0]$$

is a bijection.

- 5 Let $p: E \rightarrow B$ be a fibration and let $e_0 \in E$, $b_0 = p(e_0)$, and $F = p^{-1}(b_0)$. If B is simply connected, prove that $\pi(F, e_0) \rightarrow \pi(E, e_0)$ is an epimorphism.

6 Let $p: E \rightarrow B$ be a fibration and let $e_0 \in E$ and $b_0 = p(e_0)$. If $p^{-1}(b_0)$ is simply connected, prove that

$$\rho_{\#}: \pi(L, e_0) \simeq \pi(B, b_0)$$

7 Let $p: E \rightarrow B$ be a fibration and $b_0 \in p(E)$. If E is simply connected, prove that there is a bijection between $\pi(B, b_0)$ and the set of path components of $p^{-1}(b_0)$.